

KALAUPAPA NATIONAL HISTORICAL PARK (KALA) MOLOKA'I, HI

**REPLACE HVAC AND HUMIDITY CONTROL SYSTEM
IN THE PARK'S CURATORIAL FACILITY**

PMIS No. 339997

CONSTRUCTION SPECIFICATIONS



**NATIONAL PARK SERVICE
PACIFIC WEST REGION**

MARCH 2026

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DIVISION 01

GENERAL REQUIREMENTS

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SECTION 01 00 00 – SUMMARY OF WORK

PART 1 – GENERAL

1.1 SUMMARY

A. Section includes the following:

1. Work Covered by Contract Documents
2. Work Phases / Sequence
3. Work Under Other Contracts
4. Government-Furnished Items
5. Government-Furnished Documents
6. Conduct of Operations
7. Contractor Use of the Site
8. Public Use of the Site
9. Government Use of the Site
10. Occupancy Requirements for Buildings & Areas
11. Codes
12. Permits
13. Professional Services
14. Utility Interruptions
15. Vehicles & Equipment

1.2 WORK COVERED BY CONTRACT DOCUMENTS

- A. Project Location: Curatorial Facility, Kalaupapa National Historical Park (KALA), Molokai. Mobilization of materials to KALA is either by small aircraft with daily flights on Mokulele Airlines (passenger service), Kamaka Air, LLC (courier service), or on the annual barge with a sail date from Honolulu, HI (typically in mid-August). Refer to GFD #1 – Aerial Map.
- B. Project Background: In 2004, a new curatorial facility was constructed at KALA to house the park's museum collections and archive documents. Refer to applicable sheets of GFD #2 – Dwg. No. 491-41016 (New Curatorial Facility). A central air-conditioning & ventilation system was installed in the south wing of the curatorial facility to include a Carrier Model 38HDC024 (ACCU-1) and Carrier Model FC48N030 (FCU-1). Note that due to budget limitations, the north wing of the curatorial facility was not finished until 2011 (Refer to GFD #3 below).

In 2011, the north wing of the curatorial facility was renovated and finished. Refer to GFD #3 – Dwg No. 491-80037 (North Wing Renovations). The construction plans indicate that a Carrier Model 38HDR030 (ACCU-2) and Carrier Model FV4BNF003 (FCU-2) were installed. However, a service contract in 2022 later replaced a Rheem Model RANL-031JAZ (ACCU-2) in the north wing of the curatorial facility, so there may be some discrepancies between what is documented in the construction drawings and what was installed in the field.

Recently, a new electrical transformer was installed within the courtyard of the curatorial facility. Refer to GFD #4 – New Electrical Transformer at Curatorial.

Refer also to GFD #5 – Photos of Curatorial Facility (Exterior & Interior).

The existing air-conditioning system in the curatorial facility is beyond its service life and needs to be replaced. In addition, the park would like to install portable commercial dehumidifiers in each of the rooms that store museum collections. To properly care for the museum collections, the temperature in these rooms must be maintained throughout the year between 55-68 degrees Fahrenheit with a tolerance of +/- 5 degrees, and relative humidity must be maintained between 40-55% with a tolerance of +/- 5%.

C. Project Scope of Work: The Work consists of the following:

1. Furnish and install new mini-split air-conditioning systems in the curatorial facility as outlined in the equipment schedule below.

FAN COIL UNITS (FCU)				
ID	LOCATION	EQUIP TYPE	NOMINAL COOLING (BTU/HR)	MANUFACTURER & MODEL
FCU 1-1	South Storage	Wall Mounted	30,000	Mitsubishi MSZ-GX30NL or equal
FCU 1-2	South Storage	Wall Mounted	30,000	Mitsubishi MSZ-GX30NL or equal
FCU 2-1	North Storage	Wall Mounted	30,000	Mitsubishi MSZ-GX30NL or equal
FCU 2-2	North Storage	Wall Mounted	30,000	Mitsubishi MSZ-GX30NL or equal
FCU 3-1	South Receiving	Wall Mounted	24,000	Mitsubishi MSZ-WR24NA-U1 or equal
FCU 4-1	Administration	Wall Mounted	24,000	Mitsubishi MSZ-GX24NL or equal
FCU 4-2	Administration	Wall Mounted	24,000	Mitsubishi MSZ-GX24NL or equal
FCU 5-1	North Receiving	Wall Mounted	24,000	Mitsubishi MSZ-WR24NA-U1 or equal

AIR COOLED CONDENSING UNITS (ACCU)				
ID	ZONE	POWER	NOMINAL COOLING (BTU/HR)	MANUFACTURER & MODEL
ACCU-1	South Storage	208/230V 1-Phase	60,000	Mitsubishi MXZ-SM60NL or equal
ACCU-2	North Storage	208/230V 1-Phase	60,000	Mitsubishi MXZ-SM60NL or equal
ACCU-3	South Receiving	208/230V 1-Phase	24,000	Mitsubishi MUZ-WR24NL-U1 or equal
ACCU-4	Administration	208/230V 1-Phase	48,000	Mitsubishi MXZ-SM48NL or
ACCU-5	North Receiving	208/230V 1-Phase	24,000	Mitsubishi MUZ-WR24NL-U1 or equal

The Contractor shall develop / submit shop drawings for the work. Refer to PART 1.14.B. The work will be installed per the approved shop drawings.

The air-conditioning systems shall be complete and operable, installed per the manufacturer's instructions, to include all recommended components and appurtenances.

The concrete pads shall be flat and level. The concrete pads shall be reinforced with welded wire mesh. The concrete pads will be 4-inches thick and extend at least 1-foot beyond the ACCU unit on all sides. The Contractor shall use Quikrete 5000 Concrete Mix due to the prohibitions of transporting gravel between islands due to the coconut rhinoceros beetle.

To save time and money on the project, the park could form and pour the reinforced concrete pads for each of the five (5) ACCU prior to Contractor mobilization, that way the concrete has time to cure before Contractor arrival. This work is presented as a contract deduct in Section 01 27 00, Option A in the event the park chooses to complete this work.

The Contractor shall furnish and install the preferred models indicated in the schedule above. If the Contractor submits a substitution for the manufacturer or model, the units must:

- Meet or exceed the salient characteristics of the preferred model;
- Be the same manufacturer for all applications;
- Meet the Buy American Act (or exceptions);
- Must use the same refrigerant type;
- Replacement parts and service must be readily available from a service dealer on Oahu.

The Contractor shall install the FCU and ACCU units in the locations indicated on GFD #6. If alternative locations are recommended, show the proposed locations in the shop drawings.

The FCU units shall be wall-mounted, installed in areas and heights that do not conflict with existing shelves and rolling storage units. The park POC can advise on preferred locations.

For exterior conduit that houses wiring and refrigerant lines, consider non-metallic conduit due to the highly corrosive environment (where allowed by code).

Properly seal all penetrations through building elements. If element is fire rated, provide UL approved fire stop seals. Patch and paint all new work, color to match existing surfaces.

Once the new mini-split air-conditioning systems are connected and operational, disconnect the existing centralized air-conditioning systems from the electrical panels. The park will complete all demolition, removal, and disposal of existing air-conditioning systems.

In addition, all refuse from this work will be disposed of by the park on the subsequent barge. Ensure all refuse is placed in government-furnished containers.

The park POC will indicate the system set point for each of the FCU units.

2. Furnish and deliver four (4) new commercial dehumidifier units. The humidifiers shall be PERFECT AIRE, Portable Dehumidifier, Model 3PACD250, Std Refrigerant, R-32, 33-inch x 23-inch x 22-inch units. The park will install these dehumidifiers.

3. Project will be constructed under a single prime contract. Refer also to Section 01 27 00 – Definition of Contract Line Items.

- D. Project Mitigations: The Contractor shall comply with Section 01 11 40 – Work Restrictions & Resource Protection.

1.3 WORK PHASES / SEQUENCE

- A. Work Phases: The Work will be completed in one (1) single work phase.

1.4 WORK UNDER OTHER CONTRACTS

- A. General: Cooperate with separate contractors so that all contracts may be carried out smoothly without interfering with or delaying Work under this or other contracts. If another contract is awarded that may have an impact on this contract, the Contractor will be promptly notified.

1.5 GOVERNMENT-FURNISHED ITEMS

- A. General: The Government may furnish items such as materials, equipment, facilities, services, or housing for Contractor use on this contract. The government may furnish these items to lower the overall project costs, make use of existing item(s), or out of necessity in a remote location. The Contractor can assume that all items offered below are at no direct cost to the Contractor.

- B. Government-Furnished Materials: The following materials will be furnished to the Contractor for use on this contract:

1. Material #1: Concrete Pads for ACCU units (if Option A deduct is exercised).

In the event government-furnished materials are offered (at the outset or furnished later):

- a) The Government will arrange and pay for delivery of government-furnished materials to the project site or agreed upon location based on the contractor's approved construction schedule.
- b) The Government will inspect the delivered items for damage after delivery. The Contractor shall be present for, and assist in, the Government's inspection.
- c) The Government will arrange for the replacement of materials if the government-furnished materials are damaged, defective, or missing.
- d) The Contractor shall do their due diligence to protect government-furnished materials from damage during storage and handling, including damage from exposure to the elements.
- e) The Contractor shall install or otherwise incorporate government-furnished materials into the Work of this contract, as outlined in the contract documents.

- C. Government-Furnished Equipment / Transportation: The following equipment will be furnished to the Contractor for use on this contract at no cost to the Contractor.

1. Equipment: Heavy Equipment, operator, and fuel needed to transport items from the KALA pier to the curatorial facility and into position, as needed.
2. Ladders: Any ladders or scaffolding can be provided by the park.
3. Vehicles: Vehicle (Pickup or Crew cab) to use for Work while in the settlement.
4. Barge Delivery: Space on the mid-August barge for all materials and items. If a small item does not make the 2026 barge, Kamaka Air can transport items.

In the event government-furnished equipment is offered (at the outset or furnished later):

- a) The Government will make the equipment available for Contractor use based on the Contractor's approved construction schedule.
- b) The Government will inspect the equipment and document its condition prior to Contractor use. The Contractor shall be present for, and assist in, the Government's inspection.
- c) The Contractor shall provide all licensed, certified equipment operators.
- d) The Government will provide an orientation and training on the equipment prior to use.
- e) The Contractor shall provide all fuels / oils to operate the equipment, unless stated otherwise.
- f) The Government will arrange for the replacement or repair of equipment, or provide an alternative, if the government-furnished equipment breaks during use.
- g) The Contractor shall do their due diligence to protect the government-furnished equipment from damage, including damage from exposure to the elements.

D. Government-Furnished Facilities: The following facilities will be available to the Contractor for use on this contract. Refer also to Section 01 50 00 Temporary Facilities & Controls.

1. Restroom: The Contractor may use the restroom in the curatorial facility.
2. Field Office: NOT AVAILABLE. The Contractor may use their housing unit as an office. Note that internet is not available in the settlement, and phone service is limited.
3. Storage Space: The park will store items under cover or in a warehouse, as needed.

E. Government-Furnished Utilities: The following utilities will be available to the Contractor for use on this contract. Refer also to Section 01 50 00 Temporary Facilities & Controls.

1. Potable Water: Available at the curatorial facility.
2. Non-Potable Water: NOT AVAILABLE.
3. Wastewater Disposal: NOT AVAILABLE.
4. Hydrotesting & Disinfection Water Disposal: NOT AVAILABLE.
5. Electrical Power: Available at the curatorial facility.

F. Government-Furnished Housing: Housing will be available to the Contractor at no cost:

1. Kalaupapa National Historical Park: House, Basic Furnishings & Cookware.

1.6 GOVERNMENT-FURNISHED DOCUMENTS (GFD)

A. General: The Government may furnish reports or other documents for informational purposes to supplement the contract documents (labeled as GFD). In cases of conflict between a GFD and the drawings and specifications, the drawings and specifications will govern.

1.7 CONDUCT OF OPERATIONS

A. Park Regulations: The Contractor shall conduct their operations in conformance with the rules and regulations promulgated by the Secretary of the Interior for the National Park Service (NPS), and applicable park rules and regulations prescribed by the Park Superintendent. Law enforcement rangers will intervene if they perceive any unsafe or unlawful activity.

For projects at Kalaupapa National Historical Park (KALA), the Contractor shall also comply with the attached *Essential Service Provider Permit & Rules (rev 9/8/2025)* issued by the State of Hawaii, Department of Health (DOH).

- B. Work Hours: Regular work hours shall be limited to daylight hours between **6:00 a.m. to 4:00 p.m.** Monday through Friday, unless stated otherwise. No work shall be performed on weekends, state or federal holidays. Work outside regular hours requires advance approval from the Contracting Officer to ensure key NPS personnel are present in the event of an emergency.
- C. Advertisement: No signs or advertisements shall be displayed on the construction site or within the park unless approved by the Contracting Officer.
- D. COVID-19 (Coronavirus Disease 2019): The Contractor shall comply with the latest guidance on Covid-19. The information provided below is not intended to provide a complete analysis of requirements for Contractor and is provided as a courtesy.
 - 1. Occupational Safety & Health Administration (United States Department of Labor). www.osha.gov/coronavirus.
 - 2. Center for Disease Control (CDC) - www.cdc.gov/coronavirus/2019-ncov/index.html
 - 3. Department of the Interior (DOI) - [Our Response to COVID-19 | U.S. Department of the Interior \(doi.gov\)](https://www.doi.gov/coronavirus).

1.8 CONTRACTOR USE OF THE SITE

- A. Park Entry: Vehicles associated with the contract shall be waived the entrance fee upon entering in marked company vehicles or by providing the full names, vehicle type, and the license plate numbers of employees entering in personal vehicles. **NOT APPLICABLE.**
- B. Site Access: The Contractor shall access the site as outlined in the drawings or attached map.
- C. Project Area Limits: The Contractor will have limited use of the site for construction operations. Limit operations to the project area limits as outlined in the drawings or attached map. Do not disturb portions of the park or site beyond areas where Work is indicated. Refer to GFD #1.
- D. Storage of Materials: Confine storage of materials to the areas as outlined in the drawings or attached map. If no specific site is shown, assume storage of materials is allowed within the project area limits and arrange for storage of materials with the park facility manager so as not to disrupt normal operations. The park will attempt to provide a covered area, if available. The Contractor may also furnish conex storage containers for material / tool storage as long as they are placed on a hardened surface within the project area limits or other approved location, and as long as the Contractor protects all surfaces from damage. Damage to surfaces shall be restored to pre-construction conditions. The Contractor will also be responsible for the protection and security of all materials stored in the park.
- E. Staging of Materials and Equipment: Confine staging of materials and equipment to the areas as outlined in the drawings or attached map. If no specific site is shown, assume staging of materials and equipment is allowed within the project area limits and arrange for staging of materials and equipment with the park facility manager so as not to disrupt normal operations. Staging of heavy equipment at the jobsite overnight is permitted if the equipment remains within the designated locations within the project area limits, is checked for leaks and has the proper spill prevention measures (refer to PART 1.16.F), and has adequate surface protection to prevent permanent track marks, spalls, cracks, ruts, depressions, and other damage (refer to PART

1.16.K). The Contractor is responsible for the protection and security of all materials and equipment staged at the jobsite.

- F. Stockpiling: Confine stockpiling of excavated soils to the area immediately adjacent to the trench or excavation, and, remove stockpiled materials from the jobsite if the materials will not be reused as part of the project. Confine imported soils and/or aggregate to the area(s) outlined in the drawings or attached map. If no specific site is shown, assume imported soils or aggregate is allowed within the project limits; however, consult with the park facility manager for an optimal location so as not to disrupt normal park operations. Protect all surfaces from damage. Damage to surfaces shall be restored to pre-construction conditions. **PARK WILL DISPOSE OF GREEN WASTE OR EXCAVATED MATERIALS.**
- G. Rock Removal: When encountering large rock or a significant volume of rock during excavation, follow the guidance outlined in Section 1.8.F for initial stockpiling. If the contract documents involve rock work, use the excavated rocks before importing rock. If the contract documents do not specify instructions regarding rock removal, relocation, or disposal, please submit a Request for Information (RFI) to the NPS for further instructions (with an estimated quantity of rock). All rock excavated at project sites on Molokai, Maui, and the Big Island of Hawai'i will remain within the project area limits or be relocated to a different location within the park. **NOT APPLICABLE.**
- H. Field Office: Establishment of a construction camp within the park will not be permitted. The park will make every attempt to furnish the contractor with indoor office space for up to two (2) contractor personnel; however, if the park cannot accommodate the Contractor with office space, the Contractor will be allowed to setup a temporary field office within the project area limits (final location to be approved by the park facility manager). Note that connection to NPS internet or computer networks is prohibited for security purposes. Refer also to Section 01 50 00.
- I. Property Damage: The Contractor is required to report missing, lost, stolen, or damaged property occurring within the park. This report must be made to a Law Enforcement Ranger or NPS Dispatch at the first available opportunity at: **(808) 985-6170.**
- J. No Smoking Policy: Smoking is prohibited inside or outside within 25 feet of any buildings in the park. This applies to existing buildings and buildings under construction or renovation. Discarding of tobacco products or debris anywhere other than into designated tobacco receptacles is considered littering and subject to fines.

1.9 PUBLIC USE OF THE SITE

- A. General: Cooperate with NPS and partner staff so that the Work may be carried out smoothly without interfering with or delaying the Work under this or other contracts.
- B. Park Visitation: The project site will be **CLOSED** to park visitors during construction.
- C. Park Events: The following park events are anticipated during the Work: **NONE SCHEDULED.**

1.10 GOVERNMENT USE OF THE SITE

- A. General: Cooperate with NPS and partner staff so that the Work may be carried out smoothly without interfering with or delaying the Work under this or other contracts.

- B. Hawaiian Blessing: Prior to field construction and/or immediately following all construction activities, the NPS may conduct a Hawaiian Blessing Ceremony to consecrate the site. A determination for the need of a blessing ceremony will be made by the Park Superintendent and discussed during the pre-construction meeting. The Contractor may attend the blessing ceremony, but no compensation will be made for attendance.

1.11 OCCUPANCY REQUIREMENTS FOR BUILDINGS & AREAS

- A. Government Occupancy: The Government will **OCCUPY** the building(s) or area(s) under construction until the construction phase is complete and final acceptance has been issued. Cooperate with the Government during construction activities to minimize conflicts. Perform the Work so as not to interfere with the Government's day-to-day operations. Maintain existing building ingress, egress, doorways, corridors, access walkways, access roads, and public space unless the Contractor obtains written permission from the Contracting Officer to close, partially close, or temporarily close specific areas.
- B. Government Occupancy of Completed Areas: The Government reserves the right to occupy portions of buildings or areas and resume operations where the Work has been completed, provided such occupancy does not interfere with the completion of the remaining Work.
 - 1. The Contractor shall ensure all Work is completed; all systems are operational; all testing, balancing, and inspections have been performed; and all final cleaning is completed prior to Government occupancy.
 - 2. The Contracting Officer will prepare a Letter of Beneficial Use / Occupancy for the portion(s) of the building or areas to be re-occupied by the Government.
 - 3. Upon Government occupancy, the Government will assume responsibility for the maintenance and custodial services for occupied portions of the building or areas.

1.12 CODES

- A. General: The Contractor shall complete all Work in accordance with the latest versions of applicable federal, state, and local codes, standards, and regulations. Per Policy Memorandum 23-01 approved on May 9, 2023, the National Park Services (NPS) has adopted the International Code Council's (ICC) suite of codes and standards effective January 1, 2024. Additionally, the latest NFPA 70 (National Electric Code) is also adopted effective January 1, 2024. All work shall comply with the ICC suite of codes and standards and NFPA 70, where applicable. Additional codes, standards, and regulations may also apply. In the event of a conflict between codes, standards, regulations, and the contract documents, submit a Request for Information (RFI) with code references / citations to the Contracting Officer for a determination.

1.13 PERMITS

- A. General: The Contractor shall comply with all permit conditions, whether they are from permits furnished by the government (PART 1.13.D) or permits obtained as part of this contract (PART 1.13.E). Identify the start dates and durations to obtain all permits in the construction schedule. Note: A general building permit is not required for work on NPS land; however, the Contractor must still comply with all state and local building code provisions.
- B. Hot Work Permit: Hot work is defined as work activities (such as welding, cutting, blazing, and related work) that can be a source of ignition when flammable materials are present. All hot work

operations shall be conducted in accordance with the procedures contained in RM-58, NPS policy on Hot Work Permits (see Appendix 35.8), and 29 CFR 1910.252(a). If the project requires any activities considered "hot work," submit a NPS Hot Work Permit (Form HW-1) application to the Contracting Officer at least 48 hours prior to the scheduled hot work. The NPS will then review and complete Form HW-1 and return a copy to the Contractor with the required conditions or precautions. No hot work shall occur without a completed and signed Form HW-1 and all the required precautions in place. See attached Form HW-1.

C. Confined Space Entry Permit: The Contractor and/or Employer shall adhere to, and implement, the permit-required confined space requirements of the Authority Having jurisdiction. Before each entry into a permit-required confined space, a permit must be completed, hazards must be assessed and controlled, proper ventilation and air-monitoring ensured, and personnel trained and equipped, with an attendant(s) stationed outside the space for constant communication and rescue readiness. A copy of the permit shall be furnished to the NPS for situational awareness.

D. Government-Furnished Permits: **NONE.**

E. Other Required Permits: **NONE.**

1.14 PROFESSIONAL SERVICES

A. General: Professional services include any business, department, or individual whose core output is a service or expertise rather than a manufactured product.

B. Shop Drawings: The Contractor shall submit shop drawings per Section 01 33 23, PART 1.4.F for the following components:

1. Component #1: Layout of air conditioning equipment, electrical power and appurtenances, refrigerant and condensate lines showing the coordination of all work with other trades.

C. Professional Land Surveyor (PLS): A professional land surveyor is NOT REQUIRED for this project.

D. Professional Engineer (P.E.): A professional engineer is NOT REQUIRED for this project.

1.15 UTILITY INTERRUPTIONS

A. General: Utilities include a connection or service to or between facilities involving Potable Water, Non-Potable Water, Fire Protection, Wastewater, Electrical Power, HVAC, Security, Alarms & Notifications, Internet and Telecommunications.

B. Existing Utilities: Notify the Contracting Officer (and all applicable utility companies) of the Work being performed on a utility prior to excavation or disruption. The Contractor shall be responsible for locating and preventing damage to known utilities. If damage occurs to a known utility, repair the utility at no additional expense to the Government. If damage occurs to an unknown utility: 1) submit a Request for Information (RFI) to the NPS for further instructions, or,

if time is of essence, 2) repair the utility / restore service and an equitable adjustment will be made in accordance with the Changes Clause of the contract.

C. Utility Disruptions: Do not disrupt utility service to Government or NPS partner buildings or facilities unless required under the contract documents. If utility disruptions are required:

1. All Work shall be performed during normal business hours. If it is advantageous to perform the utility disruption after normal business hours, submit an RFI requesting allowance.
2. Notify the Contracting Officer (and all applicable utility companies) not less than seven (7) calendar days in advance of the utility disruption(s).
3. Identify the affected services, location(s), start date, and duration of the disruption.
4. For disruptions or outages lasting more than four (4) hours, or, for systems that cannot be disrupted due to critical security or public health requirements, install temporary utility services. Refer to Section 01 50 00 (Temporary Facilities & Controls) for more information.
5. For disruptions or outages lasting more than four (4) hours that cannot be mitigated with temporary utility services, notify the Contracting Officer (and all applicable utility companies) not less than fourteen (14) calendar days in advance of the utility disruption(s).
6. Do not proceed with utility disruptions without written permission from the Contracting Officer and the local utility service provider (where applicable).

1.16 VEHICLES & HEAVY EQUIPMENT – **NOT APPLICABLE**

- A. General: The Contractor shall comply with the following work restrictions for vehicles and heavy equipment.
- B. Rules of the Road: Vehicles and heavy equipment shall comply with posted traffic regulations, including speed limits and double-yellow center lane markings. Vehicles and heavy equipment shall take every safe opportunity to use paved pullouts to allow faster-moving traffic to pass. Failure to comply with traffic regulations will result in a citation being issued to the driver.
- C. Vehicle Parking: Confine parking of Contractor vehicles to the areas as outlined in the drawings or attached map. Vehicles are not permitted to park along park roads or in overlooks / pullouts. Vehicles are not permitted to park overnight in the park.
- D. Driveways and Entrances: Keep driveways and entrances serving the premises clear and available to Government employees, vendor deliveries, and emergency vehicles at all times.
- E. Vehicle Exhaust: Vehicles shall be properly maintained to ensure the exhaust produced is within legal limits. The effects of excessive vehicle exhaust may be subject to enforcement of State and Federal laws regarding emissions as well as those regarding threatened and endangered species.
- F. Spills: The Contractor shall be responsible for prevention of spills and/or cleanup efforts required in accordance with 40 CFR and all applicable state environmental quality laws regarding disposal and cleanup of hazardous / industrial wastes. The Contractor must immediately notify a Law Enforcement Ranger or NPS Dispatch of any spills at: **(808) 985-6170**.
- G. DOT Compliance: Vehicles and heavy equipment shall be compliant with all DOT requirements including but not limited to drivers with appropriate license class, possessing a medical examiner card with license class as appropriate, carrying three (3) triangles or three (3) flares, carrying a mounted fire extinguisher, and possessing a valid DOT safety sticker, as applicable.

- H. Hauling & Load Restrictions: All vehicle load, weight, and size limitations, as established by State law, apply to vehicles operated on the park road (36 CFR § 4.11). No loads heavier than the current load rating for park roads and bridges will be permitted. Load restrictions on park roads are identical to state load restrictions with such additional regulations as may be imposed by the Park Superintendent. Information regarding rules and regulations for vehicular traffic on park roads may be obtained from the Office of the Park Superintendent.
- I. Oversized Vehicles: Oversized vehicles are defined as vehicles where the width of the vehicle or load exceeds one lane (i.e., beyond the center line of the paved road). Oversized vehicles must not exceed the clearances along the park road corridor. Oversized vehicles will require an escort, and drivers must avoid driving on the shoulder of the road. Contractor shall notify the Contracting Officer of oversized vehicles, scheduled arrival dates & times at the park entrance, and the identity of who is escorting the vehicle (e.g., police, contractor, etc.). The Contractor shall avoid transporting loads using oversized vehicles during peak visitation periods in the park. These peak visitation periods may vary between parks and seasons. Submit an RFI at least seven (7) days prior to entering the park with an oversized vehicle to obtain the optimal travel times.
- J. Slow Vehicles: The Contractor is to notify the Contracting Officer of slow vehicles that may cause traffic delays or extended traffic lines.
- K. Tracked Equipment: Heavy equipment with rubber tires is preferred on projects within the national park. Heavy equipment with metal tracks may be allowed on paved asphalt surfaces, concrete surfaces, or ceremonial lawn areas only with adequate surface protection to prevent permanent track marks, spalls, cracks, ruts, depressions, and other damage. The means and methods of surface protection (such as plywood sheets or steel plates) shall also not damage surfaces. The exceptions listed below may apply. If the Contractor anticipates one of these exceptions, submit a Request for Information (RFI) to the NPS for further instructions.
1. Tracked equipment may be allowed on paved asphalt or concrete surfaces in lieu of surface protection if the Contractor repairs all the affected surfaces to pre-construction condition or replaces the surface with new asphalt or new concrete, as applicable.
 2. Tracked equipment may be allowed on older paved asphalt surfaces if the asphalt surface is scheduled for pavement preservation (by others) within one (1) year of contract completion.
 3. Tracked equipment may be allowed on un-paved gravel roads if the tracked equipment stays on the road and does not create ruts, damage, or impacts the surface water flow. If damage occurs, the Contractor shall re-grade / restore the terrain to pre-construction conditions.
 4. Tracked equipment may be allowed on grass surfaces within the project area limits with adequate surface protection or the terrain is re-graded / restored to preconstruction conditions. Reseeding of re-graded / restored areas is not required unless otherwise specified.

PART 2 – PRODUCTS (Not Used)

PART 3 – EXECUTION (Not Used)

END OF SECTION 01 11 00

Application for Service Provider Permit

To: Administration Office, **Kalaupapa Settlement**

DATE: _____

Note: Permits must be submitted at least three (3) BUSINESS DAYS prior to visitor arrival.

The DOH administration cannot guarantee flight availability for settlement access on chosen dates.

I, _____, on behalf of our organization _____,

request permission to have the following **Service Provider(s)** at Kalaupapa Settlement.

TRAVEL BY: ☐ PLANE ☐ TRAIL ☐ OCEAN [submit with approved NPS permit]

FROM: _____ TO: _____ STAYING AT: _____

Date of Arrival

Date of Departure

Tag	Full Name of Visitor (please print)	Home address & Emergency Phone contact #
		Address: _____ Phone: _____
		Address: _____ Phone: _____
		Address: _____ Phone: _____
		Address: _____ Phone: _____
		Address: _____ Phone: _____
		Address: _____ Phone: _____
		Address: _____ Phone: _____

By signing below as a representative of my agency, I certify that our service provider(s) will be fully informed of, and will comply with, the Kalaupapa Settlement Rules & Regulations during their visit. I further confirm that I have provided each individual, prior to arrival, with a copy of Attachment B, which outlines these Rules & Regulations for visitors. I understand that, as a sponsor, I am responsible for ensuring compliance with these rules and that the Kalaupapa Administration reserves the right to investigate potential violations by visitors and/or their sponsors. I acknowledge that penalties may include temporary suspension of sponsorship privileges, permanent banning of offenders, or other sanctions as deemed appropriate by the Kalaupapa Administrator.

Sponsor/Sponsoring Agency Representative

Sponsor's e-mail address

Above request is granted on _____,

SUBJECT TO RULES & REGULATIONS GOVERNING KALAUPAPA SETTLEMENT AND REVOCABLE.

Kalaupapa Administrator, **Kalaupapa Settlement**

By signing below or proceeding with entry, you confirm that you have read, understood, and voluntarily accept the terms of the “KALAUPAPA RISK ACKNOWLEDGMENT AND LIABILITY WAIVER”.

SPONSOR NAME:			
SPONSOR SIGNATURE:			
SPONSOR E-MAIL:			
DATE OF ARRIVAL		DATE OF DEPARTURE:	

By signing below or proceeding with entry, you confirm that you have read, understood, and voluntarily accept the terms of the “KALAUPAPA RISK ACKNOWLEDGMENT AND LIABILITY WAIVER”.

Tag #	Visitor(s) Full name	Signature	Date

ATTACHMENT A

SPONSOR RESPONSIBILITIES AND PENALTIES

Who can sponsor visitors? All patients, all employees, certain DOH volunteers and some representatives of ecclesiastical organizations that have a current agreement with NPS may sponsor visitors.

Each sponsor may sponsor **up to six (6) visitors at a time**, subject to the availability of accommodations.

Deadline: Visitor permits **MUST** be submitted to the DOH Administration Office at least three (3) State Office business days in advance of scheduled visit. **No exceptions will be allowed.**

Sponsor's Availability: A sponsor **MUST** be present in the settlement for the entire duration of their sponsored visitors' stay. More than one sponsor is allowed to apply and sign an Application for Permit, should the primary sponsor need to be away from the settlement at any time during their visitors' stay. Sponsor(s) assume all sponsorship responsibilities during any absences of the primary sponsor.

Visitors **must be accompanied by their sponsor(s) at all times when they are outside of the settlement** – specifically:

- Beyond the cattle guard located on Damien Road leading east toward Judd Park;
- Beyond the cattle guard on the dirt road leading toward the light house and beaches (dirt road adjacent to Kamehameha Street near the airport).

Settlement Access: There are three ways that visitors may access the Settlement:

Plane Access: The DOH permit approval doesn't guarantee that a visitor will be able to secure a flight reservation.

Trail Access: Approved visitors may enter and exit the settlement by trail, without their sponsor(s), **at their own risk**. To best ensure safety, all hikers are strongly encouraged to travel with a "buddy" who has experience hiking

the Trail. The DOH administration office provides a weekly sponsored guest list to the NPS rangers to monitor and enforce permit requirements.

Boat Access: Safe sea access to the park is limited by the location and configuration of the Kalaupapa pier and seasonality of ocean conditions. Taking into consideration public safety, protection of park resources, and weather, **NPS requires a Special Use Permit for the use of a vessel within the boundary of the park. DOH will NOT approve a visitor permit application for entry from the ocean without an APPROVED NPS Special Use Permit. For more information on the Special Use Application, visit (www.nps.gov/KALA/planyourvisit/sup).**

Permit Application: Sponsors are required to fill out the full names of all visitors legibly, and exactly as the names appear on each visitor's state or government ID. Applications are not complete without sponsor name and signature. If you seek to request a change of travel dates for your visitor AFTER an application has been submitted or approved, you are required to re-submit a new application.

Once a permit application is approved, you will be notified within three business days. Sponsors are required to confirm they have received notification of permit approval.

Upon Visitor Arrival: Sponsors are required to present all visitors at the DOH Administration office immediately upon their arrival. All visitors must present a government or state-issued identification which includes their date of birth. For visitors that intend to operate a motor vehicle during their visit, a copy of a current driver's license must also be presented at sign-in. Visitors are not permitted to operate government (i.e. State or Federal) vehicles. "Service Providers" that require self-driving ground transportation for their work duties are required to confirm any special driving arrangements with the Administrator prior to arrival.

If a visitor arrives outside of office hours, sponsors are responsible to pick up visitor tags and sign-in sheets **in advance** of their visitor's arrival. Tags and sign-in sheets can be picked up during office hours, i.e. before 2:00 pm Mondays-Thursdays and before 11:00 am on Fridays.

Visitor sign-in sheets must be completed and brought back to the Administration Office on the first business day after the visitor's arrival. **Visitor tags must be returned immediately upon visitor(s)' departure.** After-hours drop box is available at the DOH Admin Office. Sponsors will incur a \$5 state property loss charge per visitor tag and will not be able to sponsor visitors until payment of outstanding fines.

Accommodation Fees: Visitor(s) staying at VQ, Wilcox and VET's Cottage are expected to **pay accommodation fees upon arrival**, or at the latest, on the first business day after arrival if arrival has occurred outside of office hours.

VQ, Wilcox and VET's Cottage Accommodation: Do not remove linen, towels, dishes, cutlery, toilet paper, paper towels, cleaning supplies and other DOH property from accommodation premises. If housekeeping finds items missing or vandalized and/or excessive and unsorted trash upon visitor(s)' departure, sponsors may be fined and/or temporarily denied sponsorship privileges upon the discretion of the Kalaupapa Administration.

Important: Sponsors are responsible to **provide a copy of *Attachment B, Visitor Rules & Regulations*, to each sponsored visitor prior to their arrival.**

Sponsor(s) are responsible for all actions of their visitors, including any violations of the Visitor Rules & Regulations. Infractions of sponsor and/or visitor rules will result in the revocation of sponsorship privileges, as follows:

- **First Offense:** Resident will not be allowed to sponsor any visitors for 6 months;
- **Second Offense:** Resident will not be allowed to sponsor any visitors for 9 months;
- **Third Offense:** Resident will not be allowed to sponsor any visitors for 12-24 months;

Additional and/or more severe sanctions may be imposed based on the violation(s), and the discretion of the Kalaupapa Administration.

ATTACHMENT B
VISITOR RULES & REGULATIONS FOR KALAUPAPA SETTLEMENT

SPONSORS:	All patients, all employees, certain DOH volunteers and some representatives of ecclesiastical organizations that have a current agreement with NPS may sponsor visitors.
VISITORS:	All visitors must present a state or government-issued photo identification upon sign-in at the State Administration Office, i.e., Driver's License, State ID, etc. ALL IDs MUST HAVE BIRTH DATE ON THEM. The current daily visitor limit is ONE HUNDRED (100) sponsored guests within Kalawao County at any given time. All visitors to the Settlement are required to display their respective assigned visitor tag on their person at all times.
REGISTRATION:	Registration at the DOH Administration Office is required of ALL visitors immediately upon arrival. Each visitor receives their assigned visitor tag upon registration. For visitor arrivals outside of office hours, sponsors are required to pick up the sign-in sheet and the assigned visitor tag(s) in advance. The visitor permit sign-in sheet with all visitor signatures must be submitted to the admin office on the next immediate business day. Please, note that an emergency phone number of an individual outside of the settlement is required on the sign-in sheet. Applicable accommodation fees are due upon arrival. Anyone visiting the Settlement without a sponsor and signed permit is TRESPASSING and subject to legal action.
MAXIMUM STAY:	NO VISITOR will be permitted to spend more than thirteen (13) days per calendar quarter in Kalaupapa Settlement. The maximum "continuous" stay during any visit at Visitors' Quarters, Wilcox and VET's Cottage is limited to seven (7) days and six (6) nights per individual.
HOSTS/SPONSORS:	Sponsors MUST be present in the settlement for the entire duration of their sponsored visitor's stay. Should a sponsor need to be away from the settlement at any time during their visitors stay, an additional sponsor may/can be allowed to sign the permit and assume all sponsorship responsibilities during any absences of the original sponsor.

	Visitors must be accompanied by their sponsor(s) at all times outside of the settlement – specifically: • Beyond the cattle guards located on Damien Road leading east toward Judd Park; • Beyond the cattle guard on the dirt road leading toward the light house and beaches (dirt road adjacent to Kamehameha Street near the airport)
VISITATION OF MINORS:	Children under the age of sixteen (16) are not permitted in Kalaupapa Settlement.
PHOTOGRAPHY/VIDEO:	By law, photographs and video of patients and/or their property may NOT be taken without their written permission. Photograph and video permits are available at the State of Hawaii Administration Office and must be completed and submitted ahead of time. Photo/Video permits require patients' signature, prior to being reviewed/approved by the DOH administrator. Recording, Filming or Still Photography of any kind (still, video and audio) may require a NPS Special Use Permit. For more information visit: https://www.nps.gov/kala/planyourvisit/index.htm
HOSPITAL:	THERE ARE NO MEDICAL OR EMERGENCY SERVICES AVAILABLE IN KALAUPAPA. If you are sick, or have flu-like symptoms, such as fever, body ache, runny nose, nausea, coughing, etc., do not visit the Settlement. If you require medical attention or medical services while in Kalaupapa, inform your sponsor. An emergency medical response via helicopter from a neighboring island may take hours. <u>In case of emergency, call 911 immediately.</u> Be aware that the scarcity of cell phone reception on the peninsula may make calling for help difficult.
CARE HOME	The Kalaupapa Care Home exists solely for the care of our remaining patients that live in Kalaupapa. Visitors requesting to meet with individual patients in the Care Home should have their sponsor call the Care Home prior to their arrival to inquire about a potential appointment. An approved Visitor Permit does NOT include permission to enter the Kalaupapa Care Home. All requests for Care Home visits are subject to the review and decision of the Care Home's Nursing Supervisor.
RESTROOMS:	Public restrooms are located at the Airport Terminal, the DOH Administration Building, Slaughterhouse, and Judd Park.

VISITORS' QUARTERS:	Check-In 2:00PM, Check-Out 12:00 Noon Please inform Administration staff if you need an early check-in and/or late check-out time. If you fail to notify Admin office staff of a late check-out time, our housekeepers have the right to gather all your personal belongings and set them aside in order to proceed with their duties. Visitors are solely responsible for the safety of their belongings. Please DO NOT REMOVE any State-owned items from Visitors' Quarters, Wilcox or VET's cottage. Observe cleanliness and recycling instructions in common spaces, such as shared kitchens and bathrooms. If the visitor(s) leave(s) excessive and/or improperly sorted trash behind, their sponsor will be held liable and sponsorship rights withdrawn temporarily at the discretion of the Kalaupapa Administration. Female hygiene products: Please, dispose of these in the available trash cans. Our plumbing CANNOT handle flushed tampons and sanitary pads/napkins.
FOOD:	There are no restaurants in Kalaupapa, so please, plan for your needs and bring your own food, beverages and sundry items. Visitors may purchase a single soda, and a candy or a bag of chips per day at the Kalaupapa Store, but shopping for groceries is PROHIBITED.
LIBRARY:	The library is NOT open to the public. Please have your sponsor check with National Park Service if you wish to use library resources. Removal of materials from the library is strictly prohibited.
DRIVING IN THE SETTLEMENT:	If you intend to drive during your visit, please exercise the utmost caution and move onto the road shoulder if you see incoming vehicles – many of our residents have visual and hearing impairments, deer are pervasive within settlement bounds, and the occasional wild boar are unpredictable. Please, always be vigilant and aware of your surroundings.
SPEED LIMITS:	All drivers are expected to comply with the following speed limits: • <i>Hospital Zones</i> 15 MPH • <i>Unit Home Drives</i> 15 MPH • <i>Within the settlement proper, incl. the inner cattle guards</i> 25 MPH • <i>Outside the settlement proper</i> 35 MPH

AUTOMOBILE PASSENGERS:	All vehicles and operators must comply with all state and federal traffic regulations. Failure to comply with federal and state regulations can result in penalties, fines, and/or arrest. REMINDER: State law (HRS §291-14) only permits passengers to ride in the back of pickup trucks when: • There is no seating available in the cab • The tailgate is securely closed, and side racks are securely attached • Every passenger in the bed is seated and does not attempt to control unlash cargo.
HUNTING:	Visitors are NOT PERMITTED to hunt or accompany a resident hunter on a hunt. Visitors are NOT PERMITTED to carry firearms.
CAMPING:	Camping is NOT ALLOWED anywhere.
USE OF NATURAL RESOURCES:	All applicable state and federal fish and game rules and regulations must be respected. Any violations of the Settlement’s fish and game rules will result in the visitor being declared unwelcome and banned from future visits to Kalaupapa. Visitors may only fish with poles. No NET fishing is allowed. No opihi picking. No fishing equipment or the cleaning of fish is allowed in any building, except for Lion’s Club “Ocean View” Pavilion when accompanied by a sponsor. Harvesting of sea salt is to be done under the sponsor’s supervision. Please, take only what <i>you</i> need. Be mindful of others. Taking excessive amounts of sea salt will result in the visitor being declared unwelcome and, based on the severity of the violation, temporarily or permanently banned from future visits to Kalaupapa. Note: No Trespassing on DOT property, including runway area. Violations of the Kalaupapa fishing and sea salt harvesting policy should be reported to the Kalaupapa Settlement Administrator immediately following the alleged violation. For safety concerns, no diving tanks or scuba are allowed. No surf boarding or boogie boarding. No diving off the pier.
NO ANIMALS:	Visitors are not permitted to bring animals of any species into the Settlement. If a visitor requires the assistance of a service animal, their sponsor must formally request permission from the Administrator in advance.

NATIONAL HISTORIC LANDMARK PROTECTION	The entire peninsula is designated as a National Historic Landmark (NHL). Respect archeological and cultural sites. Do not remove artifacts or alter the landscape.
INHERENT RISKS	Kalaupapa Settlement, located within Kalawao County on the Mānālua Peninsula, is a remote and historically significant area overseen jointly by the Hawai'i State Department of Health (DOH) and the U.S. Department of the Interior, National Park Service (NPS). The Settlement is managed by DOH primarily to provide care and residence for individuals previously institutionalized due to Hansen's Disease and is not designed or operated as a conventional tourist destination. • Visitors understand that Kalaupapa is a remote and undeveloped environment, with limited infrastructure and emergency services. • Activities undertaken during the visit may expose individuals to serious hazards, including but not limited to: - o Wildlife encounters - o Natural disasters (e.g., tsunamis, landslides, hurricanes) - o Sudden weather changes - o Rugged terrain and limited accessibility Visitors voluntarily assume all risks of injury, illness, property damage, or death, regardless of cause or circumstance. Since the entire peninsula is designated as a National Historic Landmark (NHL) additional oversight and permitting of the construction, management and maintenance of facilities (e.g., road paving, building modifications, street lighting) is required. All improvements are guided by DOH's goal to care for patients and NPS's efforts to balance the goals of protecting, preserving and interpreting the cultural, historic and natural resources of Kalaupapa. The entirety of Kalawao County/Kalaupapa National Historical Park is therefore NOT designed for, or operating as a typical tourist destination. It is the responsibility of all Sponsors and Visitors to be aware of the inherent risks which may contribute to injury or harm during their stay. By agreeing to visit, all visitors understand and willingly assume these risks, regardless of the cause and circumstances.
WILDLIFE	When you visit national parks, you are entering animals' habitat and should behave like a polite guest. In fact, your safety and health, and that of wildlife depends on the thoughtful choices park visitors make. Risks from wildlife in national parks include potential attacks or injuries to visitors from animals that are startled, aggressive, or food-conditioned, as well as diseases transmitted between humans/pets and wildlife.

	Additionally, human activities like feeding wildlife, improper trash disposal, and getting too close can lead to <u>food conditioning</u> and increased aggression, ultimately resulting in harmful interactions for both people and animals.
WALKING IN THE SETTLEMENT AT NIGHT, OR WHEN LIGHTING/VISIBILITY IS POOR	Should pedestrians choose to move about the Settlement at night, or when lighting/visibility is poor they should: 1) use a lighting device that can: - illuminate their immediate surroundings, so that they may spot potential trip or fall hazards such as curbs, potholes, uneven pavement, debris or steep downward slopes, - allow drivers and other pedestrians to see them from a distance - act as an additional deterrent to both wildlife and potential human threats. 2) wear light colored and/or reflective gear.
LITTERING:	Please be respectful of this peninsula, known as Makanalua in ‘Ōlelo Hawai‘i. Be aware of and fulfill your responsibility to maintain its pristine beauty as a steward of the ‘aina (land) and all that lives and grows here.
ATTITUDE:	Honor and respect the history of those who once lived here and contribute to the protection of this unique place which those who still reside here unequivocally call "Home."
NO TRESPASSING:	Trespassing onto off-limit administrative buildings, personal residences and yards is strictly prohibited, including to pick fruit, vegetables or flowers.
BEFORE YOU LEAVE:	Make sure you return all assigned visitor tags and have paid your accommodation fees if applicable. Do not leave unsorted or excessive trash behind.

ANY VIOLATION OF THESE RULES GOVERNING ALL VISITORS WILL RESULT IN DISCIPLINARY ACTION AND SUSPENSION OF VISITATION AND SPONSORSHIP PRIVILEGES.

KALAUPAPA RISK ACKNOWLEDGMENT AND LIABILITY WAIVER

By entering Kalaupapa, all visitors acknowledge and accept the following:

1. Assumption of Risk

- Visitors understand that Kalaupapa is a **remote and undeveloped environment**, with **limited infrastructure and emergency services**.
- Activities undertaken during the visit may expose individuals to **serious hazards**, including but not limited to:
 - o Wildlife encounters
 - o Natural disasters (e.g., tsunamis, landslides, hurricanes)
 - o Sudden weather changes
 - o Rugged terrain and limited accessibility
- Visitors voluntarily assume **all risks of injury, illness, property damage, or death**, regardless of cause or circumstance.

2. Limited Services and Emergency Response

- Medical, transportation, and communication services are **extremely limited**.
- Emergency response times may be **delayed significantly** due to geographic isolation.
- Visitors are responsible for **their own safety and preparedness**, including bringing necessary supplies and understanding the terrain.

3. Historic and Cultural Protections

- Kalaupapa is designated as a **U.S. and State Historic Landmark**.
- All facilities and infrastructure are managed with the intent to **preserve cultural, historical, and natural resources**.
- Construction, modification, or unauthorized activity is strictly prohibited without appropriate permits.

4. Purpose of the Settlement

- Per Hawai'i Revised Statutes §326, Kalaupapa Settlement is maintained by the DOH for the **care and treatment of persons with Hansen's Disease**.
- Visitor access is **restricted and regulated** by the Department of Health.

5. Release of Liability

By entering Kalaupapa, visitors hereby release and hold harmless the State of Hawai'i, the Department of Health, the National Park Service, and any affiliated agencies, employees, or representatives from any and all liability for personal injury, death, property loss, or other damages incurred during their visit.

Hot Work Permit (Form HW-1)



Hot Work Permitting

Hot work shall only be performed in **designated** or **permit-required** areas. Before hot work operations begin in a non-designated location, a signed Hot Work Permit, issued by the Permit Authorizing Individual or his/her designee, shall be issued.

Designated Area

A designated area shall be a specific area designed for, or approved by the Permit Authorizing Individual or his/her designee for, such work. A designated area can be a maintenance shop or a detached outside location that is of noncombustible or fire-resistive construction, essentially free of combustible and flammable contents, and suitably segregated from adjacent areas. All designated areas shall have been inspected and approved by the Permit Authorizing Individual or his/her designee.

Permit-Required Area

A permit-required area shall be an area that is made fire safe and authorized by the issuance of a signed NPS Hot Work Permit. The Hot Work Permit shall be signed by Permit Authorizing Individual or his/her designee.

Non-permissible Areas

Hot work **shall not be allowed** in the following areas:

- (a) In areas not authorized by the Permit Authorizing Individual or his/her designee
- (b) In sprinklered buildings while such protection is impaired
- (c) In the presence of explosive atmospheres (that is, where mixtures of flammable gases, vapors, liquids, or dusts with air exist)
- (d) In explosive atmospheres that can develop inside unclean or improperly prepared drums, tanks, or other containers and equipment that have previously contained such materials
- (e) In explosive atmospheres that can develop in areas with an accumulation of combustible dusts

GENERAL

Park: _____ District: _____ Building: _____ Floor: _____

Work to be done: _____

Special precautions: _____

PERSON PERFORMING WORK

Contractor Business Name: _____

Contractor Representative: _____

Contractor Contact Number: _____

NPS Employee's Name: _____

NPS Office: _____

NPS Employee's Contact Number: _____

CHECK

Work area and all adjacent areas to which sparks and heat might have spread (including floors above and below and on opposite side of wall(s)) were inspected every 20 minutes for 1 hour after the work was completed and were found fire safe.

ATTENTION

Before approving any hot work permit, the Permit Authorizing Individual or his/her designee shall inspect the work area and confirm that the proper precautions have been taken to prevent fire, in accordance with the NPS Hot Work Permit Form (Form HW-1) and the NPS Hot Work Policy.

PRECAUTIONS

- ☐ Sprinklers in service
- ☐ Hot work equipment in good repair

WITHIN 35 FEET OF HOT WORK AREA

- ☐ Floors swept, clean of combustibles
- ☐ Combustible floors wetted down, covered w/ damp sand or other shields in place
- ☐ All wall and floor openings covered

WORK ON WALLS OR CEILINGS

- ☐ Combustibles moved away from opposite side of wall

WORK ON ENCLOSED EQUIPMENT (tanks, containers, ducts, dust collectors etc.)

- ☐ Equipment cleaned of all combustibles
- ☐ Containers purged of all flammable vapors

FIRE WATCH

- ☐ To be provided during and 60 minutes after operation
- ☐ Supplied with a fully charged and operable fire extinguisher
- ☐ Trained in use of equipment and activating fire alarm system

FINAL CHECK

- ☐ To be made every 20 minutes for 60 minutes after completion of any operation unless fire watch is provided

Is a fire watch required? Yes ☐ No ☐

The location where the work is to be done has been examined, necessary precautions taken, and permission is granted for this work.

Permit Issue Date: _____ Permit Expires: _____

Permit Authorizing Individual or Designee: _____

Signature: _____ Date: _____

SECTION 01 11 40 – WORK RESTRICTIONS & RESOURCE PROTECTION

PART 1 – GENERAL

1.1 SUMMARY

- A. General: The NPS is the steward of unique resources found within the national parks. The NPS is committed to protecting natural, cultural, historical, archaeological, and other resources found within the parks and the adjacent lands.
- B. Section includes work restrictions for protecting resources in the national parks including, but not limited to the following:
 - 1. Preservation of Natural Features
 - 2. Invasive Species Prevention
 - 3. Invasive Species Inspections
 - 4. Threatened & Endangered Species (T&E Species)
 - 5. Archaeological Resources
 - 6. Soundscape / Noise
 - 7. Air Quality (Indoor & Outdoor)

PART 2 – PRODUCTS (Not Used)

PART 3 – EXECUTION

3.1 PRESERVATION OF NATURAL FEATURES

- A. General: The Contractor shall comply with the following work restrictions:
 - 1. Prevent damage to natural surroundings.
 - 2. Provide temporary barriers to protect existing trees, tree roots, plants, grass areas and lawns that are directly impacted by construction operations. Temporary barriers should be free-standing orange traffic delineator posts with grabber / looper tops and plastic rails.
 - 3. Do not fasten ropes, cables, or guys to existing trees, rock outcrops, or other natural features.
 - 4. Do not remove, injure, or destroy existing trees, tree roots, or plants without approval by the Contracting Officer or unless it's explicitly required in the contract documents.
 - 5. If a tree, tree branch, root, or mass of vegetation is preventing the Contractor from completing the work, notify the Contracting Officer immediately for a determination. Trees may need to be surveyed by park staff for bird nests prior to removal. Tree removal / trimming may be subject to additional restrictions if project is in Hawaiian hoary bat habitat.
 - 6. Carefully supervise the excavation, grading, filling, equipment movement, and other construction operations near trees and tree branches to prevent damage.

3.2 INVASIVE SPECIES PREVENTION

- A. General: The introduction of non-native species including invasive weeds and plants; invasive pests such as coqui frogs and frog eggs, little fire ants (LFA), and insects including termites; other vertebrate species such as rats, mice, and reptiles; and diseases such as Rapid 'Ōhi'a Death (ROD) spread through contaminated soil can threaten the park's rare and endangered plants and animals or can create new populations of non-native species not traditionally found in the park. Ensure all vehicles, equipment, machinery, cutting tools, base yards, staging areas, materials, material

packaging, material deliveries, material storage, and personal protective equipment (PPE) are clean and free of invasive weeds and plants, invasive pests, and ROD spread through soil.

B. Vehicles, Equipment, & Machinery: In order to prevent the spread of invasive species, the Contractor shall comply with the following:

1. Vehicles, equipment, and machinery must be thoroughly pressure washed and visibly free of dirt, mud, plant debris, and invasive pests. Areas of particular concern include bumpers, grills, hood compartments, under the battery, wheel wells, undercarriage, cabs and truck beds.
2. The interior and exterior of vehicles, equipment and machinery must be free of rubbish. The interiors of vehicles and the cabs of machinery must be vacuumed clean. Floor mats must be sanitized with a solution of >70% isopropyl alcohol.
3. All vehicles, equipment, and machinery shall park on pavement and not under trees (where possible).
4. All vehicles, equipment, and machinery will be subject to inspection by NPS personnel.
5. All vehicles, equipment, and machinery that do not pass inspection will be turned away. The vehicle will not be allowed into the park for at least 48 hours.
6. All vehicles, equipment, and machinery leaving the park for any reason must be cleaned prior to re-entry into the park and will require re-inspection.
7. All vehicles, equipment, or machinery being stored at a private residence are to be kept in a pest free area, preferably in a garage, on pavement, and not under trees. Such vehicles, equipment, or machinery will be subject to additional inspection.

C. Cutting Tools: In order to prevent the spread of invasive species (most notably, the spread of Rapid 'Ōhi'a Death) the Contractor shall comply with the following:

1. Cutting tools including handsaws, machetes, chainsaws, and loppers must be sanitized to remove visible dirt, contaminants, and potential pathogens prior to entry into the park and when moving to a new project area. Tools must be sanitized using a solution of >70% isopropyl alcohol. One minute after sanitizing, you may apply an oil-based lubricant to chainsaw chains or other metallic parts to prevent corrosion. This applies to all cutting tools for all types of trees (including coconut trees).
2. All cutting tools will be subject to inspection by NPS personnel.
3. Cutting tools that do not pass inspection shall not be used until they pass inspection.

D. Contractor Base Yards, Staging Areas, and Project Areas: In order to prevent the spread of invasive species, the Contractor shall comply with the following:

1. Contractor base yards and staging areas, both inside and outside the park, shall be kept clean and free of invasive pests.
2. Base yards and staging areas may be inspected by NPS personnel for invasive pests.
3. Any invasive pests found at base yards or staging areas shall be removed immediately.
4. If new infestations or populations of invasive pests are detected in or near Contractor base yards, staging areas, or project areas, the NPS may dispatch personnel to conduct control measures. Work on the project may be temporarily delayed until the infestation or population no longer poses a threat of spreading (as determined by the NPS).

E. Materials, Packaging, Deliveries, and Storage: In order to prevent the spread of invasive species, the Contractor shall comply with the following:

1. Materials, packaging, deliveries, and storage containers shall be kept clean and free of invasive pests. This includes soil, aggregate, and rock material from quarries or the source location, as well as temporary facilities and best management practice (BMP) items employed to manage surface water drainage and erosion.
 2. Materials, packaging, deliveries, and storage containers will be inspected by NPS personnel for invasive pests. This includes soil, aggregate, and rock material from quarries or the source location, as well as temporary facilities and best management practice (BMP) items employed to manage surface water drainage and erosion.
 3. Any item that does not pass inspection will be turned away and not allowed into the park.
 4. The pest control records from quarries, source locations, or Contractor base yards may be requested.
 5. Fresh crushed aggregate is preferred (if readily available) over aggregate that has been stockpiled at a quarry for a lengthy period.
- F. Personal Protective Equipment (PPE): In order to prevent the spread of invasive species, the Contractor shall comply with the following:
1. Before going into the field or the jobsite, visually inspect and clean all personal protective equipment (PPE) to include boots, clothes, hard hats, harnesses, belts, and equipment for dirt, mud, seeds, plant debris, and insects. Soles of shoes must be regularly sanitized using a solution of >70% isopropyl alcohol.

3.3 INVASIVE SPECIES INSPECTIONS

- A. Inspections for Invasive Species: In order to prevent the spread of invasive species within the park and adjacent lands, NPS personnel from the park will perform inspections of vehicles, equipment, machinery, cutting tools, base yards, staging areas, materials, material packaging, material deliveries, material storage, and personal protective equipment (PPE). Please permit safe unfettered access to inspection sites or items being inspected for the NPS inspectors. Inspections for invasive species may include the following:
1. NPS inspectors will perform visual inspections for signs of invasive weeds and plants; invasive pests such as coqui frogs and frog eggs, little fire ants (LFA), and insects including termites; other vertebrate species such as rats, mice, and reptiles; and mud and dirt that may carry Rapid 'Ōhi'a Death (ROD).
 2. NPS inspectors will test vehicles, equipment, and machinery for little fire ants (LFA) by placing chopsticks baited with a dab of peanut butter for 30 minutes to 1 hour. About six (6) chopsticks (baits) will be used to test personal vehicles and trucks. Larger vehicles, equipment, and machinery may require up to 10-20 baits. Baits will be placed in the shade, inside the cab, engine compartment, and truck bed if applicable. Please note that baits are only good for detection and do not control ants. Baits will not be left in the vehicle for more than 1 hour as this may attract ants from beyond the area of concern.
 3. NPS inspectors may employ the same method of testing (as outlined above in G.2) on materials, material packaging, material deliveries, and material storage to include soil, aggregate, and rock material from quarries or source locations, temporary facilities and BMP items employed to manage surface water drainage and erosion.
 4. Any LFA found will be collected, bagged, and labeled for identifications by NPS inspectors (or NPS partners) and the vehicle will not be allowed to enter the park until it is cleaned, sanitized, and re-tested following a minimum 48-hour resting period.
 5. Infested vehicles will be sanitized following recommendations by the Hawaii Ant Lab (<http://www.littlefireants.com>) or other ant control expert and in accordance with all State

and Federal laws and baited with a granular or liquid insecticide bait prior to re-entry to the park. Control records will be required to confirm treatment. Treatment is the responsibility of the Contractor.

6. The park reserves the right to conduct additional tests for little fire ants (LFA) at any time during the project or require deployment of preventative granular or liquid insecticide baits.
7. Note that vehicles, equipment, machinery, and cutting tools previously used on the Big Island of Hawai'i will not be approved for entry into Haleakala National Park on Maui, and vice versa (items used on Maui will not be permitted into parks on the Big Island).

B. Invasive Species Disposal / Removal Recommendations: For current disposal / removal recommendations, refer to the following website links:

1. Little Fire Ants: <http://www.littlefireants.com>
2. Coqui Frogs: <http://www.biisc.org>
3. Rapid ōhi'a death (ROD): <https://cms.ctahr.hawaii.edu/rod>
4. All taxa: <http://www.biisc.org>

C. Inspection Scheduling / Protocol: If inspections for invasive species are required, the Contractor shall adhere to the following protocol:

1. The Contractor's construction schedule shall identify the date of project mobilization as well as the preferred date(s) for invasive species inspections in advance of mobilization.
2. The name and contact information for the NPS primary and backup inspectors will be identified at the time of the preconstruction meeting and documented in the meeting notes.
3. Formal requests for inspections shall be made by the Contractor directly to the NPS inspector(s) via the inspector's contact information.
4. Requests for the initial inspection(s) prior to mobilization shall be made at least three (3) days in advance of the inspection date.
5. Once the project is underway, inspections will be performed daily as needed (typically early morning) at the park's preferred inspection site for the duration of the contract.
6. Inspection requests outside the pre-established days and times will be performed at the discretion and availability of the inspectors.
7. Inspections shall occur at the park's preferred inspection sites identified below:
 - a. At *Kalaupapa National Historical Park* – At the Kalaupapa pier after barge delivery of items. If invasive species are found, they may be quarantined until such time that they can be treated by the Contractor per Section 01 11 40, PART 3.3.B, or, sent back to Oahu on the barge at the Contractor's expense.
8. Inspections of soil, aggregate, or rock materials may occur at the quarry or source location at the park's discretion.
9. If a re-inspection is required, the Contractor shall clean and sanitize the item thoroughly. A re-inspection for a rejected item cannot be scheduled for at least 48 hours.

3.4 THREATENED & ENDANGERED SPECIES (T&E SPECIES)

- A. General: Threatened & Endangered Plant and Animal Species are protected by Federal law. Workers must obey all park regulations and drivers must obey all traffic laws and watch for animals on or alongside roads or within the project area. Contact NPS Dispatch at: **(808) 985-6170** immediately if an animal is hit or injured or will not move off the road.
- B. General Mitigations: Within the project area, the following mitigation measures are required to minimize disturbance to T&E species:

1. The project area shall be cleaned-up at the end of each work shift so that tools, materials, debris, and trash are not left out in a manner that could be a hazard to T&E species.
2. Noise and vibrations from tools and equipment shall be kept to a minimum when working in the vicinity of a listed T&E species.
3. Hazmat spill prevention protocols shall be employed when using gas-powered equipment.
4. Tarps shall be used to contain all paint chips and building debris from exterior surfaces as these can be hazards to listed T&E species.
5. Ensure no standing water on tarps or other construction surfaces as it may be a breeding ground for mosquitoes which may carry Dengue fever, avian malaria / flu, and other diseases.
6. Ensure loose nails, screws, and fasteners do not fall on the ground when working on exterior surfaces as these can be a hazard to T&E species. If they do land on the ground, pick them up immediately and dispose of them properly.

3.5 ARCHAEOLOGICAL RESOURCES

- A. General: Archaeological resources are physical evidence of past human activity, including evidence of the effects of that activity on the environment. Archaeological resources are found above and below ground and underwater. Known archeological resources will be marked and flagged by the park. Do not disrupt known archaeological resources or flagged areas during construction activities. If required in Section 01 11 00, PART 1.2.D, the Contractor shall erect temporary barriers around the archaeological resource to help ensure avoidance. Temporary barriers shall be free-standing orange traffic delineator posts with grabber / looper tops and plastic rails (or a comparable barrier that does not involve digging but is an effective barrier). Submit a CM-16 with the proposed temporary barrier product for NPS approval.

The Contractor does not need to provide an Archaeological Monitor; however, the Contractor, or Contractor's designee, will observe all ground-disturbing activities during earthwork and other construction operations. If the Contractor discovers archaeological resources, the Contractor shall stop work, notify the Contracting Officer immediately, and relocate to a non-sensitive area. The park may have an Archaeological Monitor present at the jobsite who is authorized to stop work upon discovery of archaeological resources. Archaeological resources typically found in the park or at the project site will be discussed at the preconstruction meeting.

3.6 SOUNDSCAPE / NOISE

- A. General: The Contractor shall minimize noise levels at the project site to protect the soundscape of the park and minimize noise to staff, visitors, neighbors, and habitat. Perform operations to minimize noise. Perform noise-producing work in less sensitive hours of the day or week or as directed by the Contracting Officer. Repetitive and/or intermittent, high-level noise is permitted only during daytime hours. If the Contractor exceeds the thresholds below, the Contractor may be required to stop work, temporarily relocate to a non-sensitive area, or adjust the construction means and methods at no additional expense to the Government.

1. Maximum permissible construction equipment noise at 50 feet:

<u>EARTHMOVING</u>	<u>dB(A)</u>	<u>MATERIALS HANDLING</u>	<u>dB(A)</u>
Front Loaders	75	Concrete Mixers	75
Backhoes	75	Concrete Pumps	75
Dozers	75	Cranes	75
Tractors	75	Derricks Impact	75

Scrapers	80	Pile Drivers	95
Graders	75	Jack Hammers	75
Trucks	75	Rock Drills	80
Pavers, Stationary	80	Pneumatic Tools	80
Pumps	75	Saws	75
Generators	75	Vibrators	75
Compressors	75		

2. Do not exceed the following dB(A) limitations at 50 feet:

<u>TIME DURATION OF IMPACT NOISE</u>	<u>dB(A)</u>
More than 12 minutes in any hour	70
More than 3 minutes in any hour	80

3.7 AIR QUALITY (INDOOR & OUTDOOR)

- A. General: The Contractor shall minimize the negative impacts to indoor and outdoor air quality as a result of their operations. This may include:
 1. Control of emissions from vehicles or heavy equipment.
 2. Control of fumes, vapors, or gases from products or packaging.
 3. Control of particulates and dust from indoor operations such as cutting and sanding.
 4. Control of particulates and dust from outdoor operations.
 5. Control of moisture during operations that may lead to damage, fumes, or mold.
- B. Mitigations: The Contractor shall mitigate for negative impacts to indoor and outdoor air quality as a result of their operations. This may include:
 1. Ensure vehicles and heavy equipment do not idle when not in use.
 2. Ensure proper ventilation and air movement of indoor spaces.
 3. Install temporary mechanical ventilation where appropriate.
 4. Ensure tools are equipped with the proper guards or particulate catchment devices.
 5. Ensure particulates and debris are collected and disposed of on a regular basis.
 6. Consider water trucks or temporary irrigation devices for dust control on unpaved surfaces.
 7. Ensure products are properly dried or cured prior to advancing the work.
 8. Ensure materials are stored in a dry location.
 9. Ensure wet / damp materials are not installed or covered.

END OF SECTION 01 11 40

SECTION 01 27 00 – DEFINITION OF CONTRACT LINE ITEMS (CLIN)

PART 1 – GENERAL

1.1 SUMMARY

- A. This section explains, in general, what is and is not included in a contract line item and defines where one item ends and another begins.
- B. If no contract line item exists for a portion of work, include costs in a related item.

PART 2 – PRODUCTS (Not Used)

PART 3 – EXECUTION

3.1 LIST OF CONTRACT LINE ITEMS

A. Contract Line Item (CLIN) 00010

- 1. This item consists of furnishing and installing new mini-split air-conditioning systems in the curatorial facility. This item also consists of furnishing new commercial dehumidifiers.
- 2. Payment will be made at the contract lump sum price.

B. Option A (Deduct)

- 1. This item consists of a contract deduction if the park completed the earthwork and furnished and installed the concrete pads to support the ACCU units prior to contractor mobilization.
- 2. Payment will be made at the contract lump sum price.

END OF SECTION 01 27 00

SECTION 01 31 00 – PROJECT MANAGEMENT & COORDINATION

PART 1 – GENERAL

1.1 SUMMARY

- A. This section includes administrative provisions for coordinating construction operations on the Project including, but not limited to, the following:
 - 1. Construction Coordination
 - 2. Requests for Information (RFI)
 - 3. Project Meetings

1.2 CONSTRUCTION COORDINATION

- A. Contractor Coordination: Coordinate construction operations included in different drawing sheets and/or specification sections to ensure efficient and orderly installation of each part of the Work.
 - 1. Schedule construction operations in the sequence required to obtain the best results where installation of one part of the Work depends on installation of other components.
 - 2. Coordinate installation of different components with other Contractors to ensure maximum access for required maintenance, service, and repair.
 - 3. Plan and coordinate construction operations to incorporate applicable permit requirements.
- B. Administrative Procedures: Coordinate scheduling of administrative procedures with other construction activities to avoid conflicts and ensure orderly progress of the Work.

1.3 REQUEST FOR INFORMATION (RFI)

- A. General: Upon discovery of the need for additional information or an interpretation of the contract documents, the Contractor shall prepare and submit an RFI. The Contracting Officer will only respond to RFIs submitted by the prime Contractor. Coordinate and submit RFIs in a prompt manner to avoid delays in the work.
- B. RFI Form: Include a detailed, legible description of the subject or item needing more information or interpretation on the NPS RFI Form with the information below. Submit the RFI form in MS Word format to allow the NPS to provide a response.
 - 1. RFI number, numbered sequentially.
 - 2. Date of RFI.
 - 3. RFI subject.
 - 4. Specification Section, section title, and related paragraphs, if applicable.
 - 5. Drawing number and detail references, if applicable.
 - 6. Description of the issue or question.
 - 7. Photographs or sketches, if applicable.
 - 8. Contractor's suggested resolution. State whether the suggested resolution results in "no additional time / cost" to the contract or if it will result in "additional time / cost." Do not provide a detailed price proposal unless requested to do so by a Contracting Officer.
- C. RFI Response: The Contracting Officer, or authorized representative, will review each RFI, determine the action required, and respond accordingly.

1.4 PROJECT MEETINGS

- A. Preconstruction Meeting: Before the start of construction, the Contracting Officer will arrange for a preconstruction meeting (typically by MS Teams or teleconference). A meeting agenda will be forwarded to the group prior to the meeting. During the meeting, the agenda will be updated by the Contracting Officer or designee and distributed to the group as the meeting minutes.
- B. Progress Meetings: The NPS may schedule periodic meetings to discuss the progress of the Work with the Contractor, subcontractor(s), and key NPS staff. The need for and frequency of these meetings will be discussed at the preconstruction meeting. Progress meetings are typically held once per week via teleconference and last no more than 30 minutes. These weekly meetings are mainly to discuss progress for team situational awareness. Meeting minutes are not required.

PART 2 – PRODUCTS (Not Used)

PART 3 – EXECUTION (Not Used)

END OF SECTION 01 31 00

REQUEST FOR INFORMATION (RFI)

CONTRACTOR SECTION					
Park Name & Project Title:					
NPS Solicitation / Contract No:					
Contractor Name:					
Subject of RFI:					
RFI Number:		Date Submitted:		Response Required By:	
<i>Please provide additional information or clarifications to the questions below:</i>					
<i>Contractor Recommendation, if applicable</i>					
NPS SECTION					
Name of NPS Respondent:				Date of Response:	
<i>Below is additional information or clarifications in response to the Contractor RFI:</i>					

SECTION 01 32 16 – CONSTRUCTION SCHEDULE

PART 1 – GENERAL

1.1 SUMMARY

- A. This section consists of Construction Schedule requirements including:
 - 1. Schedule of Values
 - 2. Construction Baseline Schedule
 - 3. Construction Schedule Updates
 - 4. Look-Ahead Schedule
- B. Purpose: The construction schedule ensures adequate planning, coordination, scheduling, and reporting during execution of the Work by the Contractor. The construction schedule will assist the NPS and the Contractor in the monitoring the progress of the work, evaluating the proposed changes, and the processing the Contractor's progress payments.

1.2 DEFINITIONS

- A. Schedule of Values (SOV): A document that helps define the payment process on contracts. The SOV allocates the entire awarded contract sum to various portions or activities of the Contractor's work. The value for each portion of work in the SOV shall generally coincide with the value(s) outlined in the accepted price proposal (where applicable).
- B. Activity: A discrete part of a project identified for planning, scheduling, monitoring, and controlling the construction project. Activities included in a construction schedule consume time and resources. Critical activities are activities on the critical path. They start and finish on the planned early start and finish times. Predecessor activities are activities that precede another activity in the schedule network. Successor activities are activities that follow another activity in the schedule network.
- C. Critical Path: Longest connected chain of interdependent activities through the schedule network that establishes the minimum overall project duration and contains no float.
- D. Float: Measure of leeway in starting and completing an activity.

1.3 SUBMITTALS

- A. Schedule of Values (SOV): Submit a schedule of values (SOV) with a breakdown of all major work activities for which progress payments will be requested. The SOV should identify the park, project title, contract number, and Contractor name. In addition, the SOV must also include the following columns (at a minimum): Item #, Activity Description, Total Contract Value (\$), Previous Pay Applications (\$), Work this Period (\$), % Complete (%), and Balance to Complete (\$). The SOV must also have a total value for all (\$) columns.

Ensure each portion of work or activity is assigned to the appropriate Contract Line Item (CLIN), Bid Option, or Contract Modification. Ensure an updated SOV is submitted with each invoice.

Ensure the SOV is updated with each contract modification. Failure to submit an accurate SOV reflecting the work completed at the time of invoice submission may delay invoice approval.

B. Construction Baseline Schedule: Submit a construction baseline schedule with all major work activities. At a minimum, the schedule must have the following information:

- Milestone: Contract Award (actual date)
- Activity: Permit Applications & Approvals, if applicable
- Activity: Division 01 Plan Preparation & Review
- Milestone: Preconstruction Meeting & Notice to Proceed (anticipated)
- Activity: Material Submittal Preparation & Review
- Activity: Material Purchasing & Delivery
- Activity: Initial Invasive Species Inspections & Field Mobilization
- Activities: All Major Work Activities (with start and end dates) with critical path identified
- All Federal and State Holidays.
- Activities: Testing and Service Interruptions
- Milestone: Pre-Final Inspection (at Substantial Completion)
- Activity: Punchlist Completion (within 30 days after Substantial Completion)
- Milestone: Final Inspection (within 30 days after Substantial Completion)
- Activity: Field Demobilization (after all field work is complete and accepted)
- Activity: Preparation of Final Closeout Documents
- Milestone: End Contract Period of Performance

MS Project is the preferred computer software for developing a construction schedule; however, the NPS will accept any format that relays the requested information and can be easily updated and understood. No work shall be performed or shown on weekends, state, and federal holidays. Ensure coordination of the work with subcontractors and vendors in the development of the construction schedule and schedule updates.

C. Construction Schedule Updates: Submit updates to the construction baseline schedule: 1) when requested, 2) when there is a change to the critical path, 3) when the previously approved schedule is no longer accurate, or 4) when there is a change to the contract such as a modification. Construction schedule updates must be approved in advance of the Work.

D. Look-Ahead Schedule (Optional): The Contractor may submit a 1-3 week look-ahead schedule at the beginning of each week to notify the NPS of planned work and potential disruption to park operations. The look-ahead schedule typically shows proposed work planned for each day of the week. The look-ahead schedule does not need to be approved, but it must be consistent with the approved schedule. If the look-ahead schedule is not consistent with the approved schedule, submit a construction schedule update for approval. The look-ahead schedule can be submitted in any format, but weekly / monthly calendar view seems to work well.

PART 2 – PRODUCTS (Not Used)

PART 3 – EXECUTION (Not Used)

END OF SECTION 01 32 16

SECTION 01 33 23 – SUBMITTAL PROCEDURES

PART 1 – GENERAL

1.1 SUMMARY

A. Section includes administrative and procedural requirements for submittals including:

1. Submittal Procedures
2. Submittals
3. Submittal Review

1.2 DEFINITIONS

- A. Action Submittals: Written, graphic information, and physical samples that require the Government's approval.
- B. Informational Submittals: Written information that does not require the Government's approval. Submittals may be rejected for not complying with the requirements.
- C. Portable Document Format (PDF): An open standard file format licensed by Adobe Systems used for representing documents in a fixed-layout document format.
- D. File Transfer Protocol (FTP): Communications protocol that enables transfer of files to and from another computer over a network and that serves as the basis for standard Internet protocols. An FTP site is a portion of a network located outside of network firewalls within which internal and external users are able to access the files.

1.3 SUBMITTAL PROCEDURES

- A. General: Prepare and submit all submittals required by individual specification section, contract clause, or as requested in the drawings. The Contracting Officer reserves the right to require submittals in addition to those called for in individual specification sections. Review submittals for legibility, accuracy, completeness, and compliance with the contract documents prior to submittal to the Contracting Officer.
- B. Submittal Format: All submittals shall be transmitted using the NPS CM-16 form via email (or approved FTP / Sharepoint Site). Each submittal item shall have its own CM-16 form. Complete all applicable fields of the CM-16 form; provide a submittal number not previously used; and attach all supporting documentation. Resubmittals shall retain the original submittal number followed by a letter starting with "a, b, and so forth. The completed submittal may be submitted in PDF format (or optimized PDF format if file size is > 10 mb). Ensure all pages of the submittal are positioned upright (rotate page if needed), and all blank pages removed. No action will be taken on a submittal item unless it is accompanied by a properly formatted CM-16 form.
- C. File Transfer: For attachments with file size < 10 mb, transmit file(s) via email. For attachments with file size > 10 mb, transmit file(s) via FTP or secure Sharepoint Site. Note that Dropbox and similar commercial file share services are not secure and not allowed on the NPS network.

1.4 SUBMITTALS

- A. General: At a minimum, prepare and submit the following submittals for the project:
- B. Submittal Log (Optional): The Contractor may submit and update a submittal log to help keep track of the progress of all action and informational submittals.
- C. Contract Submittals: The following contract submittals shall be submitted within 7 days after contract award. Failure to submit these submittals may delay the notice to proceed. Note that the Contracting Officer (CO) will typically review and approve these submittals.
 - 1. Satisfactory evidence of Liability Insurance Coverage and Workers Compensation for the Contractor and all Subcontractors (per contract clause).
 - 2. Performance and Payment Bonds.
- D. Division 01 Informational Submittals: The following Division 01 informational submittals shall be received within 14 calendar days after contract award and at least 7 calendar days prior to the preconstruction meeting. If any of these documents have not been received within 7 calendar days of the preconstruction meeting, the meeting may be cancelled and the Notice to Proceed (NTP) may be delayed. Work shall not commence until written NTP has been issued by the Contracting Officer. Submit each document separately with its own CM-16 form for acceptance. Refer to applicable specification sections for the required content of each submittal. Note that the Contracting Officer's Representative (COR) will typically review and accept these submittals.
 - 1. Letter designating the Job Superintendent (comply with Section 01 73 40, PART 1.2.A).
 - 2. List of Subcontractors / Written Statement & Acknowledgement (SF1413 Form) for all Subcontractors certifying compliance with applicable labor clauses (per contract clause).
 - 3. Construction Schedule (comply with Section 01 32 16, PART 1.3.B).
 - 4. A comprehensive Schedule of Values (comply with Section 01 32 16, PART 1.3.A).
 - 5. Accident Prevention Plan (comply with Section 01 35 23, PART 1.2.A).
 - 6. Quality Control Plan (comply with Section 01 40 00, PART 1.4.A).
 - 7. Under-an-Acre Pollution Prevention Plan (UPPP) (comply with Section 01 57 23, PART 1.4).
 - 8. Construction Waste Management Plan (comply with Section 01 74 19, PART 1.4.A).
- E. Division 01-49 Action Submittals: All Division 01-49 action submittals shall be received and approved prior to construction start (i.e. field mobilization). If any of the required documents have not been received and approved prior to construction start, the start date could be delayed. Items shall not be purchased by the Contractor until written NTP has been issued by the Contracting Officer and the applicable Division 01-49 action submittals have been approved. Submit each submittal separately with its own CM-16 form for approval. Refer to applicable specification sections for the required content of each submittal. Note that the Contracting Officer's Representative (COR) will typically review and approve these submittals with assistance from key park personnel regarding colors, styles, or substitutions.
 - 1. Product Data - Submit product cut sheets / data on all materials & products to be used on the project or as required in the Division 01-49 specification sections. Highlight the size, model, or type of each required or recommended product. Ensure all products meet the Buy American Act provisions and are commercially available in the United States.

2. Samples – If applicable, submit a physical sample of a product and/or an array of color choices for approval.
- F. Shop Drawings – A shop drawing is a drawing or set of drawings produced by the Contractor, supplier, manufacturer, subcontractor, or fabricator. Shop drawings are typically required for prefabricated components or to illustrate the coordination work between trades. The shop drawing's primary emphasis is on the particular product or installation. Note that the Contracting Officer's Representative (COR) will typically review and approve these submittals.
1. Shop Drawings shall be developed in AutoCAD or similar drafting software.
 2. Shop Drawings shall include details, dimensions, notational information and other data to ensure the Contractor has a complete understanding of the work.
 3. Shop Drawings shall be submitted in DRAFT to the Contracting Officer along with a CM-16 form. Allow 7 calendar days for the drawings to be reviewed by the NPS. The Contractor shall correct any issues with the shop drawings as identified in the review comments and resubmit on a CM-16 form for final approval.
 4. Update shop drawings during and after installation if changes are necessary, and submit the final as-built shop drawings as part of the Closeout Documents.
- G. Permit Applications & Permit Approvals: All applicable permit applications and permit approvals shall be received prior to construction start (comply with Sections 011100, PART 1.13; Section 015723, PART 1.4; and Section 017340, PART 3.2.A). Note that the Contracting Officer's Representative (COR) will typically review and approve these submittals.
- H. Submittals (During Construction):
1. Construction Schedule Updates (comply with Section 01 32 16).
 2. Quality Control (QC) Daily Reports (comply with Section 01 40 00, PART 1.4.B).
 3. Testing & Inspection Data, if applicable (comply with Section 01 40 00, PART 1.4.C).
 4. Field Surveying Data, if applicable (comply with Section 01 73 40, PART 1.2.B).
 5. Field Engineering Data, if applicable (comply with Section 01 73 40, PART 1.2.C).
 6. Product Data for Newly Introduced Products (comply with Section 01 33 23, PART 1.4.D).
- I. Closeout Submittals: All closeout submittals shall be received and accepted prior to contract final acceptance. If any of these documents have not been received within 30 days after issuance of the Letter of Substantial Completion, final acceptance may not be issued. Note that the Contracting Officer's Representative (COR) will typically review and approve these submittals.
1. Project Record Drawings (comply with Section 017700, PART 1.2.B).
 2. Warranties (comply with Section 017700, PART 1.2.C)
 3. Operation & Maintenance (O&M) Manual (comply with Section 017700, PART 1.2.D).
 4. Project Photos (comply with Section 017700, PART 1.2.E)
 5. Project Closeout Binders (comply with Section 017700, PART 1.2.F).
- ## 1.5 SUBMITTAL REVIEW
- A. General: All submittals will be reviewed by the Contracting Officer, or authorized representative, to ensure compliance with the contract documents. Acceptance or approval of a Contractor's submittal does not relieve a Contractor from complying with the contract requirements.

- B. Informational Submittals: The Contracting Officer, or authorized representative, will review the informational submittal or resubmittal and indicate the appropriate action on the CM-16 Form as follows:
1. ACCEPT: Acceptable with no corrections.
 2. REJECT: Rejected as not in accordance with the contract or requiring major corrections, clarifications, or missing information. Contractor shall revise and resubmit the submittal.
- C. Action Submittals: The Contracting Officer, or authorized representative, will review the action submittal or resubmittal and indicate the appropriate action on the CM-16 Form as follows:
1. APPROVED: Acceptable with no corrections.
 2. APPROVED WITH NOTATIONS: Minor corrections or clarifications required. Comments are clear and no further review of resubmittal is required. Contractor shall address the review and comply with the comments when proceeding with the work.
 3. DISAPPROVED – RESUBMIT: Rejected as not in accordance with the contract or requiring major corrections, clarifications, or missing information. Contractor shall revise and resubmit the submittal with changes clearly identified.
- D. Brand Name or Equal: In accordance with Federal Acquisition Regulation Clause 52.211-6, items in this solicitation identified as “Brand Name or Equal,” include salient characteristics required to satisfy the Government’s needs. When proposing “equal” products, the Contractor [in addition to the requirements identified in Section L] shall submit evidence that “equal” products meet the salient physical, functional, or performance characteristics specified with their proposal for technical evaluation purposes. For each item proposed as an “approved equal” submit supporting data to include the following:
1. Name, address, and telephone number of the product vendor.
 2. Manufacturer’s literature / product cut sheets.
 3. Comparison of the salient characteristics between the proposed item and the item specified.
 4. Compliance with the Buy American Act / Commercially available in the United States.
 5. Difference in cost between the proposed item and the item specified.
- E. Partial or Incomplete Submittals: The Contracting Officer, or authorized representative, will review all submittals. If it appears the submittal is incomplete, it will be considered non-responsive and returned without further review.
- F. Processing Time: Allow at least 7 calendar days for the Contracting Officer, or authorized representative, to review a submittal and 3 calendar days to review a resubmittal.

PART 2 – PRODUCTS

PART 3 – EXECUTION

END OF SECTION 01 33 23

CM-16 TRANSMITTAL FORM - NATIONAL PARK SERVICE (NPS)

CONTRACT INFORMATION													
Park:		Contract No.:		Project Title:			Date:		Contractor Name:			Submittal No.	
CONTRACTOR ACTION								NPS ACTION					
								Action			Informational		
Item No.	Action Submittal	Informational Submittal	Specification Section	PART, Paragraph	Drawing No. (if applicable)	Description of Item (Attach Supporting Documentation)	Meets Buy American Act?	Alternative Material Proposed?	Approved	Approved with Notations	Disapproved - resubmit	Accepted	Rejected
NPS SUBMITTAL REVIEW COMMENTS							SIGNATURES						
							Contractor's Project Manager:					Date:	
							I certify that this submittal has been reviewed for accuracy, completeness, and compliance with contract requirements.						
							NPS CO / COR:					Date:	
							Approval of this submittal is subject to the contract terms and conditions, drawings and specifications. Review of submittals will be general and shall not be construed as permitting any departure from contract requirements.						

STATEMENT AND ACKNOWLEDGMENT

OMB Control Number: 9000-0066

Expiration Date: 4/30/2022

Paperwork Reduction Act Statement - This information collection meets the requirements of 44 U.S.C. § 3507, as amended by section 2 of the Paperwork Reduction Act of 1995. You do not need to answer these questions unless we display a valid Office of Management and Budget (OMB) control number. The OMB control number for this collection is 9000-0066. We estimate that it will take .05 hours to read the instructions, gather the facts, and answer the questions. Send only comments relating to our time estimate, including suggestions for reducing this burden, or any other aspects of this collection of information to: U.S. General Services Administration, Regulatory Secretariat Division (M1V1CB), 1800 F Street, NW, Washington, DC 20405.

PART I - STATEMENT OF PRIME CONTRACTOR

1. PRIME CONTRACT NUMBER		2. DATE SUBCONTRACT AWARDED		3. SUBCONTRACT NUMBER	
4. PRIME CONTRACTOR				5. SUBCONTRACTOR	
a. NAME				a. NAME	
b. STREET ADDRESS				b. STREET ADDRESS	
c. CITY		d. STATE	e. ZIP CODE	c. CITY	
				d. STATE	
				e. ZIP CODE	
6. The prime contract ☐ does, ☐ does not contain the clause entitled "Contract Work Hours and Safety Standards Act -- Overtime Compensation."					
7. The prime contractor states that under the contract shown in Item 1, a subcontract was awarded on the date shown in Item 2 to the subcontractor identified in item 5 by the following firm:					
a. NAME OF AWARING FIRM					
b. DESCRIPTION OF WORK BY SUBCONTRACTOR					

8. PROJECT		9. LOCATION	
10a. NAME OF PERSON SIGNING		11. BY (Signature)	
10b. TITLE OF PERSON SIGNING		12. DATE SIGNED	

PART II - ACKNOWLEDGMENT OF SUBCONTRACTOR

13. The subcontractor acknowledges that the following clauses of the contract shown in Item 1 are included in this subcontract:

Contract Work Hours and Safety Standards Act - Overtime Compensation (If included in prime contract see Block 6)	Construction Wage Rate Requirements
Payrolls and Basic Records	Apprentices and Trainees
Withholding of Funds	Compliance with Copeland Act Requirements
Disputes Concerning Labor Standards	Subcontracts (Labor Standards)
Compliance with Construction Wage Rate Requirements and Related Regulations	Contract Termination - Debarment
	Certification of Eligibility

14. NAME(S) OF ANY INTERMEDIATE SUBCONTRACTORS, IF ANY

A		C	
B		D	
15a. NAME OF PERSON SIGNING		16. BY (Signature)	
15b. TITLE OF PERSON SIGNING		17. DATE SIGNED	

SECTION 01 35 23 – SAFETY REQUIREMENTS

PART 1 – GENERAL

1.1 SUMMARY

- A. This section includes establishing an effective accident prevention program and providing a safe working environment for Contractor personnel, NPS staff, and park visitors.

1.2 SUBMITTALS

- A. Accident Prevention Plan (APP) – After contract award and before the Pre-Construction Meeting, submit a “site specific” accident prevention plan with the following information (at a minimum).
 - 1. Name of the onsite manager responsible for implementing the accident prevention plan.
 - 2. Emergency contact phone numbers (include NPS dispatch at: **808-985-6170**).
 - 3. Map of the local area emergency medical facilities relative to the jobsite.
 - 4. Contractor’s general company safety plan & protocol.
 - 5. Contractor’s first aid and emergency medical procedures & accident reporting forms.
 - 6. Contractor’s COVID-19 (Coronavirus) disease prevention protocols.
 - 7. Completed activity hazard analysis (AHA) completed for each major work activity.

1.3 QUALITY ASSURANCE

- A. General: Comply with the contract documents and contract clauses related to safety and health. In case of conflicts between Federal, State, and local safety and health requirements, the most stringent shall apply. Failure to comply with safety and health requirements may result in suspension of work. In the event of a perceived safety violation, NPS staff may stop work until the safety violation is addressed. In the event that NPS staff requests that Contractor personnel stop work due to a safety violation, contact the Contracting Officer immediately for a resolution.
- B. Onsite Safety Manager: Designate an authorized representative responsible for the following work. The authorized representative responsible for safety may also perform other duties.
 - 1. Implementing the accident prevention plan.
 - 2. Conducting daily safety meetings.
 - 3. Conducting safety inspections and reviews.
 - 4. Compiling and reviewing product safety data sheets (SDS) with personnel.
 - 5. Ensuring personnel are equipped with and wear proper personal protective equipment (PPE).
 - 6. Ensuring personnel comply with safety and health requirements.
 - 7. Ensuring personnel comply with the latest COVID-19 protocols.
 - 8. Investigation, reporting, and tracking accidents, injuries, or safety violations.
- C. Qualifications of Employees: Ensure the following for all Contractor personnel:
 - 1. Ensure personnel are physically able to perform their assigned duties in a safe manner.
 - 2. Do not allow personnel whose ability or alertness is impaired because of prescription or illegal drug use, fatigue, illness, intoxication, or other conditions that may expose themselves or others to injury to perform work.

3. Ensure operators of vehicles, hoisting equipment, and hazardous equipment shall be able to understand signs, signals, operating instructions, and be fully capable of operating such equipment. Retain copies of operator licenses and certifications onsite.
4. Provide adequate training to personnel to ensure prompt and efficient application of first aid.

PART 2 – PRODUCTS

2.1 FIRST AID

- A. General: Provide adequate first aid equipment for all contract employees for the hazards associated with the types of ongoing construction work at the site. NPS approval of first aid equipment is not required.

2.2 PERSONAL PROTECTIVE EQUIPMENT (PPE)

- A. General: Provide personal protective equipment (PPE) that meets applicable OSHA and ANSI standards for all contract employees. NPS approval of PPE is not required.

PART 3 – EXECUTION

3.1 DAILY SAFETY INSPECTIONS

- A. General: Conduct daily safety inspections and maintain safety reports.

3.2 EMERGENCY INSTRUCTIONS

- A. General: Display telephone numbers and reporting instructions for ambulance, physician, hospital, fire department, and police in conspicuous locations at work site.

3.3 FIRE & LIFE SAFETY

- A. General: Comply with requirements of the National Fire Protection Association (NFPA) 241 (Standards for Safeguarding Construction, Alteration, and Demolition Operations).

3.4 HAZARDOUS MATERIALS

- A. Hazardous Materials: Explosive, flammable, poisonous, corrosive, oxidizing, irritating, or otherwise harmful substances that could cause death or injury.
- B. Store hazardous materials in accordance with manufacturer's and OSHA Subpart D requirements. Display Material Safety Data Sheets (MSDS) in conspicuous locations at work site.
 1. Immediately report spills of hazardous materials to NPS Dispatch at: **808-985-6170**.
 2. Maintain a spill emergency response kit.
 3. Train employees how to respond to a spill and use emergency response kit.

3.5 PROTECTIVE EQUIPMENT

- A. General: Inspect personal protective equipment (PPE) daily and maintain in a serviceable condition. Clean, sanitize, and repair personal items as appropriate before issuing.
- B. Hard Hats: A hard hat use area shall be designated by the Contractor, as applicable. Hard hats shall be worn by all personnel in hard hat use areas. It is a Contractor's responsibility to require persons working at or visiting the jobsite to wear hard hats at all times.

3.6 SAFETY MEETINGS

- A. General: At a minimum, conduct weekly, if not daily, 15-minute "toolbox" safety meetings. These meetings shall be conducted by the Job Superintendent or a crew leader and be attended by all construction personnel at the worksite. Topics to coincide with work scheduled for the week. Document all safety meetings in the Quality Control (QC) Daily Report Form.

3.7 EMERGENCY MEDICAL & ACCIDENT REPORTING

- A. Emergency Medical & Accident Reporting: In the event of an emergency / medical incident, the Contractor shall **Dial 911** as needed. The Contractor shall also contact NPS Dispatch at: **808-985-6170** for any medical emergency, traffic accident, or personal injury occurring within the Park. NPS Dispatch is called so that the entrance station will be prepared should any emergency vehicles need to enter the park and so that any available Emergency Medical Services personnel in the park might respond until such time as the ambulance arrives.

END OF SECTION 01 35 23

SECTION 01 40 00 – QUALITY REQUIREMENTS

PART 1 – GENERAL

1.1 SUMMARY

- A. This section includes administrative and procedural requirements for quality assurance and quality control. The Contractor is responsible for all quality assurance and quality control to guard against defects and deficiencies of the work.

1.2 DEFINITIONS

- A. Quality Assurance: Activities, actions, and procedures performed before and during execution of the work to guard against defects and deficiencies and to substantiate that the proposed construction will comply with the requirements.
- B. Quality Control: Tests, inspections, procedures, and related actions during and after execution of the work to evaluate the actual products incorporated into the Work and to ensure that the completed construction complies with the requirements.
- C. Product Testing: Tests and inspections performed by 1) a Nationally Recognized Testing Laboratory (NRTL), 2) a National Voluntary Laboratory Accreditation Program (NVLAP), 3) a testing agency qualified to conduct product testing, or 4) a Contractor qualified to conduct product testing to establish product performance and compliance with industry standards.

1.3 CONFLICTING REQUIREMENTS

- A. Reference Standards: If compliance with two or more standards is specified, and those standards establish different or conflicting requirements, refer the conflicting requirements or uncertainties to the Contracting Officer for a determination before proceeding with the Work.
- B. Minimum Quality Level: Quality level shown or specified shall be the minimum provided or performed. Actual installation may comply exactly with the minimum quality specified, or it may exceed minimum quality within reasonable limits. Refer all uncertainties to the Contracting Officer for a determination before proceeding with the Work.

1.4 SUBMITTALS

- A. Quality Control Plan (QCP) – After contract award and before the Pre-Construction Meeting, submit a “site specific” quality control plan with the following information (at a minimum).
 - 1. Name of the onsite manager responsible for implementing the plan and ensuring quality.
 - 2. Organizational flow chart showing key contractor / subcontractor personnel.
 - 3. Qualifications of key staff with applicable training certificates and licenses.
 - 4. Contractor communications and reporting procedures / forms (to also include the NPS daily report form, RFI form, and CM-16 form).
 - 5. A list of project specific testing and inspection procedures at all phases of the work (i.e. preparatory, initial, follow-up) to ensure quality, with steps for correction of work / re-work.
 - 6. Qualifications of laboratory, agency, or Contractor performing product testing.

- B. Quality Control (QC) Daily Reports: Submit quality control daily reports documenting daily progress and the quality of work. The daily report shall be typed, legible, and complete and include representative and legible photos of that day's work. The daily report shall be submitted daily no later than the business day after the work was performed. The NPS prefers that the Contractor use the NPS daily report form. If the Contractor prefers to use their own form, it will be allowed if it contains all major sections / information from the NPS form.
- C. Testing and Inspection Data / Reports: Submit all completed testing and inspection data / reports for approval prior to advancing the Work on subsequent phases. If a testing or inspection report indicates the Work does not meet the requirements, submit remedial actions for approval within one (1) calendar day from receiving the results, and do not advance the Work until this issue is resolved in writing.

1.5 QUALITY ASSURANCE

- A. General: Comply with the contract documents and contract clauses related to quality. In cases of conflicts between Federal, State, and local quality requirements, the most stringent shall apply. Failure to comply with quality requirements may result in suspension of work.
- B. Onsite Quality Manager: Designate an authorized representative responsible for the following work. The authorized representative responsible for quality may also perform other duties.
 - 1. Implementing the quality control plan.
 - 2. Conducting quality inspections and reviews.
 - 3. Compiling and reviewing product testing and inspections with quality personnel.
 - 4. Ensuring all personnel comply with quality requirements.
 - 5. Completing and submitting QC Daily Reports in a timely manner.
 - 6. Completing and submitting all other submittals in a timely manner.

1.6 QUALITY CONTROL

- A. General: The Contractor is responsible for testing and inspections as identified in the contract documents. Test and inspect the work as needed to ensure that the quality of materials, workmanship, construction, finish, and functional performance comply with applicable specifications, drawings, permit requirements, and as required by the local Building Code.

PART 2 – PRODUCTS (Not Used)

PART 3 – EXECUTION

3.1 OFF-SITE CONTROL

- A. General: Items fabricated off-site shall be inspected for quality control at place of fabrication.

3.2 ON-SITE CONTROL

- A. Preparatory Phase: Prior to beginning each feature of the work:
 - 1. Review submittal requirements with personnel directly responsible for quality assurance and quality control of the work.

2. Review applicable specification sections and drawings related to the featured work.
3. Ensure copies of referenced standards related to testing and inspections are available onsite.
4. Ensure provisions have been made for field control testing.
5. Examine work area to ensure preliminary work has been completed.
6. Verify field dimensions and advise the Contracting Officer of any uncertainties or discrepancies with the contract documents.
7. Ensure necessary equipment and materials are at the project site and they comply with the approved material submittals and shop drawings (if applicable).
8. Document preparatory phase activities in the Quality Control (QC) Daily Report.

B. Initial Phase: As soon as work begins:

1. Test and inspect representative portions of a particular feature of work for quality of workmanship. Document type and locations of all tests and inspections.
2. Review control testing procedures to ensure compliance with the contract requirements.
3. Document initial phase activities in the Quality Control (QC) Daily Report.

C. Follow-Up Phase: As work progresses:

1. Ensure compliance with contract requirements until completion and acceptance of work.

3.3 ENFORCEMENT

- A. The Contractor shall stop work on any item or feature pending the satisfactory correction of the deficiency noted by the Contracting Officer.

3.4 REPAIR AND PROTECTION

- A. General: Upon completion of field testing or sampling, repair damaged construction and restore substrates and finishes. Protect construction exposed by or for quality control activities.

END OF SECTION 01 40 00

QC DAILY REPORT FORM - NATIONAL PARK SERVICE (NPS)

Date of Report: Day, Date

Report Submitted By: Name of Person

CONTRACT INFORMATION						
Park:		Contract No.		Project Title:		
Prime Contractor:		Onsite Job Superintendent:		Subcontractor(s) Onsite:		Vendor Deliveries:
NPS Contracting Officer:		Contracting Officer's Rep (COR):		NPS Inspector(s) / Monitor(s):		Official Visitor(s) to Site:
WEATHER REPORT						
Conditions (AM)	Conditions (PM)	Min. Temp	Max. Temp	Rainfall (in.)	Weather Delays or Disruptions:	
SITE CONDITIONS						
Jobsite / Housekeeping:		Conflict(s) with NPS Operations:		Utility Disruptions:		Subsurface Conditions / Digging:
LABOR POOL / PERSONNEL				HEAVY EQUIPMENT		
Employees by Job Category or by Crew:	Prime	Sub.	Hours	Heavy Equipment by Type:	Hours Operating	Hours Idle
Total Labor Hours:				Total Equipment Hours:		
SUMMARY of DAILY WORK (Attach Representative Photos of the Daily Work Progress)						
INSPECTIONS / TESTING						
PLANNED WORK / UPCOMING MILESTONES						
URGENT ITEMS for DISCUSSION						

SECTION 01 50 00 – TEMPORARY FACILITIES & CONTROLS

PART 1 – GENERAL

1.1 SUMMARY

- A. This section includes requirements for temporary facilities & controls.

1.2 QUALITY ASSURANCE / REGULATORY

- A. General: Comply with the latest edition of applicable codes, regulations, and standards when installing temporary facilities & controls.
- B. Temporary Water & Wastewater Services: Comply with:
 - 1. Hawaii: State of Hawaii, Department of Health (DOH), Hawaii Administrative Rules (HAR).
 - 2. Pacific Island Territories (Guam, Saipan, American Samoa): EPA Region 9.
- C. Temporary Potable Water Services: All materials in contact with potable water must be NSF/ANSI-61 certified.
- D. Hydrotesting & Disinfection Water: For hydrotesting and disinfection of potable water systems, comply with applicable sections from the American Water Works Association (AWWA) Standards as well as the Authority Having Jurisdiction (AHJ) regarding pollution prevention.
- E. Temporary Erosion and Sediment Control: Refer to Section 01 57 23.
- F. Temporary Electrical Service: Install service to comply with the International Electric Code.
- G. Temporary Fire Protection Services: Install service to comply with the National Fire Protection Association (NFPA) Code.
- H. Telecommunications: Install service to comply with regulations from the local service provider.
- I. Pedestrian Access Routes: Install temporary routes to comply with the Architectural Barriers Act Accessibility Standards (ABAAS) if the affected pedestrian access route is currently accessible.
- J. Temporary Traffic Control Devices: Install temporary traffic control devices to comply with the Manual on Uniform Traffic Control Devices (MUTCD).

1.3 SUBMITTALS

- A. Specified / Proposed Products: All products specified or proposed by the Contractor for use as temporary facilities & controls shall be submitted for approval per Section 01 33 23.

PART 2 – PRODUCTS

2.1 TEMPORARY MATERIALS

- A. General: Temporary materials may be new or used but they must be adequate for the usage; they must not create an unsafe condition; and they must not violate the requirements of applicable codes, regulations, and standards. Submit temporary materials for approval per Section 01 33 22.
- B. Temporary Utilities (Potable Water, Non-Potable Water, Fire Protection, Wastewater, Electrical Power, HVAC, Security, Alarms, Notifications, Internet and Telecommunications): Varies depending on utility type. Comply with PART 1.2 Quality Assurance / Regulatory. Refer also to Divisions 03-48, where applicable.
- C. Telecommunications: Varies depending on local service provider.
- D. Temporary Erosion and Sediment Control Devices: Refer to Section 01 57 23.
- E. Temporary Safety Barrier Fence: Orange plastic fence, 4-feet height minimum, secured to self-supporting posts or steel posts driven into the ground no greater than 6-feet apart. If posts are driven into the ground, comply with Section 01 11 40, PART 3.4 (Archaeological Resources).
- F. Temporary Barrier Tape: Yellow tape imprinted with "CAUTION: CONSTRUCTION AREA," manufactured by Reef Industries, Inc. Houston, Texas, or approved equal.
- G. Temporary Site Enclosure Fence: Chain link, galvanized steel, 6-feet height minimum, secured to top and bottom rails and self-supporting posts with concrete bases.
- H. Temporary Barrier or Partition: Plywood secured to lumber with nails, screws, or fasteners.
- I. Temporary Traffic Control Devices: Varies depending on project scope and site. Comply with PART 1.2 Quality Assurance / Regulatory.
- J. Lumber and Plywood: Refer to Division 06.
- K. Concrete: Refer to Division 03.
- L. Asphaltic Concrete (AC) Pavement: Refer to Division 32.

2.2 TEMPORARY EQUIPMENT

- A. General: Temporary equipment may be new or used but it must be adequate for the usage; it must not create an unsafe condition; and it must not violate the requirements of applicable codes, regulations, and standards. Submit temporary equipment for approval per Section 01 33 22.
- B. Fire Extinguishers: Portable, UL rated, with class and extinguishing agent as required by locations and classes of fire exposures.

2.3 TEMPORARY FACILITIES

- A. General: Temporary facilities may be new or used but they must be adequate for the usage; they must not create an unsafe condition; and they must not violate the requirements of applicable codes, regulations, and standards. Submit temporary materials for approval per Section 01 33 22.

- B. Temporary Toilets: Sufficiently lighted, ventilated, weatherproof, ADA-accessible toilet facilities with sturdy enclosures, privacy locks, and anti-bacterial hand sanitizer.

2.4 TEMPORARY ITEMS (GOVERNMENT-FURNISHED)

- A. General: Refer to Summary of Work Section 01 11 00, PART 1.5 for a list of government-furnished items.

PART 3 – EXECUTION

3.1 INSTALLATIONS

- A. General: Install temporary facilities & controls as noted in the contract documents, per regulatory guidance, or as instructed by the Contracting Officer.
- B. Temporary Utilities (Potable Water, Non-Potable Water, Fire Protection, Wastewater, Electrical Power, HVAC, Security, Alarms, Notifications, Internet and Telecommunications): Install temporary utility service in compliance with applicable codes, regulations, and standards.
- C. Temporary Erosion and Sediment Control Devices: Refer to Section 01 57 23.
- D. Temporary Toilets: Provide a sufficient number of toilets for Contractor personnel, NPS staff, and park visitors, as applicable. If toilets are to be used by NPS staff and/or park visitors, provide separate toilet facilities for men and women. The Contractor shall pump the toilets frequently and keep the toilets clean so as not to attract insects or animals.
- E. Temporary Field Offices: Establishment of a construction camp within the park will not be permitted. The park will make every attempt to furnish the contractor with indoor office space for up to two (2) contractor personnel; however, if the park cannot accommodate the Contractor with office space, the Contractor will be allowed to setup a temporary field office within the project area limits (final location to be approved by the park facility manager).
- F. Temporary Storage Space: The construction of storage sheds will not be allowed at the project site. If interior storage or covered space is required to protect materials and equipment, the park will attempt to provide a space for temporary use. The use of Contractor-furnished shipping containers may be allowed at the project site at the discretion of the Contracting Officer.
- G. Temporary Safety Barriers: For controlled sites, the Contractor shall construct and maintain a safety barrier fence marked with barrier tape to discourage people from entering the project site. Note that the Contractor is responsible for safety and accident prevention at the jobsite.
- H. Temporary Site Enclosures: For uncontrolled sites, the Contractor shall construct and maintain a site enclosure fence, barrier, or partition to prevent people and animals from entering the project site except by locked gates or doors. Note that the Contractor is responsible for safety and accident prevention at the jobsite.
- I. Pedestrian Access Routes: Construct and maintain temporary pedestrian access routes as outlined in the contract documents while the primary pedestrian access routes are under construction.

- J. Vehicle Traffic Access Routes: Construct and maintain temporary roads and paved areas as outlined in the contract documents while the primary vehicle access routes are under construction.
- K. Temporary Traffic Controls: The Contractor shall erect and maintain barricades, danger signals, and warning signs in compliance with applicable codes and standards. The Contractor shall provide adequate traffic control, including flaggers on either side of any work performed in the road. Any workers on or near the road must wear highly visible, bright clothing.
- L. Construction Waste Management: Refer to Section 01 74 19.

END OF SECTION 01 50 00

SECTION 01 57 23 – TEMPORARY STORMWATER POLLUTION PREVENTION

PART 1 – GENERAL

1.1 SUMMARY

- A. This section includes requirements for temporary stormwater pollution prevention and implementing measures to prevent the discharge of pollutants during construction activities.

1.2 REGULATIONS

- A. The Clean Water Act prohibits anybody from discharging “pollutants” through a “point source” into a “water of the United States” unless they have a NPDES permit. The permit will contain limits on what you can discharge, monitoring and reporting requirements, and other provisions to ensure that the discharge does not hurt water quality or people’s health.
- B. The National Park Service (NPS) standards and guidelines requires that water quality be protected at all times to ensure compliance with the Organic Act.

1.3 DEFINITIONS

- A. Pollutant - The term pollutant is defined very broadly in the Clean Water Act. It includes any type of industrial, municipal, and agricultural waste discharged into water. Some examples include dredged soil, solid waste, incinerator residue, sewage, garbage, sewage sludge, munitions, chemical wastes, biological materials, radioactive materials, heat, wrecked or discarded equipment, rock, sand, cellar dirt, and industrial, municipal, and agricultural waste.
- B. Point Source – The term point source is also defined very broadly in the Clean Water Act. It means any discernible, confined and discrete conveyance, such as a pipe, ditch, channel, tunnel, conduit, discrete fissure, or container. It also includes vessels or other floating craft from which pollutants are or may be discharged.
- C. National Pollution Discharge Elimination System (NPDES) – A permit program that addresses water pollution by regulating point sources that discharge pollutants to waters of the United States. Created in 1972 by the Clean Water Act, the NPDES permit program is authorized to state governments by EPA to perform many permitting, administrative, and enforcement aspects of the program.
- D. Under-Acre Pollution Prevention Plan (UPPP) - A stormwater management plan that is developed and implemented to protect waters of the United States from pollutants during construction activities that disturb less than one (1) acre of total land area.
- E. Stormwater Pollution Prevention Plan (SWPPP) – A stormwater management plan that is developed and implemented to protect waters of the United States from pollutants during construction activities that disturb one (1) acre or more of total land area.

1.4 AUTHORIT(IES) HAVING JURISTICTION

- A. General: Comply with the standards and regulations of the Authority Having Jurisdiction (AHJ).
- B. Authority Having Jurisdiction: The following are the AHJ for the pacific island national parks:
 - 1. Hawaii: State of Hawaii, Department of Health (DOH), Clean Water Branch (CWB).
<https://health.hawaii.gov/cwb/general-permits/>
 - 2. Pacific Island Territories (Guam, Saipan, American Samoa): EPA Region 9
<https://www.epa.gov/npdes-permits>

1.5 NPDES GENERAL PERMITS OVERVIEW (STATE OF HAWAII)

- A. General Permits are NPDES permits issued by the State of Hawaii, Department of Health, Clean Water Branch for a specific type of activity or discharge. General permits can only cover the types of activities or discharges specified in that specific general permit. If a facility or project has multiple types of discharges, an applicant may separately request coverage under multiple general permits for their facility or project (e.g., applicant may request coverage for construction storm water and dewatering discharges). If a discharge or activity does not have a general permit that covers the discharge, the applicant must apply for an individual NPDES permit.
- B. List of NPDES General Permits – Below is a list of NPDES general permits in the State of Hawaii applicable to work in NPS units with a brief description of their authorized discharges:
 - 1. Discharges of Storm Water Associated with Construction Activities - NPDES permit coverage is required for activities that disturb one (1) acre or more of total land area. Land disturbance includes, but is not limited to, clearing, grubbing, grading, excavation, demolition (even if leaving foundation), uprooting of vegetation, equipment staging on grassed areas or bare ground, equipment staging on paved roadway (only if area blocked off from public usage), storage areas, and roadway work that touches the base course.
 - 2. Discharges of Hydrotesting Waters – NPDES permit coverage is required for the release or discharge of hydrotesting waters to State waters. “Hydrotesting Waters” means water used to test the integrity of a tank or pipeline, pipeline disinfection, and/or pipeline flushing.
 - 3. Discharges Associated with Construction Activity Dewatering – NPDES permit coverage is required for discharges to State waters of construction activity dewatering effluent. “Dewatering Effluent” is any type of water (e.g. ground water, storm water, stream water, ocean water, etc.) pumped from a construction area.
 - 4. Discharges of Treated Process Wastewater Associated with Well Drilling Activities – NPDES permit coverage is required for discharges to State waters of treated process wastewater associated with well drilling activities. Treated process wastewater includes well drilling slurries, lubricating fluids wastewaters, and well purge wastewaters.

1.6 SUBMITTALS / REQUIREMENTS

- A. General: Complete and submit the applicable submittals or complete the applicable requirements after contract award and prior to contract Notice to Proceed (NTP).
- B. Projects with No Land Disturbance (Indoors) – NO SUBMITTALS REQUIRED.
- C. Projects with Land Disturbance of less than one (1) acre: Complete and submit the following submittals to NPS (only) for review and acceptance.

- 1. Under-Acre Pollution Prevention Plan (UPPP). The NPS UPPP Template can be found at: https://www.nps.gov/dscw/precon_act113.htm

- D. Projects with Land Disturbance of one (1) acre or more in the State of Hawaii with an existing and valid NPDES general permit (Refer to Section 1.5 above), complete the following:

To be eligible for NPDES general permit coverage, an activity / discharge must be able to:

- 1. Meet any eligibility requirements (such as restrictions / prohibitions on coverage);
 - 2. Comply with all terms / requirements of the specific general permit;
 - 3. Comply with any pre-request requirements in the specific general permit such as the creation and completion of any required pollution plans (e.g. Storm Water Pollution Prevention Plan as required by Appendix C).
 - 4. Comply with the standard general permit conditions contained in Hawaii Administrative Rules (HAR) Chapter 11-55 Appendix A; and
 - 5. Report any unpermitted / unauthorized discharges to the Clean Water Branch and complete compliance with any enforcement action as a result, prior to seeking general permit coverage.

To request coverage under a general permit, the Contractor must complete / submit the following to the State of Hawaii, Department of Health (DOH), Clean Water Branch (CWB):

- 1. Notice of Intent (NOI) to the CWB e-Permitting Portal, and;
 - 2. Other as requested by the CWB.

- E. Projects with Land Disturbance of one (1) acre or more in the Pacific Island Territories (Guam, Saipan / Commonwealth of the Northern Marianas Islands, American Samoa), or, in the State of Hawaii (for an activity or discharge not covered by a NPDES general permit), secure a NPDES individual permit with the Authority Having Jurisdiction (AHJ). The NPS contract Period of Performance will be adjusted to reflect the additional time needed to secure a NPDES permit. A NPDES individual permit typically involves submittal of the following:

- 1. Notice of Intent (NOI).
 - 2. NPDES Individual Permit Application.
 - 3. Storm Water Pollution Prevention Plan (SWPPP).
 - 4. Notice of Termination (NOT).

1.7 QUALITY ASSURANCE

- A. General: Comply with the applicable pollution prevention plan and the requirements of the AHJ.

- B. Onsite Manager: Designate an authorized representative responsible for pollution prevention including the duties outlined below. The authorized representative may perform other duties.
1. Implementing the applicable pollution prevention plan.
 2. Ensuring all stormwater, erosion control, and pollution prevention devices and measures, aka Best Management Practices (BMP), comply with Work Restrictions Section 01 11 40.
 3. Properly installing BMP devices and measures.
 4. Conducting daily checks on BMP devices and measures and adjusting BMPs as needed.
 5. Ensuring all personnel comply with the applicable pollution prevention plan and are familiar with the BMP devices and measures.
 6. Removing all BMP devices and measures after final inspection / acceptance.

PART 2 – PRODUCTS

2.1 BEST MANAGEMENT PRACTICES (BMP) DEVICES & MEASURES

- A. General: Furnish and install the required stormwater, erosion control, and pollution prevention devices and measures, aka Best Management Practices (BMP), as outlined in the pollution prevention plan or as required by the AHJ. All devices shall comply with Section 01 11 40. Submit BMP devices for approval per Section 01 33 23.

PART 3 – EXECUTION

3.1 REGULATORY REQUIREMENTS

- A. General: Comply with the requirements of the Authority Having Jurisdiction (AHJ) and associated permit conditions.

3.2 PROTECT PARK RESOURCES

- A. General: Comply with the Work Restrictions Section 01 11 40 for all stormwater, erosion control, and pollution prevention BMP devices. If the recommended BMP devices or measures have the potential to damage the park's natural, cultural, historical, or archaeological resources, the NPS will suggest strategies to mitigate or prevent damage to these resources. The Contractor shall include these mitigation strategies in the applicable pollution prevention plan.

3.3 POLLUTION PREVENTION

- A. General: The Contractor shall execute the following:
1. Implement the applicable pollution prevention plan.
 2. Ensure all stormwater, erosion control, and pollution prevention devices and measures, aka Best Management Practices (BMP), comply with Work Restrictions Section 01 11 40.
 3. Implement mitigation strategies to prevent resource damage from BMP devices or measures.
 4. Properly install BMP devices and measures. The park's archaeological monitor may wish to be present for BMP installation involving ground disturbance.
 5. Conduct daily checks on BMP devices and measures and adjust BMPs as needed, especially after storm events.

6. Conduct daily checks on BMP devices and measures to ensure compliance with Safety Requirements Section 01 35 23.
7. Ensure all personnel comply with the applicable pollution prevention plan and are familiar with the BMP devices and measures.
8. Ensure all construction operations remain within the Project Area Limits.
9. Complete all work within the dry season or dry periods when possible.
10. If a discharge occurs or if the project receives a written notice or order from a regulatory agency, the Contractor shall immediately notify the Contracting Officer and shall file a written report to the AHJ within 7 days of the discharge event, notice, or order. Corrective measures shall be implemented immediately following the discharge, notice, or order.
11. Properly dispose of any entrapped sediment from within or behind BMP devices or measures.
12. Manage hazardous waste in accordance with Federal, State, and local regulations.
13. Remove all BMP devices and measures after final inspection.

END OF SECTION 01 57 23

SECTION 01 67 00 – PRODUCT REQUIREMENTS

PART 1 – GENERAL

1.1 SUMMARY

- A. This section includes administrative, procedural, and regulatory requirements for the selection of products for use on the project; product delivery, storage, handling, packaging; and manufacturer's standard warranties on products.

1.2 DEFINITIONS

- A. Products: Items purchased for incorporating into the Work, whether purchased for the project or taken from previously purchased stock. The term "product" includes "material," "equipment," "system," and terms of similar intent.
- B. Named Products: Items identified by manufacturer's product name, including make or model number or other designation shown or listed in manufacturer's published product literature.
- C. New Products: Items that have not previously been incorporated into another project or facility.
- D. Comparable Product: Product demonstrated and approved through the submittal process, or where indicated as a product substitution to have indicated qualities related to type, function, dimension, in-service performance, physical properties, appearance, and other salient characteristics that equal or exceed those of the specified or named product.
- E. Chain-of-Custody: Process whereby a product is maintained under physical possession or control during its entire life cycle.
- F. Government-Furnished Materials: Refer to Section 01 11 00, PART 1.5.

1.3 REGULATORY

- A. Buy American Products: Comply with contract clause 52.225-1.
[52.225-1 Buy American-Supplies | Acquisition.GOV](#)
- B. Environmentally Preferable Purchasing: Comply with the USEPA, Final Guidance on Environmentally Preferable Purchasing.
[Final Guidance on Environmentally Preferable Purchasing | US EPA](#)

1.4 SUBMITTALS

- A. Specified Products: All products specified in the contract documents or as a result of the Request for Information (RFI) process shall be submitted for approval per Section 01 33 23.
- B. Material Safety Data Sheets (MSDS): For each product required by OSHA to have MSDS, ensure MSDS are posted during the Work.

- C. Installed Products: All approved products furnished and installed on the project shall be submitted per Section 01 77 00.
- D. Manufacturer's Warranties: All manufacturer's warranties for products furnished and installed on the project shall be submitted per Section 01 77 00.

1.5 QUALITY ASSURANCE

- A. Compatibility: If the Contractor is given the option of selecting between two or more products for use on the project, the product selected shall be the most compatible with the existing conditions or systems. Refer all conflicting requirements or uncertainties to the Contracting Officer for a determination before proceeding with the Work.

1.6 PRODUCT DELIVERY, HANDLING, AND STORAGE

- A. General: Deliver, handle, and store products using means and methods that prevent damage, deterioration, loss, and theft during the entire chain-of-custody period.
- B. Delivery and Handling:
 - 1. Comply with manufacture's written instructions on delivery and handling of products.
 - 2. Schedule delivery of products to minimize long-term storage at the project site.
 - 3. Coordinate delivery with installation time to ensure minimum holding time for items that are flammable, hazardous, easily damaged, or vulnerable to deterioration, loss, or theft.
 - 4. Comply with Section 01 11 40 for inspection of products for invasive species.
 - 5. Deliver products to the project site in an undamaged condition; in the manufacturer's original sealed container or other packaging system; complete with labels and instructions for handling, storing, unpacking, protecting, and installing.
 - 6. Inspect products upon delivery to ensure compliance with the Contract Documents. Ensure products are undamaged and properly protected.
- C. Storage:
 - 1. Comply with manufacture's written instructions on the storage of products.
 - 2. Store products vulnerable to deterioration, loss, or theft in secure, weather tight enclosures above ground with ventilation adequate to prevent condensation.
 - 3. Store products vulnerable to damage from insects and vermin in insect-proof enclosures.
 - 4. Store cementitious products on protected and elevated platforms away from moisture.
 - 5. Comply with manufacture's written instructions for temperature, humidity, ventilation, and weather-protection requirements for storage of products.
 - 6. Store loose granular products in well-drained areas on solid surfaces to prevent mixing with foreign matter and debris.

1.7 PRODUCT PACKAGING

- A. General: Minimize packaging as defined in ASTM D5834. Maximize recycling procedures as outlined in ASTM D5834. Comply with Section 01 74 19.

PART 2 – PRODUCTS

2.1 SPECIFIED PRODUCTS:

- A. General: Furnish and install all products specified in the contract documents or as a result of the Request for Information (RFI) process. All products shall be new unless otherwise specified.

2.2 COMPARABLE PRODUCTS:

- A. General: The Contracting Officer will consider a Contractor's request for comparable products if the following conditions are satisfied.
 - 1. The comparable product does not require revisions to the contract documents.
 - 2. The comparable product is similar or better in quality to the specified product.
 - 3. The comparable product is compatible with other portions of the Work.
 - 4. The comparable product meets the Buy American Act provisions.
 - 5. The comparable product provides a manufacturer's warranty.
 - 6. The comparable product is new.

2.3 GOVERNMENT-FURNISHED MATERIALS

- A. General: Refer to Summary of Work Section 01 11 00, PART 1.5.

PART 3 – EXECUTION

3.1 INSTALLATION:

- A. General: Install all products and materials per the contract documents and the manufacturer's recommendations. Refer all conflicting requirements or uncertainties to the Contracting Officer for a determination before proceeding with the Work.

3.2 PROTECTION:

- A. General: Provide adequate coverings and accommodations as necessary to protect installed products and materials from damage by natural elements, traffic, and subsequent construction. Remove coverings when no longer needed.

END OF SECTION 01 67 00

SECTION 01 73 40 – EXECUTION

PART 1 – GENERAL

1.1 SUMMARY

- A. This section includes general procedural requirements governing execution of the Work including:
 - 1. Examination.
 - 2. Coordination with Local Utility Service Providers.
 - 3. Preparation.
 - 4. Construction Layout and Documentation.
 - 5. Field Engineering.
 - 6. Installation.
 - 7. Starting and Adjusting.
 - 8. Protection of Installed Construction.
 - 9. Correction of the Work.
 - 10. Progress Cleaning.

1.2 SUBMITTALS

- A. Job Superintendent: The Contractor shall submit the names, duties, and responsibilities of the primary and backup Job Superintendent for approval per Section 01 33 23.
- B. Field Surveying Data: For field surveys conducted to confirm compliance with the contract documents, submit field surveying data per Section 01 33 23.
- C. Field Engineering Data: For field engineering conducted to confirm compliance with the contract documents, submit field engineering data per Section 01 33 23.

1.3 QUALITY ASSURANCE

- A. Job Superintendent: The Contractor shall appoint a Job Superintendent responsible for the overall project execution. The Job Superintendent shall be always present at the jobsite during the Work. The Contractor may appoint a backup Job Superintendent to fill in for the primary Job Superintendent during emergencies or temporary offsite duties and responsibilities.
- B. Land Surveyor Qualifications: Where applicable, the Contractor shall hire a professional land surveyor legally qualified to practice in jurisdiction where project is located for all Work.
- C. Engineer Qualifications: Where applicable, the Contractor shall hire a professional engineer legally qualified to practice in jurisdiction where project is located for all Work.

PART 2 – PRODUCTS (Not Used)

PART 3 – EXECUTION

3.1 EXAMINATION:

- A. Existing Conditions: Before beginning the Work, examine and verify that the existing conditions from the contract documents are correct.
- B. Existing Utilities: Before beginning the sitework, examine and verify the existence and location of all utilities to be impacted by the Work (including water, wastewater, stormwater, fire protection, electrical power, and telecommunications).
- C. Existing Facilities: Before beginning any work to facilities, verify condition of substrates, components, and systems for dimensions, tolerances, and other conditions affecting performance.
- D. Cutting & Patching: Before performing cutting and patching operations, examine surfaces to be cut and patched and conditions under which cutting and patching are to be performed.
- E. Acceptance of Conditions: Proceed with the Work only after unsatisfactory conditions have been corrected. Proceeding with the Work indicates Contractor acceptance of existing conditions.

3.2 COORDINATION WITH LOCAL UTILITY PROVIDERS:

- A. General: If the contract documents specify work to, or affecting, a utility that is owned and/or managed by a local utility service provider, contact the local utility service provider sufficiently in advance of the Work to arrange any necessary permits, notification, or coordination.

3.3 PREPARATION:

- A. Field Measurements: Take field measurements as required to fit Work properly. Recheck measurements before installing each product. Where portions of the Work are indicated to fit to other construction, verify dimensions of other construction before fabrication. Coordinate fabrication schedule with construction progress to avoid delaying the Work.
- B. Space Requirements: Verify space requirements and dimensions of items shown on the drawings.
- C. Cutting & Patching: Protect in-place construction during cutting and patching to prevent damage. Provide protection from adverse weather conditions for portion of the project that might be exposed during cutting and patching operations.
- D. Differing Site Conditions: Refer all conflicting requirements or uncertainties from differing site conditions to the Contracting Officer for a determination before proceeding with the Work.

3.4 CONSTRUCTION LAYOUT & DOCUMENTATION:

- A. Verification: Verify layout information shown on drawings before proceeding with the Work. Refer all discrepancies to the Contracting Officer for a determination.
- B. Construction Layout: Where precision and accuracy are vital to project execution, engage a professional land surveyor licensed to practice in the project area to layout the construction. Establish benchmarks and control points in less conspicuous locations without damaging resources. The park's archaeological monitor may wish to be present for any ground disturbance.

- C. Construction Documentation: Where precision and accuracy are vital to project documentation, engage a professional land surveyor licensed to practice in the project area to complete the as-constructed surveying and mapping required in Section 01 77 00.

3.5 FIELD ENGINEERING:

- A. Verification: Verify the technical requirements shown on the drawings before proceeding with the Work. Refer all discrepancies to the Contracting Officer for a determination.
- B. Field Engineering: Where confirmation of the requirements is vital to project execution, engage a professional engineer licensed to practice in the project area to confirm the requirements.

3.6 INSTALLATION:

- A. General: The Contractor is responsible for the correct installation of the Work.
- B. Location: Locate Work and components of Work accurately in correct alignment and elevation.
 - 1. Make vertical work plumb and make horizontal work level unless otherwise specified.
 - 2. Where space is limited, install components to maximize space available for future operation, maintenance, and component replacement.
 - 3. Conceal pipes, ducts, and wiring in finished areas, unless otherwise indicated.
- C. Instructions: Comply with the manufacturer's written instructions for installing products.
- D. Scheduling: Install products at the time, and under conditions, for the best possible results. Maintain conditions required for product performance until final acceptance.
- E. Loading: Conduct construction operations so that no part of the Work is subjected to loading in excess of that expected during normal conditions of occupancy.
- F. Tool & Equipment: Do not use tools and equipment that produce harmful noise or air quality.
- G. Anchors & Fasteners: Provide anchors and fasteners as required to anchor each component securely in place.
- H. Cutting & Patching: Employ skilled workers to perform cutting and patching. Cut using methods least likely to damage elements retained or adjoining construction. Patch as required to restore surfaces to original condition. Clean areas and spaces where cutting and patching was performed and remove all excess paint, mortar, oils, putty, and similar patching materials.
- I. Joints: Make joints a uniform width. Where joint locations in exposed areas are not indicated, arrange joints for best visual effect. Fit exposed connections together to form hairline joints.
- J. Hazardous Materials: Use products, cleaners, and installation materials not considered hazardous.

3.7 STARTING & ADJUSTING:

- A. Starting: Start equipment and operating components to confirm proper operation. Remove malfunctioning units; troubleshoot if necessary; replace with new units; and retest.
- B. Adjusting: Adjust operating components and equipment for proper operation.
- C. Testing: Test each piece of equipment to verify proper operation. Test and adjust controls and safeties. Replace damaged and malfunctioning controls and equipment.
- D. Manufacturer's Field Service: If components or equipment fail to operate correctly, or if specified in the contract documents, employ a factory-authorized service representative to inspect, troubleshoot, start, adjust, test, and repair field-assembled components and equipment.

3.8 PROTECTION OF INSTALLED CONSTRUCTION:

- A. Protection: Provide final protection and maintain conditions that ensure installed Work is without damage or deterioration at the time of final acceptance.
- B. Instructions: Comply with the manufacturer's written instructions for temperature and relative humidity.

3.9 CORRECTION OF THE WORK:

- A. Defects: Repair or remove and replace the defective construction. Restore damaged substrates and finishes.
- B. Restorations: Restore permanent facilities used during construction to the specified condition.
- C. Damaged Surfaces: Remove and replace the damaged surfaces exposed to view if surfaces cannot be repaired without visible evidence of repair.
- D. Damaged Components: Repair components that do not operate properly. Remove and replace operating components that cannot be repaired.
- E. Damaged Glass: Remove and replace chipped, scratched, and broken glass or reflective surfaces.

3.10 PROGRESS CLEANING:

- A. General: Keep project site, work areas, and common areas clean and free from waste, debris, excess materials, tools and equipment, which can negatively impact project execution. Clean work areas multiple times daily. Dispose of waste properly and regularly.

END OF SECTION 01 73 40

SECTION 01 74 19 – CONSTRUCTION WASTE MANAGEMENT & DISPOSAL

PART 1 – GENERAL

1.1 SUMMARY

- A. This section includes administrative and procedural requirements for managing, salvaging, recycling, and disposing nonhazardous waste and debris from demolition and construction operations.

1.2 DEFINITIONS

- A. Demolition Waste: Building and site improvement materials resulting from demolition or selective demolition operations.
- B. Construction Waste: Building and site improvement materials, packaging, and other solid waste resulting from construction, remodeling, renovation, or repair operations.
- C. Green Waste: Vegetative and organic matter such as grass, sod, roots, branches, and trees resulting from grubbing, clearing, cutting, and trimming operations.
- D. Garbage: Refuse and food scraps resulting from preparation, cooking, dispensing, and consumption of food.
- E. Solid Waste: Garbage, sludge, or other solid discharged material (except hazardous waste) resulting from domestic, industrial, or commercial operations.
- F. Debris: Nonhazardous waste generated during demolition or construction operations. Debris can also result from building materials and natural materials such as rock, sediment, trees, etc. dispersed on a landscape by natural processes such as floods, storms, typhoons, hurricanes, etc.
- A. Hazardous Materials: Explosive, flammable, poisonous, corrosive, oxidizing, irritating, or otherwise harmful substances that could cause death or injury. Materials regulated as a hazardous material in accordance with 49 CFR 173 (Code of Federal Regulations), requires a Material Safety Data Sheet (MSDS) in accordance with 29 CFR 1910.1200, or which during end use, treatment, handling, storage, transportation or disposal meets, has components which meet, or have the potential to meet the definition of Hazardous Waste in accordance with 40 CFR 261.
- G. Salvaging: Recovery of intact demolition or construction waste for subsequent processing in preparation for reuse, either on the subject project or a future project.
- H. Recycling: Recovery of demolition or construction waste materials or components for subsequent processing in preparation for reuse, either on the subject project or a future project.
- I. Disposal: Removal of demolition and construction waste, green waste, garbage, solid waste, and debris to an off-site landfill or waste hauling facility acceptable to authorities having jurisdiction.

1.3 PERFORMANCE REQUIREMENTS

- A. General: The Contractor shall minimize the creation of waste from the project. Factors contributing to the waste stream include over packaging, improper storage, ordering errors, poor planning, breakage, mishandling, and contamination. Minimize waste disposal in landfills.
- B. Hazardous Materials: If waste materials encountered during demolition or construction are found to contain lead, asbestos, polychlorinated biphenyls (PCBs) such as fluorescent lamp ballasts, or other harmful substances, they are to be handled and removed in accordance with local, state, and federal laws and requirements concerning hazardous waste.

1.4 SUBMITTALS

- A. Construction Waste Management Plan – After contract award and before the Pre-Construction Meeting, submit a “site specific” construction waste management plan that addresses the onsite storage, handling, salvaging, recycling, and disposal of demolition waste, construction waste, green waste, garbage, solid waste, debris, and hazardous materials generated on the project, as applicable. If the Contractor proposes to salvage existing materials and reuse on the current project, the salvageable materials and the proposed treatment to make them fit for reuse must be outlined in the construction waste management plan for consideration. For disposal items, identify the landfill or waste hauling facility where the waste items will be disposed.

1.5 QUALITY ASSURANCE

- A. Onsite Waste Manager: Designate an authorized representative responsible for the following work. The authorized representative responsible for waste may also perform other duties.
 - 1. Implementing the construction waste management plan.
 - 2. Ensuring adequate onsite storage containers protected from the weather and animals.
 - 3. Ensuring personnel are keeping the jobsite clean at all times.
 - 4. Ensuring personnel are regularly disposing of waste and debris.
 - 5. Ensuring onsite waste storage containers are regularly emptied at the waste hauling facility.
 - 6. Ensuring the proper handling of items identified for salvaging and recycling.
 - 7. Ensuring personnel are familiar with the MSDS of hazardous materials.
 - 8. Ensuring the proper storage, handling, and disposal of hazardous waste.

PART 2 – PRODUCTS

2.1 ONSITE WASTE STORAGE CONTAINERS

- A. General: Provide separate containers for the storage and handling of items to be salvaged, recycled, disposed, and composted (green waste). Storage containers shall be protected from the weather and animals. Storage containers for solid waste shall not leak. Provide containers for the storage and handling of hazardous waste in accordance with the Authority Having Jurisdiction (AHJ). Storage containers shall be located near the project site to encourage use and for equipment access, but preferably hidden from view of the park visitor where possible.

2.2 HAZARDOUS MATERIALS SPILL EMERGENCY RESPONSE KIT

- B. General: Furnish and maintain a hazardous materials spill emergency response kit at the project site at all times for known or suspected hazardous materials. NPS approval of hazardous materials spill response kit is not required.

PART 3 – EXECUTION

3.1 TRAINING

- A. General: Provide adequate training to contractor and subcontractor personnel, vendors, and suppliers on proper packaging and waste management procedures.

3.2 COORDINATION

- A. General: Coordinate the salvaging, recycling, and disposal of waste with contractor and subcontractor personnel, product suppliers, and waste disposal vendors.

3.3 IMPLEMENTATION

- A. General: Implement the waste management plan as outlined.

3.4 PACKAGING

- A. General: Minimize packaging as defined in ASTM D5834.

3.5 DEMOLITION & CONSTRUCTION WASTE

- A. General: Except for items identified to be salvaged, recycled, or otherwise reused, remove all demolition and construction waste materials, garbage, and solid waste from the project site and legally dispose in landfill or waste hauling facility acceptable to the authorities having jurisdiction. Do not burn waste on government property. Do not allow waste to accumulate at the project site. Dispose of waste in a manner that prevents spillage or tracking of waste and debris on government property or outside of government property.

3.6 SALVEGABLE OR RECYCLABLE ITEMS

- A. General: For demolition or construction waste that can be reuse, either on the subject project or a future project, salvage or recycle the items as outlined in the waste management plan.

3.7 GREEN WASTE

- A. General: For green waste constituting of vegetative and organic matter generated from construction operations, without any deleterious content (causing harm or damage to the environment), the green waste may remain onsite under the following conditions:
 1. Lawn clippings and sod may be spread out in designated areas as approved by the Contracting Officer as long as there's no negative impacts to park resources.
 2. Trees, branches, limbs and stems may be spread out in designated areas as approved by the Contracting Officer as long as there's no negative impacts to park resources or wildland fire management.

3. At sites where scattering green waste is not a viable option due to the volume or density of the green waste, a chipper may be used to compact the green waste. The chips generated may then be disposed in designated areas as approved by the Contracting Officer.
4. At sites where scattering green waste or using a chipper is not a viable or acceptable option, the green waste shall be removed and properly disposed of from the project site.
5. All storage, handling, and disposal of green waste shall comply with Section 01 11 40.

3.8 HAZARDOUS MATERIALS

- A. General: Treat, handle, store, transport, and disposal hazardous materials in accordance with manufacturer's and OSHA Subpart D requirements. Display Material Safety Data Sheets (MSDS) in conspicuous locations at work site.
 1. Immediately report spills of hazardous materials to NPS Dispatch at: **808-985-6170**.
 2. Maintain a spill emergency response kit.
 3. Train employees how to respond to a spill and use emergency response kit.

END OF SECTION 01 74 19

SECTION 01 77 00 – CLOSEOUT PROCEDURES

PART 1 – GENERAL

1.1 SUMMARY

A. This section includes administrative and procedural requirements for closeout including:

1. Closeout Submittals
2. Closeout Inspections & Acceptance of Work
3. Demonstration & Training
4. Demobilization & Final Cleaning

1.2 CLOSEOUT SUBMITTALS

- A. General – Submit final closeout submittals after project substantial completion. Contract final acceptance is contingent on timely submittal and NPS acceptance of final closeout submittals.
- B. As-Built Drawings – During the execution of the project, maintain a full-sized (22" x 34") hard copy of the construction drawings and/or shop drawings at the jobsite with clearly marked changes including deletions, additions, and special instructions or notes. Mark the changes on a regular if not daily basis to ensure the as-built markups are current and thorough. Make the As-Built Drawings available to the Contracting Officer or authorized representative for review at any time during construction to confirm progress. Upon completion of all fieldwork including the punchlist, transpose the red-lined markups / notes from the hardcopy As-Built Drawings to an electronic PDF copy of the drawings. Submit the PDF copy of the As-Built Drawings to the COR for review and approval. Then, update the full-sized (22" x 34") hard copy of the As-Built Drawings, as needed, and deliver / mail to the onsite Point of Contact (POC).
- C. Warranties - Submit all manufacturer's warranties for products furnished and installed on the project. These are usually found on the product cut sheets. In addition, submit a letter with a Contractor-issued warranty on all workmanship for a period of one (1) year starting from the date of final acceptance. The Contractor-issued warranty on workmanship is covered under the contract terms; however, inclusion of this letter in the closeout documents will permit the park to track the warranty period. Submit all warranties in PDF format to the COR.
- D. Operation & Maintenance (O&M) Manual: Submit an Operation & Maintenance (O&M) manual in PDF format to the COR with the following information:
1. Preventative Maintenance Schedule – Include a preventative maintenance schedule for all recommended work on all major products, components, systems, or facilities to maximize their lifespan. The preventative maintenance schedule is typically a one-page summary or table of all preventative maintenance recommended by the vendor or contractor. The preventative maintenance schedule shall include the item, the maintenance recommendation, and the frequency (i.e. every quarter, 5 years, etc.). The preventative maintenance schedule should be located on the first page of the O&M Manual for quick reference.

2. Product Information – Include all products approved and used on the project. Please ensure only approved products are submitted. Do not include products that were not approved. Referencing CM-16's that were submitted prior to the construction start is not acceptable.
 3. Operating Instructions – Include operating instructions for all major equipment, components, systems, or facilities.
 4. Testing & Inspection Reports – Include any testing and inspection reports completed.
- E. Project Photos – Submit clear, representative photos of the project BEFORE, DURING, and AFTER construction in PDF or JPG format to the COR. Photos shall be labeled with subject, location, and date. If photos are copied to a MS Word file, photos shall not exceed four (4) photos to a page. Referencing photos attached to daily reports is not acceptable.
- F. Project Closeout Binders – Submit two (2) D-ring binders containing hard copies of the final approved closeout submittals from Section 01 77 00, PART 1.2 (B thru E above). These binders shall include: As-Built Drawings (11x17), Warranties, Preventative Maintenance Schedule, Product Information, Operating Instructions, Testing & Inspection Reports, and Project Photos. Label the outside of each project closeout binder with the project title, contract number, contractor name, and completion month and year. Upon approval of the closeout submittals, the Project Closeout Binders shall be delivered / mailed to the onsite POC.
- G. Accessibility Inspection Report: If the project requires compliance with ABAAS for accessibility under Section 01 11 00, PART 1.12.B, fill out applicable sections of the Accessibility Inspection Form to document compliance with the Architectural Barriers Act Accessibility Standards (ABBAS). Inspect the work at various stages of construction to ensure the finished product meets the contract requirements. Failure to comply with ABAAS may result in rework. Complete the Accessibility Inspection Form in PDF format and submit to the COR for review and approval.

1.3 CLOSEOUT INSPECTIONS & ACCEPTANCE

- A. Pre-Final Inspection & Substantial Completion:
1. When the project, or a designated portion of the project, is substantially complete, the Contractor shall request a pre-final inspection in writing (via email) at least seven (7) calendar days in advance of the inspection date. The Work will be considered substantially complete when all major work elements are complete, and all components are operable.
 2. During the pre-final inspection, the Contractor and the NPS will develop a "punchlist" of remaining work or deficiencies to satisfy the contract requirements. The punchlist will be reviewed onsite with the Contractor to ensure all parties have a clear understanding of the remaining work. The basis for what may be included in a punchlist includes:
 - a. Items required as outlined in the construction documents (i.e. plans and specifications).
 - b. Items required as outlined from the responses to RFI questions.
 - c. Mitigation of a potential hazard generated from the work performed.
 - d. Correction of a workmanship issue (per industry standards, manufacturer's recommendations, or as outlined in Section 01 73 40 Execution).
 - e. Items associated with final cleanup.
 3. If the NPS confirms that the work is substantially complete at the time of the pre-final inspection, the Contracting Officer will issue a Letter of Substantial Completion with the punchlist of remaining work to be completed or corrected prior to final acceptance. The

original contract period of performance will end, and the Contractor will have thirty (30) calendar days to complete the punchlist, as weather permits.

4. If the NPS determines that the work is NOT substantially complete at the time of the pre-final inspection, the Contracting Officer will notify the Contractor in writing. The Contractor may request a new pre-final inspection; however, all re-inspection costs may be charged against the Contractor in accordance with the Inspection of Construction clause of the contract.
5. If all or a portion of the Work is determined to be substantially complete, and the Government wishes to occupy or begin using the area, the Contracting Officer may issue a Letter of Beneficial Use / Occupancy to the Contractor outlining the terms of occupancy.
6. If there are no field punchlist items identified during the pre-final inspection, and only administrative closeout submittals remain, the Contracting Officer may issue a Letter of Beneficial Use / Occupancy to the Contractor.

B. Final Inspection & Final Acceptance:

1. When all field punchlist items are completed, the Contractor shall request a final inspection in writing at least three (3) calendar days in advance of the inspection date. During the final inspection, the NPS will confirm punchlist completion. After all field punchlist items are completed, and all closeout submittals have been received and accepted, the Contracting Officer will issue a Letter of Final Acceptance.
2. If the Contractor fails to complete the field punchlist items and/or submit the closeout submittals in the allotted thirty (30) day timeframe, the Contracting Officer may either replace the work with an appropriate reduction in contract price or negotiate other considerations. In addition, re-inspection costs may be charged against the Contractor in accordance with the Inspection of Construction clause of the contract.
3. If there are no field punchlist items, a final inspection is not required.
4. Once the Letter of Final Acceptance is issued and the Contracting Officer receives the Release of Claims, the Contractor may submit final invoice.

PART 2 – PRODUCTS (Not Used)

PART 3 – EXECUTION

3.1 LABEL SWITCHES & POST OPERATING INSTRUCTIONS

- A. General: After completion of the work, label all switches and controls and post operating instructions, as applicable. All labels and posts shall be printed, clear, secure, and protected.

3.2 SPECIALTY TOOLS, SPARE PARTS, EXCESS MATERIALS

- A. General: After completion of the work, deliver specialty tools required to operate new equipment, components, systems, or facilities to the onsite POC. If the Contractor wishes to transfer spare parts or excess materials to the park, and the park is amenable, the Contractor shall document the items to transferred.

3.3 DEMONSTRATION & TRAINING

- A. General: Provide onsite demonstration and training to NPS operators and personnel on the proper and safe operation of all new systems and components. This demonstration & training can occur at the time of the pre-final or final inspections, or it can be arranged as a separate event to accommodate either park personnel or the training vendors, if applicable. However, in either case

the Contractor shall notify the NPS at least seven (7) days in advance of the proposed demonstration & training so the park can arrange for key or interested staff to attend.

3.4 REMOVAL OF TEMPORARY FACILITIES & CONTROLS

- A. General: After completion of the work, remove all temporary facilities and controls such as portable toilets, surface protections, fencing, traffic and erosion / sediment control devices.

3.5 DEMOBILIATION

- A. General: After completion of the work, remove all equipment and machinery from the site.

3.6 FINAL CLEANING

- A. General: After completion of the work, clean all surfaces and remove all waste materials. Properly dispose of all waste and debris offsite. Dispose of all food refuse in proper containers so as not to attract wildlife, vermin, or pests.

END OF SECTION 01 77 00

DIVISION 02-48

TECHNICAL REQUIREMENTS

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SECTION 03 30 00 - CAST-IN-PLACE CONCRETE

PART 1 - GENERAL

1.1 DESCRIPTION

- A. This Section specifies cast-in place concrete, including formwork, reinforcement, concrete materials, mixture design, placement procedures, and finishes; includes concrete accessories including the following:

- 1. Slab on grade concrete pads for mechanical equipment.

1.2 COORDINATION

- A. Coordinate embedded anchors.

1.3 DEFINITIONS

- A. Cementitious Materials: Portland cement alone or in combination with one or more of the following: blended hydraulic cement, fly ash and other pozzolans, ground granulated blast-furnace slag, and silica fume; subject to compliance with requirements.
- B. Form-Facing Material: Temporary structure or mold for the support of concrete while the concrete is setting and gaining sufficient strength to be self-supporting.
- C. Formwork: The total system of support of freshly placed concrete, including the mold or sheathing that contacts the concrete, as well as supporting members, hardware, and necessary bracing.
- D. Water/Cement Ratio (w/cm): The ratio by weight of water to cementitious materials.

1.4 ACTION SUBMITTALS

- A. Product Data: For each type of product indicated and required for installation. Include the following:
 - 1. Admixtures: Include limitations of use, including restrictions on cementitious materials, supplementary cementitious materials, air entrainment, aggregates, temperature at time of concrete placement, relative humidity at time of concrete placement, curing conditions, and use of other admixtures.
- B. Design Mixtures: For each concrete mixture. Submit alternate design mixtures when characteristics of materials, Project conditions, weather, test results, or other circumstances warrant adjustments.
 - 1. Indicate amounts of mixing water to be withheld for later addition at Project site.

2. Include substantiating substantial test data to show compliance with ACI 309.

1.5 INFORMATIONAL SUBMITTALS

A. Qualification Data:

1. Qualification Data: For firms and persons specified in “Quality Assurance” Article to demonstrate their capabilities and experience. Include lists of completed projects with project names and addresses, names and addresses of Architects and Owners, and other information specified.
2. Installer: Include copies of applicable ACI certificates.
3. Ready-mix concrete manufacturer.
4. Testing agency: Include copies of applicable ACI certificates.

B. Material Certificates: For each of the following, signed by manufacturers:

1. Cementitious materials.
2. Admixtures.
3. Form materials and form-release agents.
4. Steel reinforcement and accessories.
5. Curing compounds.

1.6 QUALITY ASSURANCE

A. ACI Publications: Comply with ACI 301 unless modified by requirements in the Contract Documents.

B. Installer Qualifications:

1. Concrete: A qualified installer who employs on Project personnel qualified as ACI-certified Flatwork Technician and Finisher and supervisor who is an ACI-certified Concrete Flatwork Technician.
2. Installer shall be experienced cast-in-place concrete installer, with not less than five consecutive years' experience, specializing in installing cast-in-place concrete similar in material, design, and extent to that indicated for this Project and whose work has resulted in construction with a record of successful in-service performance.

C. Manufacturer Qualifications: A firm experienced in manufacturing ready-mixed concrete products and that complies with ASTM C94/C94M requirements for production facilities and equipment.

1. Manufacturer certified according to NRMCA’s “Certification of Ready Mixed Concrete Production Facilities.

D. Concrete Testing Service: Engage a qualified independent testing agency to perform material evaluation tests and to design concrete mixtures.

E. ACI Publications: Comply with the following unless modified by requirements in the Contract Documents:

1. ACI 301, "Specification for Structural "Concrete", Sections 1 through 5.
2. ACI 117, "Specifications for Tolerances for Concrete Construction and Materials."

F. Preinstallation Conference: Conduct conference at Project site.

1. Before submitting design mixtures, review concrete design mixture and examine procedures for ensuring quality of concrete materials. Require representatives of each entity directly concerned with cast-in-place concrete to attend, including the following:
 - a. Contractor.
 - b. Entity responsible for concrete design mixes.
 - c. Ready mix concrete manufacturer or representative.
 - d. Concrete subcontractor.
2. Review form sequencing, placement and removal.

1.7 PRECONSTRUCTION TESTING

A. Preconstruction Testing Service: Engage a qualified testing agency to perform preconstruction testing on each concrete mixture.

1. Include the following information in each test report:
 - a. Admixture dosage rates.
 - b. Slump.
 - c. Air content.
 - d. Seven-day compressive strength.
 - e. 28-day compressive strength.
 - f. Permeability.

1.8 DELIVERY, STORAGE, AND HANDLING

- A. Deliver, store, and handle steel reinforcement to prevent bending and damage.
- B. Comply with ASTM C94 and ACI 301.

PART 2 - PRODUCTS

2.1 PERFORMANCE REQUIREMENTS

- A. Concrete Formwork: Design, engineer, erect, shore, brace, and maintain formwork, shores, and reshores in accordance with ACI 301, to support vertical, lateral, static, and dynamic loads, and construction loads that might be applied, until structure can support such loads, so that resulting concrete conforms to the required shapes, lines, and dimensions.
 1. Design wood panel forms in accordance with APA's "Concrete Forming Design/Construction Guide."

2.2 FORM-FACING MATERIALS

- A. Rough-Formed Finished Concrete: Plywood, lumber, metal, or another approved material. Provide lumber dressed on at least two edges and one side for tight fit.
- B. Form-Release Agent: Commercially formulated form-release agent that will not bond with, stain, or adversely affect concrete surfaces and will not impair subsequent treatments of concrete surfaces.
 - 1. Formulate form-release agent with rust inhibitor for steel form-facing materials.

2.3 STEEL REINFORCEMENT

- A. Reinforcing Bars: ASTM A615/A615M, Grade 60, deformed.
- B. Low-Alloy-Steel Reinforcing Bars: ASTM A706/ A706M, deformed.
 - 1. Provide for all reinforcing steel to be welded or bent in field.

2.4 CONCRETE MATERIALS

- A. Source Limitations:
 - 1. Obtain all concrete mixtures from a single ready-mixed concrete manufacturer for entire Project.
 - 2. Obtain each type or class of cementitious material of the same brand from the same manufacturer's plant.
 - 3. Obtain aggregate from single source.
 - 4. Obtain each type of admixture from single source from single manufacturer.
- B. Cementitious Material: Use the following cementitious materials, of the same type, brand, and source, throughout Project:
 - 1. Portland Cement: ASTM C150, Type I/II
 - a. Fly Ash: ASTM C618, Class C or F. Refer to General Structural Notes for additional information.
 - b. Ground Granulated Blast-Furnace Slag: ASTM C989, Grade 100 or 120.
- C. Silica Fume: ASTM C1240, amorphous silica.
- D. Normal-Weight Aggregate: ASTM C33, coarse aggregate or better, graded. Provide aggregates from a single source.
 - 1. Maximum Coarse-Aggregate Size: 3/4 inch.
 - 2. Combined Aggregate Gradation: Well graded from coarsest to finest with not more than 18 percent and not less than 8 percent retained on an individual sieve, except that less than 8 percent may be retained on coarsest sieve and on No. 50 sieve, and less than 8 percent may be retained on sieves finer than No. 50.
 - 3. Fine Aggregate: Free of materials with deleterious reactivity to alkali in cement.

- E. Water: ASTM C94/C94M and potable.

2.5 CURING MATERIALS

- A. General: Curing materials used shall be guaranteed by manufacturer, prior to application and use, to not affect the bond or performance of applied finishes. Named products may be used where manufacturer's documentation is provided during Submittals.
- B. Absorptive Cover: AASHTO M 182, Class 2, burlap cloth made from jute or kenaf, weighing approximately 9 oz./sq. yd. when dry.
- C. Moisture-Retaining Cover: ASTM C171, polyethylene film or white burlap-polyethylene sheet.
 - 1. Color:
 - a. Ambient Temperature Below 50 deg F: Black.
 - b. Ambient Temperature between 50 deg F and 85 deg F: Any color.
 - c. Ambient Temperature Above 85 deg F: White.
- D. Water: Potable.
- E. Clear, Waterborne, Membrane-Forming, Dissipating Curing Compound: ASTM C309, Type 1, Class B.

2.6 REPAIR MATERIALS

- A. Products to be used in locations permanently exposed to the weather shall be approved by the manufacturer for use in exterior environments.

2.7 LIQUID FLOOR TREATMENTS, FOR SEALED CONCRETE

- A. Penetrating Liquid Floor Treatment: Clear, chemically reactive, waterborne solution of inorganic silicate or silicate materials and proprietary components; odorless; that penetrates, hardens, and densifies concrete surfaces.
 - 1. Basis-of-Design Product: Euco Dimond Hard by Euclid Chemical or approved equal.
 - 2. Application: At exposed concrete floors scheduled for sealed concrete finish.

2.8 CONCRETE MIXTURES, GENERAL

- A. Prepare design mixtures for each type and strength of concrete, proportioned on the basis of laboratory trial mixture or field test data, or both, according to ACI 301 and in accordance with the following:
 - 1. Compressive Strength: minimum 4,000 psi.

2. Air Content:

- a. Structures: Add air-entraining admixture at manufacturer's prescribed rate to result in concrete at point of placement having the air content shown.
- B. Cementitious Materials: Limit percentage, by weight, of cementitious materials other than portland cement in concrete as shown.
- C. Limit water-soluble, chloride-ion content in hardened concrete to 0.15 percent by weight of cement.
- D. Admixtures, General: Use admixtures according to manufacturer's written instructions. Do not add multiple admixtures simultaneously.
 - 1. Use water-reducing, high-range water-reducing or plasticizing admixture or high-range water-reducing admixture (superplasticizer) in concrete, as required, for placement and workability.
 - 2. Use water-reducing and retarding admixture when required by high temperatures, low humidity, or other adverse placement conditions.
 - 3. Use water-reducing admixture in pumped concrete, concrete for slabs, concrete required to be watertight, and concrete with a water-cementitious materials ratio below 0.50.
 - 4. At board-formed concrete, use water-reducing and plasticizing admixtures.

3.9 CONCRETE MIXING

- E. Ready-Mixed Concrete: Measure, batch, mix, and deliver concrete according to ASTM C94/C94M and ASTM C1116. Furnish batch ticket information.
 - 1. When air temperature is between 85 and 90 deg F, reduce mixing and delivery time from 1-1/2 hours to 75 minutes; when air temperature is above 90 deg F, reduce mixing and delivery time to 60 minutes.

PART 3 - EXECUTION

3.1 FORMWORK

Design, erect, shore, brace, and maintain formwork, according to ACI 301, to support lateral, static, and dynamic loads, and construction loads that might be applied, until concrete structure can support such loads.

- A. Limit concrete surface irregularities, designated by ACI 347R as abrupt or gradual, as follows:
 - 1. Surface Finish-2.0: ACI 117 Class B, 1/4 inch for rough-formed finished surfaces and board-formed concrete.
- B. Construct forms tight enough to prevent loss of concrete mortar.

- C. Fabricate forms for easy removal without hammering or prying against concrete surfaces. Provide crush or wrecking plates where stripping may damage cast concrete surfaces. Provide top forms for inclined surfaces steeper than 1.5 horizontal to 1 vertical.
 - 1. Install keyways, reglets, recesses, and the like, for easy removal.
 - 2. Do not use rust-stained steel form-facing material.
- D. Set edge forms, bulkheads, and intermediate screed strips for slabs to achieve required elevations and slopes in finished concrete surfaces. Provide and secure units to support screed strips; use strike-off templates or compacting-type screeds.
- E. Clean forms and adjacent surfaces to receive concrete. Remove chips, wood, sawdust, dirt, and other debris just before placing concrete
- F. Retighten forms and bracing before placing concrete, as required, to prevent mortar leaks and maintain proper alignment.
- G. Coat contact surfaces of forms with form-release agent, according to manufacturer's written instructions, before placing reinforcement.

3.2 FORMS

- A. General: Formwork that does not support weight of concrete may be removed after cumulatively curing at not less than 50 deg F for 24 hours after placing concrete if concrete is hard enough to not be damaged by form-removal operations and provided curing and protection operations are maintained.
- B. Clean and repair surfaces of forms to be reused in the Work. Split, frayed, delaminated, or otherwise damaged form-facing material will not be acceptable for exposed surfaces. Apply new form-release agent.
- C. When forms are reused, clean surfaces, remove fins and laitance, and tighten to close joints. Align and secure joints to avoid offsets. Do not use patched forms for exposed concrete surfaces unless approved by the CO.

3.3 STEEL REINFORCEMENT

- A. General: Comply with CRSI's "Manual of Standard Practice" for placing reinforcement.
- B. Clean reinforcement of loose rust and mill scale, earth, ice, and other foreign materials that would reduce bond to concrete.
- C. Accurately position, support, and secure reinforcement against displacement. Locate and support reinforcement with bar supports to maintain minimum concrete cover..
 - 1. Weld reinforcing bars according to AWS D1.4, where indicated.
- D. Set wire ties with ends directed into concrete, not toward exposed concrete surfaces.

3.4 JOINTS

- A. General: Construct joints true to line with faces perpendicular to surface plane of concrete.
- B. Construction Joints: Install so strength and appearance of concrete are not impaired, at locations indicated or as approved by the CO.
 - 1. Place joints perpendicular to main reinforcement. Continue reinforcement across construction joints, unless otherwise indicated.
 - 2. Form keyed joints as indicated. Embed keys at least 1-1/2 inches into concrete.
 - 3. Locate joints complying with requirements.
 - 4. Locate horizontal joints where concealed from view.
 - 5. Space vertical joints where shown. Locate joints, near corners, and in concealed locations where possible.
 - 6. Use a bonding agent or roughen interface to 1/4-inch amplitude at locations where fresh concrete is placed against hardened or partially hardened concrete surfaces.

3.5 CONCRETE PLACEMENT

- A. Before placing concrete, verify the following:
 - 1. That installation of formwork, reinforcement, and embedded items is complete and that required inspections have been performed.
 - 2. That all standing water and pooling water on vapor retarders have been removed.
 - 3. That all vapor retarder terminations and penetrations are complete.
 - 4. That all piping penetrations have expansion material placed around items penetrating concrete, and as required by Plumbing code.

3.6

- A. Do not add water to concrete during delivery, at Project site, or during placement, unless approved by the CO.
 - 1. Do not add water to concrete after adding high-range water-reducing admixtures to mix.
- B. Adjust mix as required to maintain specified air content at the point of discharge.
- C. Deposit concrete continuously in one layer or in horizontal layers of such thickness that no new concrete will be placed on concrete that has hardened enough to cause seams or planes of weakness. If a section cannot be placed continuously, provide construction joints as indicated. Deposit concrete to avoid segregation.
 - 1. Deposit concrete in horizontal layers of depth to not exceed formwork design pressures and in a manner to avoid inclined construction joints.
 - 2. Consolidate placed concrete with mechanical vibrating equipment according to ACI 301.

3. Do not use vibrators to transport concrete inside forms. Insert and withdraw vibrators vertically at uniformly spaced locations to rapidly penetrate placed layer and at least 6 inches into preceding layer. Do not insert vibrators into lower layers of concrete that have begun to lose plasticity. At each insertion, limit duration of vibration to time necessary to consolidate concrete and complete embedment of reinforcement and other embedded items without causing mixture constituents to segregate.
- D. Deposit and consolidate concrete for slabs in a continuous operation, within limits of construction joints, until placement of a panel or section is complete.
1. Consolidate concrete during placement operations so concrete is thoroughly worked around reinforcement and other embedded items and into corners.
 2. Maintain reinforcement in position on chairs during concrete placement.
 3. Screed slab surfaces with a straightedge and strike off to correct elevations.
 4. Slope surfaces uniformly to drains where required.
 5. Begin initial floating using bull floats or darbies to form a uniform and open-textured surface plane, before excess bleedwater appears on the surface. Do not further disturb slab surfaces before starting finishing operations.
- E. Cold-Weather Placement:
1. When average high and low temperature is expected to fall below 40 deg F for three successive days, maintain delivered concrete mixture temperature within the temperature range required by ACI 301.
 2. Do not use frozen materials or materials containing ice or snow. Do not place concrete on frozen subgrade or on subgrade containing frozen materials.
 3. Do not use calcium chloride, salt, or other materials containing antifreeze agents or chemical accelerators, unless otherwise specified and approved in mixture designs.
- F. Hot-Weather Placement: Comply with ACI 301 and as follows:
1. Maintain concrete temperature below 90 deg F at time of placement. Chilled mixing water or chopped ice may be used to control temperature, provided water equivalent of ice is calculated to total amount of mixing water. Using liquid nitrogen to cool concrete is Contractor's option.
 2. Fog-spray forms, steel reinforcement, and subgrade just before placing concrete. Keep subgrade moisture uniform without standing water, soft spots, or dry areas.

3.7 FINISHING FLOORS AND SLABS

- A. General: Comply with ACI 302.1R recommendations for screeding, restrecting, and finishing operations for concrete surfaces. Do not wet concrete surfaces.
- B. Scratch Finish: While still plastic, texture concrete surface that has been screeded and bull-floated or darbied. Use stiff brushes, brooms, or rakes to produce a profile amplitude of ¼ inch (6mm) in one direction.
1. Apply scratch finish to surfaces indicated and to surfaces to receive concrete floor topping or mortar setting beds for bonded cementitious floor finishes.

- C. Float Finish: Consolidate surface by hand floating
- D. Repeat float passes and restraightening until surface is left with a uniform, smooth, granular texture.
 - 1. Apply float finish to surfaces.

3.8 MISCELLANEOUS CONCRETE ITEMS

- A. Filling In: Fill in holes and openings left in concrete structures, unless otherwise indicated, after work of other trades is in place. Mix, place, and cure concrete, as specified, to blend with in-place construction. Provide other miscellaneous concrete filling indicated or required to complete the Work.
- B. Equipment Bases: Provide 4-inch-thick concrete pads under mechanical equipment as required or thickness indicated. Reinforce as indicated. Refer to Mechanical drawings for locations. Pad size to extend 6-inches beyond edge of equipment on all sides unless otherwise indicated. Set anchor bolts for machines and equipment at correct elevations, complying with diagrams or templates of manufacturer furnishing machines and equipment.

3.9 CONCRETE PROTECTION AND CURING

- A. General: Protect freshly placed concrete from premature drying and excessive cold or hot temperatures. Comply with ACI 306.1 for cold-weather protection and with recommendations in ACI 305R for hot-weather protection during curing.
- B. Cure concrete according to ACI 308.1 by one or a combination of the following methods:
 - 1. Moisture Curing: Keep surfaces continuously moist for not less than seven days with the following materials:
 - a. Water.
 - b. Continuous water-fog spray.
 - c. Absorptive cover, water saturated, and kept continuously wet. Cover concrete surfaces and edges with 12-inch lap over adjacent absorptive covers.
 - 2. Moisture-Retaining-Cover Curing: Cover concrete surfaces with moisture-retaining cover for curing concrete, placed in widest practicable width, with sides and ends lapped at least 12 inches, and sealed by waterproof tape or adhesive. Cure for not less than seven days. Immediately repair any holes or tears during curing period using cover material and waterproof tape.
 - a. Moisture cure or use moisture-retaining covers to cure concrete surfaces to receive floor coverings.
 - b. Moisture cure or use moisture-retaining covers to cure concrete surfaces to receive penetrating liquid floor treatments.
 - c. Cure concrete surfaces to receive floor coverings with either a moisture-retaining cover or a curing compound that the manufacturer certifies will not interfere with bonding of a floor covering used on Project.

3. Curing Compound: Apply uniformly in continuous operation by power spray or roller according to manufacturer's written instructions. Recoat areas subjected to heavy rainfall within three hours after initial application. Maintain continuity of coating and repair damage during curing period.
 - a. After curing period has elapsed, remove curing compound without damaging concrete surfaces by method recommended by curing compound manufacturer unless manufacturer certifies curing compound will not interfere with bonding of floor covering used on Project.

3.10 CONCRETE SURFACE REPAIRS

- A. General: At exposed areas, do not proceed with required repairs without written permission to proceed from the CO.
 1. Repair mortar color shall be provided in mockup for review at areas scheduled to be concealed.
 2. Installer shall mockup proposed mix used for color matching at exposed areas, in an area sufficient for review and comparison and with similar conditions required for patching.
- B. Defective Concrete: Repair and patch defective areas when approved by the CO. Remove and replace concrete that cannot be repaired and patched to CO approval.
- C. Patching Mortar: Mix dry-pack patching mortar, consisting of one part Portland cement to two and one-half parts fine aggregate passing a No. 16 sieve, using only enough water for handling and placing.
- D. Perform structural repairs of concrete, subject to the CO's approval, using epoxy adhesive and patching mortar.
- E. Repair materials and installation not specified above may be used, subject to the CO's approval.

3.11 FIELD QUALITY CONTROL

- A. Testing and Inspecting: Engage a special inspector and qualified testing and inspecting agency, to perform field tests and inspections and prepare test reports.
- B. Concrete Tests: As indicated in the Statement of Special Inspections and the following;
 1. Samples for preparing strength test specimens of each concrete mixture placed each day shall be taken in accordance with (a) through (c):
 - a. At least once a day.
 - b. At least once for each 150 cubic yards of concrete.
 - c. At least once for each 5000 square feet of surface area for slabs or walls.
 2. When strength of field-cured cylinders is less than 85 percent of companion laboratory-cured cylinders, Contractor shall evaluate operations and provide corrective procedures for protecting and curing in-place concrete.

3. Strength of each concrete mixture will be satisfactory if every average of any three consecutive compressive-strength tests equals or exceeds specified compressive strength and no compressive-strength test value falls below specified compressive strength by more than 500 psi.
4. Test results shall be reported in writing to the CO, concrete manufacturer, Building Official, and Contractor within 48 hours of testing. Reports of compressive-strength tests shall contain Project identification name and number, date of concrete placement, name of concrete testing and inspecting agency, location of concrete batch in Work, design compressive strength at 28 days, concrete mixture proportions and materials, compressive breaking strength, type of break for both 7- and 28-day tests, and air content.
5. Nondestructive Testing: Impact hammer, sonoscope, or other nondestructive device may be permitted by the CO but will not be used as sole basis for approval or rejection of concrete.
6. Additional Tests: Testing and inspecting agency shall make additional tests of concrete when test results indicate that slump, air entrainment, compressive strengths, or other requirements have not been met, as directed by the CO. Testing and inspecting agency may conduct tests to determine adequacy of concrete by cored cylinders complying with ASTM C42/C42M or by other methods as directed by the CO.
7. Additional testing and inspecting, at Contractor's expense, will be performed to determine compliance of replaced or additional work with specified requirements.
8. Correct deficiencies in the Work that test reports and inspections indicate do not comply with the Contract Documents.

3.12 PROTECTION

A. Protect concrete surfaces as follows:

1. Protect from petroleum stains.
2. Diaper hydraulic equipment used over concrete surfaces.
3. Prohibit placement of steel items on concrete surfaces.
4. Prohibit use of acids or acidic detergents over concrete surfaces.
5. Protect liquid floor treatment from damage and wear during the remainder of construction period. Use protective methods and materials, including temporary covering, recommended in writing by liquid floor treatments installer.
6. Protect concrete surfaces and saw cut and exposed joint edges during construction.
7. If any penetrations are to be cored through concrete walls after curing, walls shall be scanned for rebar and penetrations shall be positioned to avoid cutting rebar.

END OF SECTION 03 30 00

SECTION 09 91 13 - PAINTING

PART 1 - GENERAL

1.1 RELATED DOCUMENTS

- A. Drawings and general provisions of the Contract including General and Supplementary Conditions and Division 01 Specification Section, apply to this Section.

1.2 DELIVERY, STORAGE, AND HANDLING

- A. Store materials not in use in tightly covered containers in well-ventilated areas with ambient temperatures continuously maintained at not less than 45 deg F.
 - 1. Maintain containers in clean condition, free of foreign materials and residue.
 - 2. Remove rags and waste from storage areas daily.

1.3 FIELD CONDITIONS

- A. Apply paints only when temperature of surfaces to be painted and ambient air temperatures are between 50 and 95 deg F.
- B. Do not apply paints in rain, fog, or mist; when relative humidity exceeds 85 percent; at temperatures less than 5 deg F above the dew point; or to damp or wet surfaces.

PART 2 - PRODUCTS

2.1 PAINTING SCHEDULE

- A. Interior Gypsum Board Ceilings:
 - 1. Primer MPI 172
 - 2. Intermediate Coat: Matching Top coat.
 - 3. Top Coat: MPI 53, Latex Exterior, Flat, MPI Gloss level 1

PART 3 - EXECUTION

3.1 EXAMINATION

- A. Examine substrates and conditions, with Applicator present, for compliance with requirements for maximum moisture content and other conditions affecting performance of the Work.

- B. Verify suitability of substrates, including surface conditions and compatibility, with finishes and primers.
- C. Proceed with coating application only after unsatisfactory conditions have been corrected.
 - 1. Application of coating indicates acceptance of surfaces and conditions.

3.2 PREPARATION

- A. Comply with manufacturer's written instructions applicable to substrates and paint systems indicated.
- B. Remove hardware, covers, plates, and similar items already in place that are removable and are not to be painted. If removal is impractical or impossible because of size or weight of item, provide surface-applied protection before surface preparation and painting.
 - 1. After completing painting operations, use workers skilled in the trades involved to reinstall items that were removed. Remove surface-applied protection.
- C. Clean substrates of substances that could impair bond of paints, including dust, dirt, oil, grease, and incompatible paints and
 - 1. Remove incompatible primers and re-prime substrate with compatible primers or apply tie coat as required to produce paint systems specified in this Section.

3.3 INSTALLATION

- A. Apply paints in accordance with manufacturer's written instructions.
 - 1. Use applicators and techniques suited for paint and substrate indicated.
 - 2. Do not paint over labels of independent testing agencies or equipment name, identification, performance rating, or nomenclature plates.
 - 3. Primers specified in the Exterior Painting Schedule may be omitted on items that are factory primed or factory finished if compatible with intermediate and topcoat coatings and acceptable to intermediate and topcoat paint manufacturers.
- B. Tint undercoats same color as topcoat, but tint each undercoat a lighter shade to facilitate identification of each coat if multiple coats of same material are to be applied. Provide sufficient difference in shade of undercoats to distinguish each separate coat.
- C. If undercoats or other conditions show through topcoat, apply additional coats until cured film has a uniform paint finish, color, and appearance.
- D. Apply paints to produce surface films without cloudiness, spotting, holidays, laps, brush marks, roller tracking, runs, sags, ropiness, or other surface imperfections. Cut in sharp lines and color breaks.

3.4 CLEANING AND PROTECTION

- A. At end of each workday, remove rubbish, empty cans, rags, and other discarded materials from Project site.
 - 1. Do not clean equipment with free-draining water and prevent solvents, thinners, cleaners, and other contaminants from entering into waterways, sanitary and storm drain systems, and ground.
 - 2. Dispose of contaminants in accordance with requirements of authorities having jurisdiction.
 - 3. Allow empty paint cans to dry before disposal.
 - 4. Collect waste paint by type and deliver to recycling or collection facility.
- B. Clean spattered surfaces. Remove spattered paints by washing, scraping, or other methods. Do not scratch or damage adjacent finished surfaces.
- C. Protect work of other trades against damage from paint application. Correct damage to work of other trades by cleaning, repairing, replacing, and refinishing, as approved by Architect, and leave in an undamaged condition.
- D. At completion of construction activities of other trades, touch up and restore damaged or defaced painted surfaces.

END OF SECTION 09 91 13

SECTION 230000 – MECHANICAL GENERAL REQUIREMENTS

PART 1 - GENERAL

1.1 SUMMARY

- A. This section applies to all sections of Division 23 – HEATING, VENTILATING, AND AIR CONDITIONING.
- B. Scope of work shall be as described in the Technical Sections of Division 23 – HEATING, VENTILATING, AND AIR CONDITIONING.
- C. Related Sections:

DIVISION 1 - GENERAL REQUIREMENTS
SECTION 099000 – PAINTING AND COATING
SECTION 23 81 00 – UNITARY AIR CONDITIONING EQUIPMENT
DIVISION 26 – ELECTRICAL

1.2 GENERAL REQUIREMENT

- A. The mechanical general requirements govern work specified under all sections of DIVISION 23 – HEATING, VENTILATING, AND AIR CONDITIONING.
- B. The Contractor shall furnish all labor, materials, tools, and equipment and perform all work and services necessary for a complete and properly operational mechanical work, equipment and systems, as shown on drawings and as specified, in accordance with provisions of the Contract Documents and completely coordinated with work of all other trades.
- C. The Contractor shall examine completely the Contract Documents and shall report to Owner's Project Manager any error, inconsistency, or omission he/she discovers.
- D. Furnish and install all supplementary of miscellaneous items, details, appurtenances and devices incidental to or necessary for a sound, secure, and complete mechanical system where work required is not specifically indicated.
- E. Drawings and specifications shall be taken together. Provide work specified and not indicated or work indicated and not specified as though mentioned in both.
- F. The Contractor shall warrant that all materials and equipment furnished under this Contract will be new and that all work will be good quality, free from faults and defects and in conformance with contract documents for a guaranteed period of one year. It shall be the Contractor's responsibility to obtain extended warranties for use of all new equipment provided by the Contractor prior to project acceptance at no additional cost to Owner.
- G. The Contractor shall maintain at the site one copy of all Drawings, Specifications, Addenda, acceptable Shop Drawings, Change Orders, and other modifications, in good order and marked to record all changes made during construction. These shall be made available to Owner's Project Manager.

- H. The Contractor shall keep the premises free from accumulation of waste materials or rubbish caused by his operations at all times. At the completion of the work, he shall remove all his waste materials and rubbish from and about the project as well as his tools, construction equipment, machinery, and surplus materials and shall clean all new equipment and accessories.
- I. The Contractor shall give Owner's Project Manager timely notice of its readiness for testing any work including the data arranged so Owner's Project Manager may observe such testing. The Contractor shall bear all cost of such tests.
- J. Owner's Project Manager shall have the right to accept or reject material, equipment, and/or workmanship and determine when the Contractor has complied with the contract documents.

1.3 INSPECTION OF SITE

- A. The Contractor shall visit the site and examine the conditions affecting his work before submitting his proposal. The submission of the proposal shall be considered evidence that the Contractor has visited the site and no extra payments will be allowed to the Contractor on account of extra work made necessary by his failure to visit the site.

1.4 SUBMITTALS

- A. Submittals required in this and other sections of Division 23, HVAC, shall conform to the General Provision and Special Provisions and the following additional requirements. Submittals shall include the manufacturer's name, trade name, place of manufacture, catalog model or number, nameplate data, size, layout dimension, capacity, project specification and paragraph reference, applicable industry, and technical society reference standards, years of satisfactory service, and other information necessary to establish contract compliance of each item the Contractor proposes to provide. Photographs of existing installations are unacceptable and will be returned without approval. Submittals for each section of Division 23, HVAC, shall be complete. Incomplete submittals will be returned without review.
- B. Manufacturer's Catalog Data: Submittals for each manufactured item shall be current manufacturer's descriptive literature of cataloged products, equipment drawings, diagrams, performance and characteristic curves, and catalog cuts. Each submittal shall clearly identify equipment to be provided. Information not pertaining to equipment shall be deleted or crossed out.
- C. Substitution Requests: No substitution is allowed during bidding. Substitutions are processed after award of the contract. Substitute materials or equipment may be used if prequalified by written permission from the Owner's Construction Manager. Submit a list of the substitutions with qualifying data for approval after the bidding.
- D. Shop Drawings:
 - 1. Provide shop drawings for each Section of Division 23, HVAC. Shop Drawings shall use a minimum scale of 1/4 inch per foot on drawing sheets the same size as contract drawings. Include floor plans, sectional views, wiring diagrams, and installation details of submitted equipment, and equipment spaces identifying and indicating proposed location, layout and arrangement of items of equipment, control panels, accessories, piping, ductwork, and other items that must be shown to ensure a coordinated installation. Wiring diagrams shall identify circuit terminals, and indicate the internal wiring for each

item of equipment and the interconnection between each item of equipment. Drawings shall indicate adequate clearance for operation, maintenance, and replacement of operating equipment devices. If equipment is disapproved, drawings shall be revised to show accepted equipment and resubmitted.

2. When shop drawings have substantial changes from the design to the point that compliance with "AS-BUILT DRAWINGS" of this Section become impractical, the approved shop drawing shall be substituted for the contract drawing and the intent of "AS-BUILT DRAWINGS" paragraph implemented using the approved shop drawings.
 3. Electronic files of mechanical design drawings may be made available if requested, provided the General Contractor and subcontractors sign an agreement satisfactory to Owner's Project Manager indemnifying him of all liabilities. The electronic files will be in AutoCAD 2013 DWG format and will contain only drawing layers pertaining to the mechanical system design.
- E. Manufacturer's Instructions: Where installation procedures or part of installation procedures are required to be in accordance with the manufacturer's instructions, submit printed copies of those instructions with product submittals. All products or items that carry manufacturer's warranty shall be installed in accordance with manufacturer's instructions. Installation of the item shall not proceed until the manufacturer's instructions are received. Failure to submit shall be cause for rejection of the equipment or material. When manufacturer's instructions and these specifications have different requirements, the more stringent requirement shall prevail. However, if the requirements are conflicting, it is the Contractor's responsibility to notify Owner's Project Manager in writing prior to procurement and installation.

All but not limited to, the following items shall be installed in accordance with the manufacturer's instructions and these bid documents.

Air Cooled Condensing Unit
Fan Coil Unit
Controls
Equipment Supports
Vibration Isolators
Pipes & Pipe Insulation

- F. Certificates of Compliance: Submit a certificate of compliance from the manufacturer for approval for products, finishes, and equipment as specified in the Technical Sections whose compliance with organizational standards or specifications is not regulated by an organization using its own listing or label as proof of compliance. The certificates shall identify the manufacturer, the products, equipment, or materials, and the referenced standard and shall simply state that the manufacturer certifies that the product conforms to the requirements specified.
- G. Reference Standards Compliance: Where equipment for materials are specified to conform to industry and technical society reference standards of organizations such as the American National Standards Institute (ANSI), American Society for Testing and Materials (ASTM), National Electrical Manufacturers Association (NEMA), American Society of Mechanical Engineers (ASME), American Gas Association (AGA), American Refrigeration Institute (ARI), and Underwriters Laboratories (UL), submit proof of such conformance. If an organization uses a label or listing to indicate compliance with a particular reference standard, the label or listing will be acceptable evidence, unless otherwise specified in the individual sections.

- H. Independent Testing Organization Certificate: In lieu of the label or listing, submit a certificate from an independent testing organization, competent to perform testing and approved by Owner's Project Manager.
- I. Operation and Maintenance Manuals: When specified in subsequent Sections of Division 23, HVAC, the Contractor shall submit 6 (six) complete sets of operating and maintenance manual on all equipment and the system as a whole. Each set shall be permanently bound with table of contents and shall have a hard cover. One complete set shall be furnished at the time the test procedure is submitted, and the remaining sets shall be furnished before the contract is completed. The following identification shall be inscribed on the covers: the words "OPERATING AND MAINTENANCE MANUALS," the project name, location of the building, name of the Contractor, telephone number, and date. The manual shall include the local representative for each item of equipment. Flysheet shall be provided for piece of equipment and each subject. The instruction sheets shall be approximately 8-1/2 by 11 inches, with large sheets of drawings folded in. The instructions shall include, but shall not be limited to the following:
1. Identify each equipment with tag number, manufacturer's name, model, serial number, capacity, location, and area or rooms served.
 2. Manufacturer's Data
 - a. Approved equipment submittal.
 - b. Manufacturer's wiring and control diagrams, with data to explain the detailed operation and control of each component.
 - c. Manufacturer's operating and maintenance manuals for each piece of equipment, including lubrication instructions.
 - d. Parts lists and recommended spare parts.
 - e. Manufacturer's bulletins, cuts and descriptive data. Applicable data shall be indicated by highlight, arrows, or underlining. Non-applicable data shall be crossed-out.
 4. System layout showing piping, valves, and controls.
 5. Source of service and replacement parts.
 6. A control sequence describing startup, operation, shutdown, restarting after power failure.
- J. Operating Instructions: Submit text of posted operating instructions for each system and principal item of equipment as specified in the technical sections.
- K. As-Built Drawings:
1. The Contractor shall maintain at the job site one (1) set of full size contract drawings, marking them in red to show all variations between the construction actually provided and that indicated or specified in the contract documents, including buried or concealed construction.
 2. Where a choice of material or method is permitted herein or where variations in scope or character of work from that of the original contract is authorized, the drawings shall be marked to define the construction actually provided.
 3. Where equipment installation is involved, the size, manufacturer's name, model number, power input or characteristic as applicable shall be shown on the as-built drawings.
 4. The representation of such changes shall conform to standard drafting practice and shall include such supplementary notes, legends, and details as necessary to clearly portray the as-built construction.

5. The drawings shall be maintained and updated on a daily basis. The Contractor shall sign, and date each sheet to certify that the dimensions and details shown on the drawings reflect the dimensions and details, and specifications as constructed in the field.
 6. The drawings shall show existing conditions and new work shown on the design drawings. Except that the new work shown shall be the as-built condition. Drawings shall include the equipment schedules with the installed equipment data.
 7. Submit the following to Owner's Project Manager for review and approval:
 - a. Job site marked-up drawings.
 - b. Incorporate into electronic drawing files all construction variations and information required by subparagraph 1 through 5 above, and submit final as-built drawings plotted with ink on 16 pound vellum.
 - c. Electronic files of as-built drawings in AutoCAD 2007 DWG format on compact disc (CD).
- L. Welders Qualifications: All welders shall be certified by an independent testing laboratory and certificates shall be submitted to Owner's Project Manager for approval. Testing of welders shall be in accordance with the welding section of ANSI B-31.09 "Building Services Piping".
- M. Balancing Report and Maintenance Manuals: After installation, the new system shall be tested, balanced, and adjusted. Submit copies of the balancing report and operating and maintenance manuals for approval before final inspection.
- N. Posted Operating Instructions: Submit in accordance to this section.
- O. Warranty: Submit in accordance to this section.
- 1.5 LAWS, REGULATIONS AND CODES
- A. All work shall be in accordance to government laws, ordinances, rules, and regulations and orders.
 - B. The following shall govern where applicable; Plumbing Code, Building Code, Fire Code, Electrical Code, Hawaii Department of Health Regulations, U.S. Department of Health and Human Services, Applicable National Fire Protection Association Standards, OSHA, Rules and Regulations and all other codes and standards referenced in these specifications. Where requirements differ in these codes and standards, the more stringent shall apply.
- 1.6 SUBSTITUTION OF EQUIPMENT
- A. The Contractor shall assume full responsibility for proper fit, performance and additional work relating to other sections of the specifications.
- 1.7 VERIFICATION OF DIMENSIONS
- A. The Contractor shall be responsible for the coordination and proper relation of his work to the building structure and to the work of all trades. The Contractor shall visit the premises and thoroughly familiarize himself with all details of the work and working conditions, to verify all dimensions in the field, and to advise Owner's Project Manager of any discrepancy before performing any work.

1.8 DISCREPANCIES

- A. The Drawings and Specifications are intended to be cooperative. Any materials, equipment, or system related to this section and exhibited on the Electrical or Mechanical Drawings but not mentioned in the Specifications are to be executed to the intent and meaning thereof, as if it were both mentioned in the Specifications and set forth on the Drawings.
- B. In case of differences between the Drawings and Specifications, the Specifications shall govern first, and then the Drawings. Large-scale details shall take precedence over small scale Drawings as to the shape and details of construction. Specifications shall govern as to materials.
- C. Drawings and Specifications are intended to be fully cooperative and to agree, but should any discrepancy or apparent difference occur between Drawings and Specifications or should error occur in the work of others affecting the work, the Contractors shall notify Owner's Project Manager at once. If the Contractor proceeds with the work affected without instructions from Owner's Project Manager, he shall make good any resultant damage or defect. All interpretations of Drawings and Specifications shall be clarified by Owner's Project Manager.

1.9 DELIVERY, STORAGE, AND HANDLING

- A. Handle, store, and protect equipment and materials to prevent damage before and during installation in accordance with the manufacturer's recommendations, and as approved by Owner's Project Manager. Replace damaged or defective items. Rusted pipe, fittings, and accessories delivered or installed at the job site will be rejected.
- B. Materials and equipment shall be stored in weather protected locations and covered to prevent dust or moisture damage. Damage to materials or equipment due to contractor's neglect shall be repaired or replace to the satisfaction of Owner's Project Manager by and at the expense of the Contractor.

1.10 TRADE NAME

- A. Mentioning of a trade name in the plans and specifications indicates that the manufacturer is acceptable to Owner's Project Manager. However, certain specified construction and details may not be regularly included in the manufacturer's catalogued product. The Mechanical Contractor shall provide the material or equipment complete as specified.

1.11 WORKMANSHIP AND MATERIALS

- A. Workmanship shall be of the best quality and none but competent mechanical workers skilled in their trades and thoroughly familiar with the work involved shall be employed. The Contractor shall furnish the services of an experienced superintendent, who will be constantly in charge of the erection of the work, until completed and accepted.
- B. Unless otherwise hereinafter specified, each article of its kind shall be the standard product of a single manufacturer.
- C. Whenever the words "or acceptable equal" or other words of similar intent or meaning are used, implying that judgment is to be exercised, it is understood that it is the judgment of Owner's Project Manager that is referred to.

- D. Owner's Project Manager shall have the right to accept or reject material, equipment and/or workmanship and determine when the Contractor has complied with the requirements herein specified.
- E. All manufactured materials shall be delivered and stored in their original containers. Equipment shall be clearly marked or stamped with the manufacturer's name and rating. Equipment and materials shall be carefully handled, properly stored, and adequately protected to prevent damage before and during installation, in accordance with the manufacturer's recommendations and as acceptable by Owner's Project Manager. Damaged or defective items, in the opinion of Owner's Project Manager, shall be replaced.
- F. Reference to standards is intended to be the latest revision of the standard specified.

1.12 MANUFACTURER'S RECOMMENDATIONS

- A. Equipment installed under this Division of the Specifications shall be installed according to manufacturer's recommendations, unless otherwise shown on the drawings or herein specified. Where installation procedures or any part thereof are required to be in accordance with the recommendations of the manufacturer of the material being installed, printed copies of these recommendations shall be furnished to Owner's Project Manager, prior to installation. Installation of the item will not be allowed to proceed until the recommendations are received. Failure to furnish these recommendations can be cause for rejection of the material.

1.13 OPENINGS, CUTTING AND REPAIRING

- A. The Mechanical Contractor shall cooperate with the work to be done under other sections in providing information as to openings required in walls and slabs for all piping including sleeves where required.
- B. It shall be the responsibility of the Contractor to ascertain that all openings are properly located. Openings shall be coordinated with structural drawings, Officer-in-Charge and Contractors. Provide sleeves and fire rated material through walls, slabs and beams as required by code.

1.14 ELECTRICAL WORK

- A. Final hook-up power wiring to main mechanical equipment will be provided under DIVISION 26 - ELECTRICAL of Specifications. Control devices requiring control and related power wiring shall be provided by the Licensed Controls Contractor and to be wired by their Electrical Contractor. Mechanical Contractor shall not segregate or delineate the controls wiring including related power from the Licensed Contractor.
- B. Electrical work under Electrical Division is based on the electrical rating of equipment indicated on the Mechanical Drawings. Additional electrical work caused by any deviation under the Mechanical Division shall be paid for by the Mechanical Contractor.
- C. All control and power wiring is included under mechanical work and shall be in accordance with Code requirements specified in DIVISION 26 - ELECTRICAL, and where specified in DIVISION 23 - HVAC.

- D. The Mechanical Contractor shall furnish all starters for installation by the Electrical Contractor. The Mechanical Contractor shall turn over these items to the Electrical Contractor at the site after receipt of notice from the Electrical Contractor that he is ready to install said items. Mechanical Contractor shall be responsible for review of starters during Bid period. After bid period mechanical contractor shall provide all starters and related control power wiring for their equipment whether or not indicated on drawings.

1.15 PAINTING

- A. Contractor shall paint all exposed work specified in DIVISION 23 – HVAC equipment piping, ducts, supports and conduits. Paint shall match adjacent wall or ceiling; verify with Officer-in-Charge exact type, color, prime and number of coats of the paint required.

1.16 LABELING

- A. Contractor shall label all mechanical equipment specified in DIVISION 23 - HVAC with month and year of installation.

1.17 SEISMIC RESTRAINTS

- A. Provide seismic restraints for mechanical systems in accordance with the Building Code.

PART 2 - PRODUCTS

2.1 MATERIALS AND EQUIPMENT

- A. Asbestos Prohibition: No asbestos containing materials shall be used under this section. The Contractor shall ensure that all materials incorporated in the project are asbestos-free.
- B. Materials and equipment shall conform to the requirements of applicable Technical Sections; publications specified therein and shall be as shown. Materials and equipment shall be new and shall be the products of manufacturers regularly engaged in the manufacture of such products. All items shall essentially duplicate materials and equipment that have been in satisfactory use at least two (2) years prior to bid opening and shall be supported by a service organization that is, in the opinion of Owner's Project Manager, reasonably convenient to the site of installation.

PART 3 - EXECUTION

3.1 PIPING IDENTIFICATION

- A. Identification of all new pipe lines shall be by means of colored, waterproof, all temperature, self-adhering labels and directional arrow. Refer to SECTION 099000 - PAINTING for color coding of existing and new piping.
- B. At Contractor's option, each and every system may be identified by painting with contrasting colors, using 3/4" high minimum stencil letters. Painting shall be done by the Mechanical Contractor or his Subcontractor.
- C. All exposed pipes, whether insulated or not shall be identified. Labels may be excluded for piping where the use is obvious, due to its connection to equipment and where the appearance would be objectionable in finished rooms, as acceptable by direction.

D. Identification labels shall be placed as follows:

1. Near each valve and branch connection
2. Wherever piping merges or disappears from view from the floor of the room in which it is installed
3. Labels shall not be more than 50 feet apart

3.2 VALVE INDEX

- A. The Mechanical Contractor shall provide brass or plastic tags on all valves with letters stamped or engraved thereon designating service of each valve.
- B. Identify electrical panel and related circuit breaker for controls power wiring.
- C. Valve, controllers and related power accessories identification shall be correlated with sequence of operation documents.

3.3 PROTECTION OF WORK IN PROGRESS

- A. Pipe and duct openings shall be closed with caps or plugs until connections are made. Equipment shall be securely covered for protection against physical or chemical damage. In areas exposed to weather, materials unused at the end of each day's work shall be stored in weather-protected locations. Damage to materials or equipment due to the Contractor's neglect shall be repaired or replaced to the satisfaction of Owner's Project Manager by, and at the expense of the Contractor. Ductwork openings shall be covered and protected from dust and moisture daily. No equipment, ductwork, piping, or insulation should be installed if contaminated by dust or moisture. Objects should be cleaned or dried before installation.

3.4 PIPING IDENTIFICATION

- A. The Mechanical Contractor shall perform all tests of the installed work and shall provide all services, labor, equipment, materials and instruments needed for the tests. During pressure tests, all items in the system to be tested not designed for test pressures shall be removed or isolated from the system and shall be reconnected or unblocked after tests are completed. Should operating tests require the presence of manufacturers' representatives, the Mechanical Contractor shall cooperate with them and shall place at their disposal all assistance, materials, and services required to perform such test. The Mechanical Contractor shall certify in writing that all work has passed all required tests.

3.5 LOCAL TECHNICAL SUPPORT

- A. The mechanical equipment suppliers that furnish equipment for this project shall have a local Hawaii sales and service office, staffed with factory trained representatives fully capable of providing instruction, routine maintenance and emergency maintenance service on all system components supplied for this project. If the maintenance service is provided by a separate company, that company shall be specifically trained by manufacturer and authorized to perform maintenance on the equipment furnished for this project.

3.6 POSTED OPERATING INSTRUCTION

- A. Furnish acceptable operation instructions for each principal item of equipment for the use of the operation and maintenance personnel. The operation instructions shall include wiring diagrams, control diagrams and control sequence for each principal item of equipment. Operating instruction shall be printed or engraved and shall be framed under glass or in acceptable laminated plastic and posted where directed by Owner's Project Manager. Operating instructions shall be attached to or posted adjacent to each principal item of equipment including start up, procedure in the event of equipment failure and other items of instruction as recommended by the manufacturer of each item of equipment. Operating instructions exposed to the weather shall be made of weather-resistant materials or shall be suitably enclosed to be weather protected. Operating instructions shall not fade when exposed to sunlight and shall be secured to prevent easy removal or peeling.

3.7 QUALITY ASSURANCE

- A. The Mechanical Contractor shall have a local Hawaii office, staffed with factory trained engineers fully capable of providing instruction, routine maintenance and emergency maintenance service on all system components. The Mechanical Contractor shall have a three year experience record in the design and installation of variable refrigerant flow systems similar in scope and performance to that specified herein, and shall be prepared to provide evidence of this history as condition of acceptance and approval.

3.8 INSTRUCTION TO PERSONNEL

- A. The Contractor shall furnish the services of competent instructors who will give full instruction to the designated personnel in the adjustment, operation and maintenance, including pertinent safety requirements, of the equipment or system specified. Each instructor shall be thoroughly familiar with all parts of the installation and shall be trained in operating theory as well as practical operation and maintenance work. Instruction shall be given during the first regular workweek after the equipment or system has been accepted and turned over to Owner for regular operation. The number of man-days (8 hours) of instruction furnished shall be as specified in other sections. When more than 4 man-days of instruction are specified, approximately half of the time shall be used for classroom instruction. All other times shall be used for instruction with the equipment or system. When significant changes or modifications in the equipment or systems are made under the term of the contract, additional instruction shall be provided to acquaint the operating personnel with the changes or modifications.

3.9 SAFETY REQUIREMENTS

- A. Belts, pulley, chains, gears, couplings, projecting setscrews, keys and other rotating parts located so that any person can come in close proximity thereto shall be fully enclosed or properly guarded. Provide positive means of locking out equipment so that equipment cannot be accidentally started during maintenance procedures. High temperature equipment and piping so located as to endanger personnel or create a fire hazard shall be properly guarded or covered with insulation of a type as specified herein. Ensure that access openings leading to equipment are large enough to carry through routine maintenance items such as filters and tools.

3.10 FINAL INSPECTION

- A. Final inspection shall be requested by the Mechanical Contractor only after submittal of all required certificates. No final inspection will be made until all moving parts of equipment are properly guarded, all controls and safety devices tested and operative, all painting required is done and the site cleaned up.

3.11 WARRANTY

- A. All air conditioning and ventilation systems, plumbing, controls, and energy management systems shall be provided with a one year warranty from the date of acceptance of the project by Owner against any defects due to faulty materials, equipment, workmanship, or installation. Upon notice of defect, the Mechanical Contractor shall correct or replace defective item at no additional cost to Owner. The Contractor shall be advised that Owner's Project Manager shall have the right for beneficial use of all new equipment prior to the project acceptance and guarantee period. The guarantee period shall be for one year after acceptance date. Start of beneficial use is not the start of the guarantee period.

END OF SECTION

SECTION 23 07 00 – THERMAL INSULATION FOR MECHANICAL SYSTEMS

PART 1 - GENERAL

1.1 DESCRIPTION

- A. Section includes the requirements for field-applied insulation and accessories on mechanical systems as specified herein and as indicated on the drawings, including but not limited to the following:
 - 1. Piping Insulation
 - 2. Equipment insulation
 - 3. Aluminum Jackets

1.2 CODES, STANDARDS, REGULATIONS

- A. All applicable codes, regulations and ordinances of public bodies having jurisdiction are considered a part of these specifications; all work installed and materials provided must comply with the current edition of such codes, regulations and ordinances.

1.3 REFERENCES

- A. ASTM International (ASTM):
 - 1. ASTM A580/A580M, (2023), Standard Specification for Stainless Steel Wire
 - 2. ASTM B209, (2014) Standard Specification for Aluminum and Aluminum-Alloy Sheet and Plate
 - 3. ASTM C534/C534M, (2023) Standard Specification for Preformed Flexible Elastomeric Cellular Thermal Insulation in Sheet and Tubular Form
 - 4. ASTM C552, (2022), Standard Specification for Cellular Glass Thermal Insulation
 - 5. ASTM C795, (2008; R 2023) Standard Specification for Thermal Insulation for Use in Contact with Austenitic Stainless Steel
 - 6. ASTM C920, (2018) Standard Specification for Elastomeric Joint Sealants
 - 7. ASTM C1136, (2023) Standard Specification for Flexible, Low Permeance Vapor Retarders for Thermal Insulation
 - 8. ASTM C1710, (2011) Standard Guide for Installation of Flexible Closed Cell Preformed Insulation in Tube and Sheet Form

9. ASTM E84, (2023) Standard Test Method for Surface Burning Characteristics of Building Materials
 10. ASTM E96/E96M, (2024) Standard Test Methods for Gravimetric Determination of Water Vapor Transmission Rate of Materials
 11. ASTM E2231, (2021) Standard Practice for Specimen Preparation and Mounting of Pipe and Duct Insulation Materials to Assess Surface Burning Characteristics
- B. FM Global (FM):
1. FM APP GUIDE, (updated on-line) Approval Guide <http://www.approvalguide.com/>
- C. Manufacturers Standardization Society of The Valve and Fittings Industry (MSS)
1. MSS SP-58, (2018) Pipe Hangers and Supports - Materials, Design and Manufacture, Selection, Application, and Installation
- D. Midwest Insulation Contractors Association (MICA):
1. MICA Insulation Stds, (8th Ed) National Commercial & Industrial Insulation Standards
- E. Underwriters Laboratories (UL)
1. UL 94, (2023; Reprint Jan 2024) UL Standard for Safety Tests for Flammability of Plastic Materials for Parts in Devices and Appliances
 2. UL 723, (2020) UL Standard for Safety Test for Surface Burning Characteristics of Building Materials

1.4 SHOP DRAWINGS

- A. Contract drawings are essentially diagrammatic, indicating general layout and approximate locations toward establishing the scope for uniform estimating basis. Equipment specified in this section, may differ in dimension and configuration by manufacturer. Equipment and piping arrangements shall fit into space allotted and shall allow adequate clearances for servicing and maintenance. Reasonable modifications to indicated locations and arrangement to suit job conditions shall not constitute a basis for requesting additional funds from the NPS.
- B. Capacities of all equipment and materials shall be not less than those indicated.
- C. Nameplate: Each major component of equipment shall have the manufacturer's name, address, and catalog number on a plate securely attached to the item of equipment.
- D. Verification of Dimensions: The Contractor shall be responsible for the proper relation of his work to the building structure and to the work of all trades. The Contractor shall visit the premises and thoroughly familiarize himself with all details of the work and working conditions, to verify all dimensions in the field, and to advise the NPS of any discrepancy before performing any work.

1.5 ACTION SUBMITTALS

- A. Submit the following in accordance with Section 01 33 23 – Submittal Procedures:
- B. Product Data: Submit product data for the following:
 - 1. Pipe Insulation Systems
 - 2. Equipment Insulation Systems
 - 3. Aluminum
- C. Shop Drawings: Submit shop drawings of plans, performance data and details showing locations and installation including but not limited to the following.
 - 1. MICA Plates
 - 2. Pipe Insulation Systems
 - 3. Equipment Insulation Systems

1.6 INFORMATIONAL SUBMITTALS

- A. Guarantee and Certificate: Submit one year guarantee and certificate.

1.7 OMISSIONS

- A. It is the intent of the plans and specifications to provide a complete installation. Should there be omissions, the Contractor shall call the attention of the NPS to such omissions fifteen (15) days advance of the date of the bid or proposal so that the necessary corrections can be made.

1.8 DELIVERY, STORAGE, AND HANDLING

- A. Deliver materials in the manufacturer's unopened containers. Protect materials delivered and placed in storage from weather, humidity, dirt, dust and other contaminants. The Contracting Officer may reject insulation material and supplies that become dirty, dusty, wet, or contaminated by some other means. Attach manufacturer's stamp or label giving the name of the manufacturer and brand, and a description of the material, date codes, and approximate shelf life (if applicable) to packages or standard containers of insulation, jacket material, cements, adhesives, and coatings delivered for use, and samples required for approval. Insulation packages and containers must be asbestos free.

PART 2 - PRODUCTS

2.1 STANDARD COMMERCIAL PRODUCTS

- A. Provide materials which are the standard products of manufacturers regularly engaged in the manufacture of such products and that essentially duplicate items that have been in satisfactory use for at least 2 years prior to bid opening. Submit a complete list of materials, including manufacturer's descriptive technical literature, performance data, catalog cuts, and installation instructions. Include the product number, k-value, thickness and furnished accessories including adhesives, sealants and jackets for each mechanical system requiring insulation. The product data must be copyrighted, have an identifying or publication number, and have been published prior to the issuance date of this solicitation. Submit materials furnished under this section together in a booklet.
- B. Provide insulation systems in accordance with the approved MICA National Insulation Standards plates as supplemented by this specification. Provide field-applied insulation for refrigerant piping systems that are located within, on, under, and adjacent to buildings as indicated on the plans. Provide CFC and HCFC free insulation.
- C. Surface Burning Characteristics: Unless otherwise specified, insulation must have a maximum flame spread index of 25 and a maximum smoke developed index of 50 when tested in accordance with ASTM E84. Determine flame spread, and smoke developed indexes, by ASTM E84 or UL 723. Test insulation in the same density and installed thickness as the material to be used in the actual construction. Prepare and mount test specimens according to ASTM E2231.

2.2 MATERIALS

- A. General: Provide insulation that meets or exceed the requirements of ASHRAE 90.1. Ensure insulation exterior is cleanable, grease resistant, non-flaking and non-peeling. Provide compatible materials that do not contribute to corrosion, soften, or otherwise attack surfaces to which applied in either wet or dry state. Use materials on stainless steel surfaces meeting ASTM C795 requirements. Use asbestos free materials. Provide product recognized under UL 94 (if containing plastic) and listed in FM APP GUIDE.
- B. Caulking: ASTM C920, Type S, Grade NS, Class 25, Use A.
- C. Corner Angles: Nominal 0.016 inch aluminum 1 by 1 inch with factory applied kraft backing. Aluminum must be ASTM B209, Alloy 3003, 3105, or 5005.
- D. Fittings: Fabricated Fittings are the prefabricated fittings for flexible elastomeric pipe insulation systems in accordance with ASTM C1710. Together with the flexible elastomeric tubes, they provide complete system integrity for retarding heat gain and controlling condensation drip from refrigerant piping. Flexible elastomeric, fabricated fittings provide thermal protection (0.25 k) and condensation resistance (0.05 Water Vapor Transmission factor). For satisfactory performance, use properly installed protective vapor retarder/barriers and vapor stops on high relative humidity and below ambient temperature applications to reduce movement of moisture through or around the insulation to the colder interior surface.

- E. Aluminum Jackets: Provide aluminum jackets consisting of embossed or smooth sheet, 0.016 inch nominal thickness; ASTM B209, Temper H14, Temper H16, Alloy 3003, 5005, or 3105. Do not use corrugated aluminum jacket outdoors. Aluminum jacket securing bands must be Type 304 stainless steel, 0.015 inch thick, 1/2 inch wide for pipe under 12 inch diameter and 3/4 inch wide for pipe over 12 inch and larger diameter. Aluminum jacket circumferential seam bands must be 2 by 0.016 inch aluminum matching jacket material. Ensure bands for insulation below ground are 3/4 by 0.020 inch thick stainless steel, or fiberglass reinforced tape. The jacket may, at the option of the Contractor, be provided with a factory fabricated Pittsburgh or "Z" type longitudinal joint. When the "Z" joint is used, use bands at the circumferential joints that are designed by the manufacturer to seal the joints and hold the jacket in place.
- F. Wire: Soft annealed ASTM A580/A580M Type 302, 304 or 316 stainless steel, 16 or 18 gauge.
- G. Insulation Bands: Provide 1/2 inch wide; 26 gauge stainless steel insulation bands.
- H. Sealants: Choose sealants from the butyl polymer type, the styrene-butadiene rubber type, or the butyl type of sealants. Provide sealants with a maximum permeance of 0.02 perms based on Procedure B for ASTM E96/E96M, and a maximum flame spread index of 25 and a maximum smoke developed index of 50 when tested in accordance with ASTM E84.

2.3 PIPING INSULATION SYSTEMS

- A. General: Conform insulation materials to Table 1 and minimum insulation thickness as listed in Table 2 and meet or exceed the requirements of ASHRAE 90.1. Limit pipe insulation materials to those listed herein and meeting the following requirements.
- B. Aboveground Cold Pipeline (-30 to 60 deg. F): Provide insulation for outdoor, indoor, exposed or concealed applications, as follows:
 - 1. Cellular Glass: ASTM C552, Type II, and Type III. Supply the insulation from the fabricator with ASJ vapor retarder and installed with all longitudinal overlaps sealed and all circumferential joints ASJ taped or supply the insulation unfaced from the fabricator and install with all longitudinal and circumferential joints sealed with vapor barrier mastic.
 - 2. Flexible Elastomeric Cellular Insulation: Closed-cell, foam- or expanded-rubber materials containing anti-microbial additive, complying with ASTM C534/C534M, Grade 1, Type I or II. Type I, Grade 1 for tubular materials. Type II, Grade 1, for sheet materials. Ensure Type I and II have vapor retarder/vapor barrier skin on one or both sides of the insulation, and require an additional exterior vapor retarder covering for high relative humidity and below ambient temperature applications.

TABLE 1
Insulation Materials for Piping

Service: Refrigerant Suction Piping, 35°F, nominal				
Material	Specification	Type	Class	Vapor Retarder/ Vapor Barrier Req'd
Flexible Elastomeric Cellular	ASTM C534/C534M	I		No
Cellular Glass	ASTM C552	II	1	Yes
Service: Condensate Drain Located Inside Building				
Material	Specification	Type	Class	Vapor Retarder/ Vapor Barrier Req'd
Cellular Glass	ASTM C552	II	2	No
Flexible Elastomeric Cellular	ASTM C534/C534M	I		No

TABLE 2
Insulation Thickness for Piping
Service: Refrigerant Suction Piping, 35°F, nominal
Condensate Drain Located Inside Building

Material	Tube and Pipe Size (inch)		
	<1	1 - <1.5	1.5 - <4
Cellular Glass	1.5"	1.5"	1.5"
Flexible Elastomeric Flexible	1"	1"	1"

2.4 EQUIPMENT INSULATION SYSTEMS

- A. Insulate equipment and accessories as specified in Tables 3. The thickness listed are for outside locations. Increase the specified insulation thickness for equipment where necessary to equal the thickness of angles or other structural members to make a smooth, exterior surface. Submit a booklet containing manufacturer's published installation instructions for the insulation systems. The instructions must be copyrighted, have an identifying or publication number, and have been published prior to the issuance date of this solicitation.

TABLE 3
Insulation Materials for Cold Equipment
Service: Condensate Drain Pumps, 35° to 60°F

Material	Thickness (Inches)
Cellular Glass	2"
Flexible Elastomeric Flexible	1.5"

PART 3 - EXECUTION

3.1 APPLICATION - GENERAL

- A. Apply insulation to uncooled piping and equipment. Do not compress flexible elastomeric cellular insulation at joists, studs, columns, ducts, and hangers. The insulation must not pull apart after a one hour period; replace any insulation found to pull apart after one hour.
- B. Installation: Except as otherwise specified, install material in accordance with the manufacturer's written instructions. Do not apply insulation materials until tests specified in other sections of this specification are completed. Remove material such as rust, scale, dirt and moisture from surfaces to receive insulation. Keep insulation clean and dry. Do not remove insulation from its shipping containers until the day it is ready to use and return to like containers or equally protect from dirt and moisture at the end of each workday. Thoroughly clean insulation that becomes dirty prior to use. If insulation becomes wet or if cleaning does not restore the surfaces to like new condition, reject the insulation, and immediately remove from the jobsite. Stagger joints on multi-layer insulation. Install insulation, jacketing and accessories in accordance with MICA Insulation Stds plates except where modified herein or on the drawings.
- C. Painting and Finishing: Paint as specified in Section 09 90 00 PAINTS AND COATINGS.
- D. Installation of Flexible Elastomeric Cellular Insulation:
 - 1. Install flexible elastomeric cellular insulation with seams and joints sealed with rubberized contact adhesive. Stagger seams when applying multiple layers of insulation. Protect insulation exposed to weather and not shown to have vapor barrier weatherproof jacketing with two coats of UV resistant finish or PVC or metal jacketing as recommended by the manufacturer after the adhesive is dry and cured.
 - 2. Adhesive Application: Apply a brush coating of adhesive to both butt ends to be joined and to both slit surfaces to be sealed. Allow the adhesive to set until dry to touch but tacky under slight pressure before joining the surfaces. Ensure insulation seals at seams and joints are not capable of being pulled apart one hour after application. Replace insulation that can be pulled apart one hour after installation.
 - 3. Adhesive Safety Precautions: Use natural cross-ventilation, local (mechanical) pickup, and/or general area (mechanical) ventilation to prevent an accumulation of solvent vapors, keeping in mind the ventilation pattern must remove any heavier-than-air solvent vapors from lower levels of the workspaces. Gloves and spectacle-type safety glasses are recommended in accordance with safe installation practices.
- E. Pipes/Equipment That Require Insulation: Insulation is required on all refrigerant pipes, and equipment, except for omitted items as specified.

3.2 PIPE INSULATION SYSTEMS INSTALLATION

- A. Pipe Insulation:

1. General: Install pipe insulation on new aboveground refrigerant pipeline systems as specified below to form a continuous thermal retarder/barrier, including straight runs, fittings and appurtenances unless specified otherwise. Install full length units of insulation using a single cut piece to complete a run. Do not use cut pieces or scraps abutting each other.
 2. Omit pipe insulation on the following:
 - a. Adjacent insulation.
 - b. ASME stamps.
 - c. Cleanouts or handholes.
- B. Pipes Passing Through Walls, and Floors:
1. Provide continuous pipe insulation through the sleeve. Provide an aluminum jacket or vapor barrier/weatherproofing self-adhesive jacket (minimum 2 mils adhesive, 3 mils embossed) less than 0.0000 permeability, greater than 3 ply standard grade, silver, white, black and embossed with factory applied moisture retarder over the insulation wherever penetrations require sealing.
 2. Penetrating Exterior Walls: Continue the aluminum jacket required for pipe exposed to weather through the sleeve to a point 2 inches beyond the interior surface of the wall.
 3. Pipes Passing Through Hangers or supports: Ensure insulation is continuous through hangers or supports. Support all horizontal pipes 2 inches and smaller on hangers with the addition of a Type 40 protection shield to protect the insulation in accordance with MSS SP-58. Whenever insulation shows signs of being compressed, or when the insulation or jacket shows visible signs of distortion at or near the support shield, install insulation inserts as specified below for piping larger than 2 inches, or factory insulated hangers (designed with a load bearing core) can be used.
 4. Horizontal Pipes Larger Than 2 Inches and Below 60 Degrees F: Supported on hangers with the addition of a Type 40 protection shield in accordance with MSS SP-58. Install an insulation insert of cellular glass, prefabricated insulation pipe hangers, or perlite above 80 degrees F above each shield. Ensure insert covers no less than the bottom 180-degree arc of the pipe. Provide inserts that are the same thickness as the insulation, and extend 2 inches on each end beyond the protection shield. When insulation inserts are required in accordance with the above, and the insulation thickness is less than 1 inch, wooden or cork dowels or blocks may be installed between the pipe and the shield to prevent the weight of the pipe from crushing the insulation, as an option to installing insulation inserts. Ensure the insulation jacket is continuous over the wooden dowel, wooden block, or insulation insert.
 5. Inserts: Covered with a jacket material of the same appearance and quality as the adjoining pipe insulation jacket, overlap the adjoining pipe jacket 1-1/2 inches, and seal as required for the pipe jacket. Use jacket material to cover inserts in flexible elastomeric cellular insulation conforming to ASTM C1136, Type 1, and is allowed to be of a different material than the adjoining insulation material.
- C. Flexible Elastomeric Cellular Pipe Insulation: Use tubular form flexible elastomeric cellular pipe insulation for pipe sizes 6 inches and less. Stagger seams when applying multiple layers of insulation. Insulate sweat fittings with miter-cut pieces the same size as on adjacent piping.

Insulate screwed fittings with sleeved fitting covers fabricated from miter-cut pieces and overlap and seal to the adjacent pipe insulation.

3.3 ABOVEGROUND COLD PIPELINES

- A. Insulate the refrigerant and condensate pipelines for minus 30 to plus 60 degrees F with the materials and thicknesses specified in Table 1 and Table 2.
- B. Factory or Field Applied Jacket:
 - 1. Cover insulation with a factory applied vapor retarder jacket/vapor barrier or with a field applied, 3-ply laminated, self-adhesive vapor barrier/weatherproofing jacket in accordance with the manufacturer's published instructions.

3.4 PIPING EXPOSED TO WEATHER

- A. Aluminum Jacket: Cover insulation with a field applied aluminum insulation jacket in accordance with the manufacturer's published instructions. Use pre-formed sections for covering fittings such as elbow and tees.

3.5 EQUIPMENT INSULATION SYSTEMS

- A. General: Provide removable insulation sections to cover parts of equipment that must be opened periodically for maintenance including vessel covers, fasteners, flanges and accessories. Omit equipment insulation on the following:
 - 1. ASME stamps.
 - 2. Manufacturer's nameplates.
- B. Insulation for Cold Equipment: Furnish insulation on equipment handling media below 60 degrees F including the following:
 - 1. Condensate water pumps
- C. Pump Insulation:
 - 1. Insulate pumps by forming a box around the pump housing. Construct the box by forming the bottom and sides using joints that do not leave raw ends of insulation exposed. Join joints between sides and between sides and bottom by adhesive with contact adhesive for flexible elastomeric cellular insulation. Ensure box conforms to the requirements of MICA Insulation Stds plate No. 49 when using flexible elastomeric cellular insulation. Ensure joints between top cover and sides fit tightly forming a female shiplap joint on the side pieces and a male joint on the top cover, thus making the top cover removable.
 - 2. Protect exposed insulation corners with corner angles.

3. Upon completion of installation of the insulation, including removable sections, apply two coats of vapor retarder coating with a layer of glass cloth embedded between the coats. The total dry thickness of the finish must be 1/16 inch. Provide a parting line between the box and the removable sections allowing the removable sections to be removed without disturbing the insulation coating. Apply flashing sealant to parting line, between equipment and removable section insulation, and at all penetrations.
- D. Vapor Retarder/Vapor Barrier: Upon completion of installation of insulation, caulk penetrations. Apply two coats of vapor retarder coating or vapor barrier jacket over insulation, including removable sections, with a layer of open mesh synthetic fabric embedded between the coats. Ensure the total dry thickness of the finish is 1/16 inch. Apply flashing sealant or vapor barrier tape to parting line between equipment and removable section insulation.
- E. Equipment Exposed to Weather: Apply two coats of weather barrier mastic reinforced with fabric or mesh for outdoor application to the entire surface. Ensure each coat of weatherproof mastic has a minimum thickness of 1/16 inch. Ensure exterior is a metal jacketing applied for mechanical abuse and weather protection, and secure with screws or vapor barrier/weatherproofing jacket less than 0.0000 permeability greater than 3 ply, standard grade, silver, white, black, and embossed or greater than 8 ply, heavy duty white and natural. Apply membrane overlapping material by 3 inches.

END OF SECTION 23 07 00

SECTION 23 81 00 – UNITARY AIR CONDITIONING EQUIPMENT

1.1 SUMMARY

- A. This section covers the furnishing, fabrication, delivery and installation of the air conditioning system complete, including but not limited to the following:
Direct Expansion Split Systems.

1.2 ACTION SUBMITTALS

- A. Product Data: Submit for each type of the following. Include dimensional drawings and locations of connections.
 - 1. Air Cooled Condensing Unit (ACCU)
 - 2. Wall Mounted Fan Coil Units
 - 3. Condensate Drain Pumps
 - 4. Central Controller
- B. Shop Drawings
 - 1. Refrigerant piping and associated accessories.
 - 2. Control Wiring diagrams

1.3 QUALITY ASSURANCE

Carefully investigate the plumbing, fire protection, electrical, structural and finish conditions that would affect the work to be performed and arrange such work accordingly, furnishing required offsets, fittings, and accessories to meet such conditions. Submit drawings consisting of:

- A. Equipment layouts which identify assembly and installation details
- B. Plans and elevations which identify clearances required for maintenance and operation.
- C. Wiring diagrams which identify each component individually and interconnected or interlocked relationships between components
- D. Foundation drawings, bolt-setting information, and foundation bolts prior to concrete foundation construction for equipment indicated or required to have concrete foundations
- E. Details, if piping and equipment are to be supported other than as indicated, which include loadings and type of frames, brackets, stanchions, or other supports
- F. Installation details which includes the amount of factory set superheat and corresponding refrigerant pressure/temperature

1.4 DELIVERY, STORAGE, AND HANDLING

- A. Protect stored items from the weather, humidity and temperature variations, dirt and dust, or other contaminants. Properly protect and care for all material both before and during installation. Submit an inventory of all the stored items. Replace any materials found to be damaged, at no additional cost to the Government. During installation, cap piping and similar openings capped to keep out dirt and other foreign matter.

1.5 WARRANTY

- A. Installing Contractor: Provide equipment with the Manufacturer's Standard Warranty.

PART 2 - PRODUCTS

- 2.1 DUCTLESS SPLIT TYPE AIR CONDITIONER: Provide split-type air source air conditioning unit consisting of indoor units and remote air-to-air outdoor units, rated in accordance with ARI 210/240. Mitsubishi Electric or approved equal.

A. OUTDOOR UNIT

1. General: The air cooled condensing unit shall be equipped with multiple circuit boards that interface to the direct digital controls system and shall perform all functions necessary for operation. The outdoor unit shall have an epoxy coating on the cabinet and a coating on the condenser coil as specified. The outdoor unit shall be completely factory assembled, piped, and wired. Each unit shall be run tested at the factory. Outdoor unit shall have a sound rating no higher than 63 dB(A). Both refrigerant lines from the outdoor unit to indoor units shall be insulated. The outdoor unit shall have an accumulator with refrigerant level sensors and controls, a high pressure safety switch, over-current protection, and DC bus protection. The outdoor unit shall have a high efficiency oil separator plus additional logic controls to ensure adequate oil volume in the compressor is maintained.
2. Unit Cabinet: The casing shall be fabricated of galvanized steel, bonderized and finished with a powder coated baked enamel and epoxy coating.
3. Fan: The outdoor unit shall be furnished with one direct drive, variable speed propeller type fan. The fan motor shall have inherent protection, have permanently lubricated bearings, and be completely variable speed. The fan motor shall be mounted for quiet operation. The fan shall be provided with a raised guard to prevent contact with moving parts.
4. Coil: The outdoor coil shall be of nonferrous construction with lanced or corrugated plate fins on copper tubing. The coil fins shall have a coating as specified. The coil shall be protected with an integral metal guard. Refrigerant flow from the outdoor unit shall be controlled by means of an inverter driven compressor.
5. Compressor: The outdoor units shall be equipped with one inverter driven scroll hermetic compressor. A crankcase heater shall be factory mounted on the compressor. The outdoor unit compressor shall have an inverter to modulate capacity. The capacity shall be completely variable down to 16 percent of rated capacity. The compressor shall be equipped with an internal thermal overload and mounted to avoid the transmission of vibration.
6. Provide coating of condenser coil and cabinet as indicated.
7. Electrical:
 - a. The outdoor unit electrical power supply shall be 208/230 volts, 1-phase, 60 hertz.
 - b. The outdoor unit shall be controlled by microprocessors located in the indoor unit and outdoor unit. A 12 to 24 volt DC data stream shall communicate between the units providing all necessary information for full function control.
 - c. The outdoor unit shall be equipped with Pulse Amplitude Modulation (PAM) compressor inverter drive control for maximum efficiency with minimum power consumption.

8. BS Coating: The outdoor unit shall be coated with the manufacturer's seacoast protective coating.
9. Coating for Printed Circuit Boards (PCBs): Apply silicone to the parts and patterns on PCBs by following manufacturer's instructions. The coating shall provide protection from salt damage and prevent short-circuit failure from entry of geckos, roaches and other small creatures.

B. INDOOR UNIT

1. General:
 - a. Indoor units shall be factory assembled, wired and run tested. Contained within the unit shall be all factory wiring, piping, electronic modulating linear expansion device, control circuit board and fan motor. The unit shall have a self-diagnostic function, 3-minute time delay mechanism, an auto restart function, and a test run switch. Indoor unit and refrigerant pipes shall be charged with dehydrated air before shipment from the factory.
2. Unit Cabinet:
 - a. Indoor Unit Cabinet
 - (1) All casings, regardless of model size, shall have the same white finish.
 - (2) Multi directional drain and refrigerant piping offering four (4) directions for refrigerant piping and two (2) directions for draining are required.
 - (3) There shall be a separate back plate which secures the unit firmly to the wall.
3. Fan:
 - a. Indoor Unit Fan
 - (1) Each indoor fan shall be statically and dynamically balanced to run on a single motor with permanently lubricated bearings.
 - (2) A manual adjustable guide vane shall be provided with the ability to change the airflow from side to side (left to right).
 - (3) A motorized air sweep louver shall provide an automatic change in airflow by directing the air up and down to provide uniform air distribution.
4. Filter: Return air shall be filtered by means of a long-life washable filter.
5. Coil:
 - a. Indoor units include factory-installed LEV/EEV.
 - b. Indoor coil shall be of nonferrous construction with smooth plate fins on copper tubing. The tubing shall have inner grooves for high efficiency heat exchange. All tube joints shall be brazed with phos-copper or silver alloy.
 - c. The coils shall be pressure tested at the factory.
 - d. The unit shall be provided with an integral condensate lift mechanism that will be able to raise drain water 33 inches above the condensate pan.
6. Electrical
 - a. The unit electrical power shall be 208/230 volts, 1-phase, 60 hertz.
 - b. The system shall be capable of satisfactory operation within voltage limits of 187-228 volts (208V/60Hz) or 207-253 volts (230V/60Hz).
7. Controls
 - a. Indoor unit shall compensate for the higher temperature sensed by the return air sensor compared to the temperature at level of the occupant when in HEAT mode. Disabling of compensation shall be possible for individual units to accommodate instances when compensation is not required.
 - b. Control board shall include contacts for control of external heat source. External heat may be energized as second stage with 1.8°F – 9.0°F adjustable deadband from set point.

- c. Indoor unit shall include no less than four (4) digital inputs capable of being used for customizable control strategies.
- d. Indoor unit shall include no less than three (3) digital outputs capable of being used for customizable control strategies.

B. Control Wiring

- 1. General: Controller power and communications shall be via a common non-polar communications bus and shall operate at 30VDC.
- 2. Wiring: Control wiring shall be installed in a daisy chain configuration from indoor unit to indoor unit, and to the outdoor unit. Control wiring to remote controllers shall be run from the indoor unit terminal block to the controller associated with that unit.
- 3. Wiring Type: Wiring shall be 2-conductor (16 AWG), twisted, stranded, shielded wire as recommended by the manufacturer.
- 4. Control Wiring Raceway: See Division 16 ELECTRICAL.

C. FCU Controls: Provide each FCU with a touch screen controller. The touch screen controller shall be Mitsubishi model PAR-40MAA or approved substitute.

D. System Controls: Provide a centralized controller for the system. The centralized controller shall be Mitsubishi model or approved substitute.

2.2 CONTROLS

- A. Control System shall be factory provided, unit mounted with wall mounted thermostat. Control system shall be microprocessor based, with LCD display and shall include cool, fan, and auto modes.

2.3 PIPING AND FITTINGS

- A. General: All pipe and pipe fittings shall be suitable for 125 psi minimum working pressure, in accordance with the latest edition of ASTM and ANSI specifications as indicated.
- B. Refrigerant Piping: Seamless copper tubing, hard drawn, Type "K", conforming to ASTM B88 with wrought copper fittings and solder joints.
- C. Service: Condensate drain, hard drawn copper tubing, type "L", conforming to ASTM B88 with cast bronze soldered joint ANSI BI 6.18 fittings, 95-5 tin-antimony solder.

2.4 REFRIGERANT VALVES AND PIPING ACCESSORIES

- A. Provide valves designed, manufactured, and tested specifically for refrigerant service. Internal parts shall be removable for inspection or replacement without applying heat or breaking pipe connections. Threaded ends of valves shall conform to ANSI B2.1. Valves shall open when turned counterclockwise.
- B. Globe and Angle Valves: Forged brass, bronze alloy, forged steel, steel alloy, or ductile iron body with packed stem and seal cap. Packless type with handwheels and forged brass or bronze alloy bodies with brazing ends may be used in sizes up to and including 7/8-inch O.D. 1-1/4-inches and larger valves shall be tongue and groove flanged and shall have bolted bonnets. Valves one-inch and smaller shall have brazed or soldered ends. Refrigerant valves shall be of

the backseating type making it possible to repack the valves under pressure without removal from the line.

- C. Solenoid Valves: Solenoid valves shall be bronze or brass body, packless type, with stainless steel trim, rated for continuous-duty service, direct or pilot operated, provided with manual lift stem, and designed for use with the type of refrigerant used. Valves shall have a safe working pressure of 400 psi and a maximum operating pressure differential of at least 200 psi at 85 percent rated voltage. Valves shall have an operating pressure differential suitable for the refrigerant used. Valves shall have adequate capacity for the installation at a pressure drop suitable for the refrigerant used. The solenoid coil shall have moisture proof insulation and shall be UL approved and shall conform to ARI 760.
- D. Filter-Driers: Filter-driers shall conform to ARI 710. Sizes 1/2-inch and larger shall be the full flow, replaceable core type. Sizes smaller than 1/2-inch shall be the sealed type. Cores shall be a desiccant that will not plug, cake, dust, channel, or break down, and shall remove water, acid, and foreign material from the refrigerant. A strainer shall be an integral part of the unit and the assembly shall be fabricated of steel with copper solder end connections. The dryer shall be constructed so that none of the desiccant will pass into the refrigerant lines. The minimum bursting pressure shall be 1500 psi.
- E. Moisture Indicator Sight Glass: Moisture indicator sight glasses shall be provided where indicated or specified. The sight glass and moisture indicating element shall be removable. The moisture indicator shall have a reversible color indicator which has an easily distinguished color change. The sight glass shall be provided with a protective cover.

2.5 PIPE HANGERS AND SUPPORT

- A. Hangers and supports shall be of manufacturer and type specified or as indicated or accepted equivalent. Pipe supports and hangers in contact with copper piping shall be copper or plastic coated. Provide hot dipped galvanized pipe hangers and supports.
- B. Shields: Provide protection shields at hanger points of insulated piping. Shields shall be Crawford Fig. 25, Elcen Fig. 219, or Fee and Mason Fig. 81 or Grinnell Fig. 167.
- C. Supports:
 - 1. Piping shall be supported from structural steel, grouted CMU walls, or concrete slab only; piping shall not be supported in any manner from the roofing.
 - 2. Drilled-In Threaded Inserts: Where supports in slabs are required after concrete has been poured, Phillips "Redhead" drilled in threaded inserts shall be provided, installed in accordance with manufacturer's recommendations.
 - 3. Expansion Anchors and Power-Actuated Fasteners and Devices: Install lead shield anchors or power actuated sleeves, fasteners, and devices in accordance with manufacturer's recommendations. Powder actuated pins are not allowed to be "shot" into the concrete structure.
- D. Schedules: Pipe support spacing and sizes of pipe hanging suspension rods shall conform to the following table, except support spacing for copper tubing shall not exceed 6 feet on centers:

PIPE SIZE	SUPPORT SPACING
1/2-inch and 3/4-inch	Not over 6-ft.-6-inch
one-inch and 1-1/4-inch	Not over 8-ft.-6-inch
1-1/2-inch	Not over 10-ft.-0-inch
2-inch and 2-1/2-inch	Not over 12-ft.-0-inch

PIPE SIZE	ROD SIZE
Up to 2-inch	3/8-inch
2-1/2-inch	1/2-inch

- E. Supplementary Steel: Provide all necessary supplementary steel for proper support or attachment of hangers. Steel shall be painted with one coat of rust inhibiting primer.
- F. Single Hangers: Unless otherwise indicated, support single pipe runs as follows:
 - 1. Pipe 2-1/2-inch and Smaller: Split ring type hanger, Grinnell Fig. 104, Crawford Fig. 104, Fee and Mason Fig. 199, Elcen Fig. 92 or adjustable clevis hanger, or accepted equivalent.
 - 2. Pipe 3-inch and Larger: Clevis hanger conforming to MSS SP-69, Type 1.
- G. Riser Clamps: Carbon steel conforming to MSS SP-69 Type 8. Grinnell Fig. 261, Fee and Mason Fig. 241, or accepted equivalent.

PART 3 - EXECUTION

3.1 INSTALLATION

- A. Install equipment and components in a manner to ensure proper and sequential operation of equipment and equipment controls. Install equipment not covered in this section, or in manufacturer's instructions, as recommended by manufacturer's representative. Provide proper foundations for mounting of equipment, accessories, appurtenances, piping, and controls including, but not limited to, supports, vibration isolators, stands, guides, anchors, clamps, and brackets. Foundations for equipment shall conform to equipment manufacturer's recommendation, unless otherwise indicated. Set anchor bolts and sleeves using templates. Provide anchor bolts of adequate length and provide with welded-on plates on the head end embedded in the concrete. Level equipment bases, using jacks or steel wedges, and neatly grout-in with a non-shrinking type of grouting mortar. Locate equipment to allow working space for servicing including shaft removal, disassembling compressor cylinders and pistons, replacing, or adjusting drives, motors, or shaft seals, access to water heads and valves of shell and tube equipment, tube cleaning or replacement, access to automatic controls, refrigerant charging, lubrication, oil draining and working clearance under overhead lines. Provide electric isolation between dissimilar metals for the purpose of minimizing galvanic corrosion.
- B. Insulation: Provide insulation for refrigerant lines and condensate drain lines as well as any equipment or valves in accordance with Section 23 01 00 – Insulation.
- C. Electrical: Installing contractor shall coordinate electrical requirements and connections for all power feeds with electrical contractor. Refer to Division 26 for additional information.

3.2 PIPING INSTALLATION

- A. Pipe shall be cut accurately to measurements established at the jobsite and worked into place without springing or forcing, properly clearing all windows, doors, and other opening. Cutting or other weakening of the building structure to facilitate piping installation will not be permitted without written approval.
- B. Pipes shall be cut square, shall have burs removed by reaming, and shall be so installed as to permit free expansion and contraction without damage to joints or hangers. Install special swing and expansion joints in pipe lines where required. Filings, dust, or dirt shall be wiped from interior of the pipe before connections are made.
- C. Changes in direction shall be made with fittings. All piping shall be installed with sufficient pitch to insure adequate drainage.
- D. Screw joints shall be made with tapered threads properly cut conforming to requirements of NBS handbook H28. Joints shall be made perfectly tight with a stiff mixture of litharge and glycerin or other approved threaded joint compound applied with a brush to the male threads only. Not more than three threads shall show after the joint is made up.
- E. Flanges shall be suitable for the required operating pressures and temperature conditions. Gaskets shall be fiber plastic or other synthetic material suitable for water service. All nuts, bolts and washers used shall be galvanized for ferrous flanges and brass for copper or bronze flanges.
- F. Pipes passing through walls above and below grade or concrete floors shall be provided with pipe sleeves fitted into place at time of construction. Sleeves shall not be installed in structural members. Each sleeve shall extend through its respective wall or floor and shall be cut flush with each surface. Unless otherwise indicated, sleeves shall be of such size as to provide a minimum of 1/4 inch all around clearance between jacket over insulation and sleeves. Space between sleeve and piping shall be packed with twisted jute packing and then sealed with waterproof sealant. Penetrations through fire rated walls shall be sealed with approved fire stopping material.
- G. All piping systems shall be thoroughly tested for leaks prior to insulating pipe and closing up ceilings, walls and floors. Piping system shall be pressure tested to maximum allowable working pressure. Correct all leaks at no cost to the Owner.

3.3 SERVICES OF MANUFACTURER'S REPRESENTATIVE

- A. Upon completion of the installation, provide the services of the Manufacturer's representative to inspect and verify proper operations of the system components and associated controls.
- B. The installing contractor shall assist the manufacturer's representative in their inspection by having an onsite technician available with appropriate diagnostic tools, materials and equipment, as required, for the duration of the inspection process.

3.4 START-UP AND INITIAL OPERATIONAL TESTS

- A. Test the air conditioning systems and systems components for proper operation. Adjust safety and automatic control instruments as necessary to ensure proper operation and sequence. Conduct operational tests for not less than 8 hours.

3.5 EQUIPMENT START-UP AND TESTING

- A. Test, adjust and balance the system in accordance with the requirements in Section 15950 Testing, Adjusting and Balancing for HVAC.
- B. Testing shall include verification that the Centralized Controller is operating in accordance with the manufacturer's specifications.

3.6 TRAINING OF OPERATING PERSONNEL

- A. Provide the services of a manufacturer's representative to train the facilities personnel in the operation of the HVAC system. Training shall be a minimum of 4 hours on-site and shall include operation of the system's central controller.

END OF SECTION 23 81 00

SECTION 26 00 00 - COMMON WORK RESULTS FOR ELECTRICAL

PART 1 - GENERAL

1.1 SUMMARY

- A. Section Includes: Basic Electrical Requirements specifically applicable to Division 26 Sections.

1.2 SCOPE OF WORK

- A. The specifications describe the quality and character of the materials and methods of installation.
- B. The drawings and these specifications are complementary to each other in that all apparatus, materials and equipment outlined in the Drawings and/or specified herein shall be considered essential to the contract.
- C. The drawings include plans of the building, with diagrammatic layouts showing approximate locations of equipment and devices. Before installing, study adjacent architectural features, and make installation in the most logical manner in accordance with Code and Regulatory Requirements.
- D. The electrical symbols, notes, instructions and schedules on the drawings are included as part of these specifications.

1.3 WORK SCHEDULE AND SEQUENCE

- A. Install work in stages to accommodate the Owner's operational requirements. Coordinate schedule and hours of operation with the Owner prior to start of construction.
- B. Coordinate installation sequence with detailed coordination shop drawings provided under Division 1.

1.4 REFERENCES

- A. ANSI/NFPA 70 - National Electrical Code
- B. National Electrical Safety Code

1.5 SUBMITTALS

- A. Conform to Division 1 requirements.
- B. Include a transmittal form clearly indicating the Project, the name of the Contractor and the contents of the submittal.

- C. Include Contractor's stamp and signature indicating that the submittal has been reviewed and conforms to Contract Documents. Submittals without Contractor's stamp will be returned without review.
- D. Identify deviations from Contract Documents, including variations and limitations. Review of a submittal does not constitute acceptance of deviations from the Contract Documents, unless such deviation is clearly indicated as such on the submittal, and specifically accepted as such.
- E. Submit shop drawings and product data, grouped to include complete systems, products and accessories in a single package.
 - 1. Reference catalog cuts and brochures of products to proper paragraph in specifications. Furnish numerical index by specification article number, listing product name, catalog number and reference to page number of submittal brochure.
 - 2. Arrange the submittals in the same sequence as the specifications and reference in the upper right-hand corner, the particular specification provision for which each submittal is intended.
 - 3. Cross reference individual catalog numbers of substitute products to number of specified materials.
 - 4. Submit manufacturer's certification that equipment meets or exceeds the minimum requirements as specified.
 - 5. Where materials, equipment and installations are specified to conform with societies or agencies such as ANSI, NECA, etc., submit certification of such compliance.
 - 6. The submittal shall be complete and with catalog data and information properly marked to show, among other things, material capacity and performance to meet capacities or performance as specified or indicated
 - 7. Mark dimensions and values in units to match those specified.
- F. Review of the submittal is only for general conformance with design concept of project and general compliance with information given in the contract documents. The Contractor is responsible for confirmation and correlation of the dimensions, quantities and sizes, for information that pertains to fabrication methods or construction techniques, and for coordination work of all trades.
- G. For items which are not manufactured and which have to be specifically fabricated, submit shop drawings and details.
- H. Ordering of equipment prior to approval of submittals is done entirely at the risk of the Contractor.

1.6 COORDINATION SUBMITTALS

- A. Develop detailed coordination shop drawings in conjunction with other trades, where required by complex and/or congested spaces, to minimize conflict, to allow for correct sequence of installation, and to provide all required clearances. See Division 1 for expanded requirements.

1.7 PROJECT RECORD DRAWINGS

- A. Conform to Division 1 requirements.

- B. Keep an accurate record of the work under this Contract, as it progresses, to be available for inspection at all times. See individual Division 26 Sections for specific requirements.
- C. Upon completion of the work, transfer all changes and information onto a new set of reproducible drawings in an orderly and legible manner.

1.8 QUALITY ASSURANCE

- A. For actual fabrication, installation and testing of the work, use only trained and experienced workers completely familiar with the equipment and materials, and the manufacturer's installation requirements.
- B. Include the services of experienced superintendents for each sub-section who shall be constantly in charge of the work, together with the qualified journeymen, helpers, and laborers, required to properly unload, install, connect, adjust, start, operate and test the work involved, including equipment and materials furnished by others.
- C. Perform the work under this section shall be in cooperation with the work of other trades to prevent conflict or interference and to aid in timely completion of the overall project.

1.9 DELIVERY, STORAGE AND HANDLING

- A. Storage and handling of equipment and/or systems for project, whether it be onsite or offsite, shall be addressed by the Contractor's IAQ Management Plan. Plan must meet or exceed the recommended control measures of the "Sheet Metal and Air Conditioning National Contractors Association (SMACNA) IAQ Guidelines for Occupied Buildings Under Construction, 2nd Edition 2007, ANSI/SMACNA 008-2008 (Chapter 3)."

1.10 REGULATORY REQUIREMENTS

- A. Conform to International Building Code, International Fire Code, and NFPA 101 Life Safety Code
- B. Electrical: Conform to NFPA 70.

1.11 FIELD CONDITIONS

- A. Visit the project site and become familiar with field conditions including accessibility and physical obstructions. Bid submission indicates familiarity with, and acceptance of, field conditions.
- B. Separate Sections cover site, architectural and mechanical Work. Study the complete set of contract documents to become familiar with the entire Project including site, architectural and structural features and systems as related to Work in this Division. Pay special attention to Divisions featuring equipment requiring electrical interface including Owner-furnished equipment, elevator equipment, and mechanical systems (plumbing, hvac, fire sprinkler, controls).

- C. Study and become familiar with any and all peculiarities and limitations of the spaces available for the installation of all work and materials furnished and installed under the contract. Exercise due and particular caution to determine that all parts of work are made quickly and easily accessible.
- D. If any conflicts occur which necessitate departures from the Drawings, submit details of departures and reasons therefore for written approval. Do not install the affected equipment or related impacted wiring until approval is received.
- E. Should there be omissions or discrepancies in the plans and specifications, or discrepancies from actual site conditions, bring them to the attention of the Engineer ten (10) working days in advance of the date of bid opening so that corrections or clarifications can be made.
- F. Install Work in locations shown on Drawings, unless prevented by Project conditions. Coordinate work with that of other trades. Verify that adjacent and related construction conforms to contract documents and to coordination shop drawings.
- G. If Project conditions, including changes initiated by other trades or discovery of conditions unknown at time of bid, require rearrangement of work, mark such changes on as-built drawings. If Project conditions require unspecified materials and methods, submit Request For Information (RFI) to the Architect with drawings showing the proposed alternative materials or methods. Obtain permission of the Architect before proceeding.
- H. All RFIs must include a proposed solution. RFIs submitted without proposed solutions will be returned without review.

1.12 COORDINATION

- A. Provide and coordinate all information, drawings or layouts of equipment or work under this section which affect the work of the other trades.
- B. In case changes in the indicated locations or arrangements are necessary due to developed conditions in the construction, or rearrangement of furnishings, or equipment, these changes shall be made without extra cost to the Owner, provided the change is ordered before work directly connected is installed, and no extra materials are required.

1.13 EXISTING UTILITIES

- A. The location of utilities shown on the plans are the best known information available at time of design. Contact the appropriate agencies and confirm the information and make arrangements for connection thereto, prior to excavation and installation of any piping or systems.
- B. Perform exploratory excavation and/or use locate service as needed to confirm locations of existing underground utilities.

1.14 PROJECT SITE VISITS

- A. Periodic visits to the project site by the Architect/Engineer are for the express purpose of verifying compliance with the contract documents. Such site visits shall not be construed as

construction supervision, i.e., the Architect/Engineer assumes no responsibility for providing a safe place for the performance of the work by the Contractor or the Contractor's employees or the safety of the supplies of the Contractor. Neither shall such site visits relieve the Contractor of the responsibility for the discovery of his own errors and the correction of them, nor of the responsibility of properly performing the work.

PART 2 - PRODUCTS

2.1 MATERIALS AND EQUIPMENT

- A. See Drawings and individual Sections of Division 26.

2.2 SUBSTITUTIONS

- A. Conform to Division 1 requirements.
- B. Conformance to construction documents is the responsibility of the party making the substitution, regardless of approval.
- C. Layout on drawings, including space allotted for clearances, access, etc., is based on performance and physical attributes of equipment specified and/or scheduled on plans. Coordinate with other systems, subsystems and trades as required when using substituted materials or equipment.
- D. If the use of substitute materials or equipment requires alternate arrangement of equipment, fixtures, devices, wiring or accessories, prepare drawings showing proposed changes. Obtain permission of the Architect before proceeding.
- E. If the use of substitute materials or equipment results in different performance than that provided by the specified materials or equipment, adjust Work as required to provide parity performance, at no additional cost to the Government. Obtain permission of the Engineer before proceeding.
- F. If the use of substitute materials or equipment results in an increase in the cost, including changes to the Work of other trades, pay for any said increase in cost.
- G. See Drawings and individual Sections of Division 26 for further specific information required for substitutions.

2.3 VALUE ENGINEERING

- A. Conform to SUBSTITUTIONS above.

2.4 REFERENCE TO SCHEDULES AND DRAWINGS

- A. Refer to schedules on drawings for equipment identification, features, ratings, configuration, performance and design requirements.

PART 3 - EXECUTION

3.1 INSTALLATION

- A. See Drawings and individual Sections of Division 26. In addition the following general requirements shall apply:
 - 1. Obtain Manufacturer's printed installation instruction to aid in properly executing work of installing equipment whenever such instructions are available. Submit copies of such instructions to the Architect prior to time of installation.
 - 2. Install equipment in a neat and workmanlike manner. Align, level and adjust for satisfactory appearance and operation. Install so that connection and disconnection of wiring and accessories can be made readily, and so that all parts are easily accessible for inspection, operation, maintenance and repair.

END OF SECTION 26 00 00

SECTION 260519 - LOW-VOLTAGE ELECTRICAL POWER CONDUCTORS AND CABLES

PART 1 - GENERAL

1.1 SUMMARY

- A. Section Includes:
 - 1. Copper building wire.
 - 2. Connectors and splices.

1.2 ACTION SUBMITTALS

- A. Product Data:
 - 1. Copper building wire.
 - 2. Connectors and splices.
- B. Product Schedule: Indicate type, use, location, and termination locations.

1.3 INFORMATIONAL SUBMITTALS

- A. Field quality-control reports.

PART 2 - PRODUCTS

2.1 COPPER BUILDING WIRE

- A. Description: Flexible, insulated and uninsulated, drawn copper current-carrying conductor with an overall insulation layer or jacket, or both, rated 600 V or less.
- B. Standards:
 - 1. Listed and labeled as defined in NFPA 70, by a qualified testing agency, and marked for intended location and use.
 - 2. Conductor and Cable Marking: Comply with wire and cable marking according to UL's "Wire and Cable Marking and Application Guide."
- C. Conductors: Copper, complying with ASTM B3 for bare annealed copper and with ASTM B8 for stranded conductors.
- D. Conductor Insulation:
 - 1. Type THHN. Comply with UL 83.
 - 2. Type THW. Comply with NEMA WC-70/ICEA S-95-658 and UL 83.

3. Type XHHW-2. Comply with UL 44.

2.2 CONNECTORS AND SPLICES

- A. Description: Factory-fabricated connectors, splices, and lugs of size, ampacity rating, material, type, and class for application and service indicated; listed and labeled as defined in NFPA 70, by a qualified testing agency, and marked for intended location and use.
- B. Jacketed Cable Connectors: For steel and aluminum jacketed cables, zinc die-cast with set screws, designed to connect conductors specified in this Section.
- C. Lugs: One piece, seamless, designed to terminate conductors specified in this Section.
 1. Material: Copper or Bronze.
 2. Type: One or Two hole with standard or long barrels.
 3. Termination: Compression.

PART 3 - EXECUTION

3.1 CONDUCTOR MATERIAL APPLICATIONS

- A. Feeders:
 1. Copper; solid for No. 10 AWG and smaller; stranded for No. 8 AWG and larger.
- B. Branch Circuits:
 1. Copper:
 - a. Solid for No. 10 AWG and smaller; stranded for No. 8 AWG and larger.
 - b. Solid for No. 12 AWG and smaller; stranded for No. 10 AWG and larger.

3.2 CONDUCTOR INSULATION AND MULTICONDUCTOR CABLE APPLICATIONS AND WIRING METHODS

- A. Feeders: Type XHHW-2, single conductors in raceway.
- B. Branch Circuits: Type THHN/THWN-2, single conductors in raceway.

3.3 INSTALLATION, GENERAL

- A. Conceal cables in finished walls, ceilings, and floors unless otherwise indicated.
- B. Complete raceway installation between conductor and cable termination points in accordance with Section 260533.13 "Conduits for Electrical Systems" prior to pulling conductors and cables.

- C. Use manufacturer-approved pulling compound or lubricant where necessary; compound used must not deteriorate conductor or insulation. Do not exceed manufacturer's recommended maximum pulling tensions and sidewall pressure values.
- D. Use pulling means, including fish tape, cable, rope, and basket-weave wire/cable grips, that will not damage cables or raceway.
- E. Install exposed cables parallel and perpendicular to surfaces of exposed structural members, and follow surface contours where possible.
- F. Support cables according to Section 260529 "Hangers and Supports for Electrical Systems."
- G. Complete cable tray systems installation according to Section 260536 "Cable Trays for Electrical Systems" prior to installing conductors and cables.

3.4 CONNECTIONS

- A. Tighten electrical connectors and terminals according to manufacturer's published torque-tightening values. If manufacturer's torque values are not indicated, use those specified in UL 486A-486B.
- B. Make splices, terminations, and taps that are compatible with conductor material and that possess equivalent or better mechanical strength and insulation ratings than unspliced conductors.

3.5 IDENTIFICATION

- A. Identify and color-code conductors and cables according to Section 260553 "Identification for Electrical Systems."
- B. Identify each spare conductor at each end with identity number and location of other end of conductor, and identify as spare conductor.

3.6 SLEEVE AND SLEEVE-SEAL INSTALLATION FOR ELECTRICAL PENETRATIONS

- A. Install sleeves and sleeve seals at penetrations of exterior floor and wall assemblies. Comply with requirements in Section 260544 "Sleeves and Sleeve Seals for Electrical Raceways and Cabling."

3.7 FIRESTOPPING

- A. Apply firestopping to electrical penetrations of fire-rated floor and wall assemblies to restore original fire-resistance rating of assembly according to Section 078413 "Penetration Firestopping."

3.8 FIELD QUALITY CONTROL

A. Tests and Inspections:

1. Perform each of the following visual and electrical tests:
 - a. Inspect exposed sections of conductor and cable for physical damage and correct connection according to the single-line diagram.
 - b. Test bolted connections for high resistance using one of the following:
 - 1) A low-resistance ohmmeter.
 - 2) Calibrated torque wrench.
 - 3) Thermographic survey.
 - c. Inspect compression-applied connectors for correct cable match and indentation.
 - d. Inspect for correct identification.
 - e. Inspect cable jacket and condition.
 - f. Insulation-resistance test on each conductor for ground and adjacent conductors. Apply a potential of 500 V(dc) for 300 V rated cable and 1000 V(dc) for 600 V rated cable for a one-minute duration.
 - g. Continuity test on each conductor and cable.
 - h. Uniform resistance of parallel conductors.

B. Cables will be considered defective if they do not pass tests and inspections.

C. Prepare test and inspection reports to record the following:

1. Procedures used.
2. Results that comply with requirements.
3. Results that do not comply with requirements, and corrective action taken to achieve compliance with requirements.

END OF SECTION 260519

SECTION 260526 - GROUNDING AND BONDING FOR ELECTRICAL SYSTEMS

PART 1 - GENERAL

1.1 SUMMARY

A. Section Includes:

1. Grounding and bonding conductors.
2. Grounding and bonding clamps.
3. Grounding and bonding bushings.
4. Grounding and bonding connectors.

B. Related Requirements:

1. Section 260553 "Identification for Electrical Systems" specifies electrical equipment labels and warning signs installed by this Section.

1.2 ACTION SUBMITTALS

A. Product Data: For each type of product.

1. Product Listing: Include copy of unexpired approval letter, on letterhead of qualified electrical testing agency, certifying product's compliance with specified listing criteria.
 - a. If listed manufacturer differs from selling manufacturer, indicate relationship between entities on submittal. Clearly indicate which entity warrants product performance and fitness for purpose.
 - b. Listing criteria identified in approval letter must match specified listing criteria. UL label indicating approval of equipment's enclosure is not considered approval of equipment for intended application.
 - c. Product identification in approval letter must match product branding and model numbers in submittal. Approval letters for discontinued or superseded products are unacceptable for submitted product.

B. Field quality-control reports.

1.3 INFORMATIONAL SUBMITTALS

A. Manufacturer's published instructions.

PART 2 - PRODUCTS

2.1 PERFORMANCE REQUIREMENTS

- A. Regulatory Requirements: Products or components listed and labeled in accordance with NFPA 70, by qualified electrical testing laboratory recognized by authorities having jurisdiction, and marked for intended location and application.

2.2 GROUNDING AND BONDING CONDUCTORS

- A. Equipment Grounding Conductor:
 - 1. Standard Features: 600 V, THHN/THWN-2, copper wire or cable, green color, in accordance with Section 260519 "Low-Voltage Electrical Power Conductors and Cables."

2.3 GROUNDING AND BONDING CLAMPS

- A. Description: Clamps suitable for attachment of grounding and bonding conductors to grounding electrodes, pipes, tubing, and rebar. Grounding and bonding clamps specified in this article are also suitable for use with communications applications.
- B. UL KDER and KDSH - Hex-Fitting-Type Pipe and Rod Grounding and Bonding Clamp:
 - 1. Source Limitations: Obtain products from single manufacturer.
 - 2. Listing Criteria: Investigated, labeled, and marked by qualified electrical testing laboratory in accordance with guide information and standards specified for the following UL product categories:
 - a. Grounding and Bonding Equipment: UL CCN KDER; including UL 467.
 - b. Grounding and Bonding Equipment for Communications: UL CCN KDSH; including UL 467.
 - 3. Standard Features:
 - a. Two pieces with zinc-plated bolts.
 - b. Clamp Material: Silicon bronze.
 - c. Listed for outdoor use.
- C. UL KDER and KDSH - U-Bolt-Type Pipe and Rod Grounding and Bonding Clamp:
 - 1. Source Limitations: Obtain products from single manufacturer.
 - 2. Listing Criteria: Investigated, labeled, and marked by qualified electrical testing laboratory in accordance with guide information and standards specified for the following UL product categories:
 - a. Grounding and Bonding Equipment: UL CCN KDER; including UL 467.
 - b. Grounding and Bonding Equipment for Communications: UL CCN KDSH; including UL 467.

3. Standard Features:
 - a. Clamp Material: Brass.
 - b. Listed for outdoor use.

D. UL KDER and KDSH - Strap-Type Pipe and Rod Grounding and Bonding Clamp:

1. Source Limitations: Obtain products from single manufacturer.
2. Listing Criteria: Investigated, labeled, and marked by qualified electrical testing laboratory in accordance with guide information and standards specified for the following UL product categories:
 - a. Grounding and Bonding Equipment: UL CCN KDER; including UL 467.
 - b. Grounding and Bonding Equipment for Communications: UL CCN KDSH; including UL 467.
3. Standard Features:
 - a. Clamp Material: Copper.
 - b. Listed for outdoor use.

E. UL KDER - Beam Grounding and Bonding Clamp:

1. Source Limitations: Obtain products from single manufacturer.
2. Listing Criteria: Investigated, labeled, and marked by qualified electrical testing laboratory in accordance with guide information and standards specified for the following UL product categories:
 - a. Grounding and Bonding Equipment: UL CCN KDER; including UL 467.
 - b. Grounding and Bonding Equipment for Communications: UL CCN KDSH; including UL 467.
3. Standard Features: Mechanical-type, terminal, ground wire access from four directions; with dual, tin-plated or silicon bronze bolts.

2.4 GROUNDING AND BONDING BUSHINGS

- A. Description: Bonding bushings connect conduit fittings, tubing fittings, threaded metal conduit, and unthreaded metal conduit to metal boxes and equipment enclosures, and have one or more bonding screws intended to provide electrical continuity between bushing and enclosure. Grounding bushings have provision for connection of bonding or grounding conductor and may or may not also have bonding screws.
- B. UL KDER - Bonding Bushing:
 1. Source Limitations: Obtain products from single manufacturer.
 2. Listing Criteria: Investigated, labeled, and marked by qualified electrical testing laboratory in accordance with guide information and standards specified for the following UL product categories:
 - a. Grounding and Bonding Equipment: UL CCN KDER; including UL 467.

3. Standard Features: Threaded bushing with insulated throat.

C. UL KDER - Grounding Bushing:

1. Source Limitations: Obtain products from single manufacturer.
2. Listing Criteria: Investigated, labeled, and marked by qualified electrical testing laboratory in accordance with guide information and standards specified for the following UL product categories:
 - a. Grounding and Bonding Equipment: UL CCN KDER; including UL 467.
3. Standard Features: Threaded bushing with insulated throat and mechanical-type wire terminal.

2.5 GROUNDING AND BONDING CONNECTORS

A. UL KDER - Crimped Pressure-Type Grounding and Bonding Cable Connector:

1. Source Limitations: Obtain products from single manufacturer.
2. Listing Criteria: Investigated, labeled, and marked by qualified electrical testing laboratory in accordance with guide information and standards specified for the following UL product categories:
 - a. Grounding and Bonding Equipment: UL CCN KDER; including UL 467.
 - b. Grounding and Bonding Equipment for Communications: UL CCN KDSH; including UL 467.
3. Standard Features: Crimp-and-compress connectors that bond to conductor when connector is compressed around conductor.
 - a. Copper, C and H shaped.

B. UL KDER - Split-Bolt Pressure-Type Grounding and Bonding Cable Connector:

1. Source Limitations: Obtain products from single manufacturer.
2. Listing Criteria: Investigated, labeled, and marked by qualified electrical testing laboratory in accordance with guide information and standards specified for the following UL product categories:
 - a. Grounding and Bonding Equipment: UL CCN KDER; including UL 467.
 - b. Grounding and Bonding Equipment for Communications: UL CCN KDSH; including UL 467.
3. Standard Features: Bolts that surround cable and bond to cable under compression when nut is tightened.
 - a. Copper.

PART 3 - EXECUTION

3.1 SELECTION OF GROUNDING AND BONDING PRODUCTS

A. Grounding and Bonding Conductors:

1. Provide solid conductor for 8 AWG and smaller, and stranded conductors for 6 AWG and larger unless otherwise indicated.
2. Bonding Cable: 28 kcmil, 14 strands of 17 AWG conductor, 1/4 inch (6 mm) in diameter.
3. Bonding Conductor: 4 AWG or 6 AWG, stranded conductor.
4. Bonding Jumper: Copper tape, braided conductors terminated with copper ferrules; 1-5/8 inch (41 mm) wide and 1/16 inch (1.6 mm) thick.

B. Grounding and Bonding Connectors:

1. Pipe and Equipment Grounding Conductor Terminations: Bolted connectors.

3.2 INSTALLATION OF GROUNDING AND BONDING

A. Comply with manufacturer's published instructions.

B. Reference Standards:

1. Electrical Construction: ICC IBC, ICC IFB, NFPA 1, NFPA 70, and NECA NEIS 1.
2. Electrical Maintenance: NFPA 70B.
3. Electrical Safety: NFPA 70E.
4. Grounding and Bonding: NECA NEIS 331 and Article 250 of NFPA 70.
5. Work in Confined Spaces: NFPA 350.

C. Special Techniques:

1. Grounding and Bonding Conductors:

- a. Route along shortest and straightest paths possible unless otherwise indicated or required by Code. Avoid obstructing access or placing conductors where they may be subjected to strain, impact, or damage.

2. Grounding and Bonding Connectors: Make connections so possibility of galvanic action or electrolysis is minimized. Select connectors, connection hardware, conductors, and connection methods so metals in direct contact are galvanically compatible.

- a. Use electroplated or hot-tin-coated materials to ensure high conductivity and to make contact points closer in order of galvanic series.
- b. Make connections with clean, bare metal at points of contact.
- c. Coat and seal connections having dissimilar metals with inert material to prevent future penetration of moisture to contact surfaces.
- d. Bonding Straps and Jumpers: Install in locations accessible for inspection and maintenance except where routed through short lengths of conduit.

- 1) Bonding to Structure: Bond straps directly to basic structure, taking care not to penetrate adjacent parts.
 - 2) Bonding to Equipment Mounted on Vibration Isolation Hangers and Supports: Install bonding so vibration is not transmitted to rigidly mounted equipment.
3. Equipment Grounding and Bonding:
 - a. Install insulated equipment grounding conductors with feeders and branch circuits.

3.3 FIELD QUALITY CONTROL FOR GROUNDING AND BONDING

A. Tests and Inspections:

1. After installing grounding system but before permanent electrical circuits have been energized, test for compliance with requirements.
2. Inspect physical and mechanical condition. Verify tightness of accessible, bolted, electrical connections with calibrated torque wrench in accordance with manufacturer's published instructions.

B. Nonconforming Work:

1. Grounding system will be considered defective if it does not pass tests and inspections.
2. Remove and replace defective components and retest.

C. Collect, assemble, and submit test and inspection reports.

3.4 PROTECTION

- A. After installation, protect grounding and bonding cables and equipment from construction activities. Remove and replace items that are contaminated, defaced, damaged, or otherwise caused to be unfit for use prior to acceptance by Owner.

END OF SECTION 260526

SECTION 260529 - HANGERS AND SUPPORTS FOR ELECTRICAL SYSTEMS

PART 1 - GENERAL

1.1 SUMMARY

A. Section Includes:

1. Support systems.
2. Mounting, anchoring, and attachment components.
3. Installation of fabricated metal supports.

B. Related Requirements:

1. Section 018116 "Facility Environmental Requirements" specifies basis-of-design environmental conditions and performance criteria that are applicable to product selection and installation of the Work on the Project.
2. Section 018123 "Facility Seismic and Wind Criteria" specifies basis-of-design seismic and wind criteria for nonstructural components on the Project.
3. Section 260548 "Vibration and Seismic Controls for Electrical Systems" specifies vibration controls, seismic restraints, and wind restraints referenced by this Section.

1.2 ACTION SUBMITTALS

A. Product Data: For each type of product.

1. Include construction details, material descriptions, dimensions of individual components and profiles, and finishes for the following:
 - a. Slotted support systems, hardware, and accessories.
 - b. Clamps.
 - c. Hangers.
 - d. Sockets.
 - e. Eye nuts.
 - f. Fasteners.
 - g. Anchors.
 - h. Saddles.
 - i. Brackets.
2. Include rated capacities and furnished specialties and accessories.

1.3 INFORMATIONAL SUBMITTALS

A. Welding certificates.

PART 2 - PRODUCTS

2.1 PERFORMANCE REQUIREMENTS

- A. Surface-Burning Characteristics: Comply with ASTM E84; testing by a qualified testing agency. Identify products with appropriate markings of applicable testing agency.
 - 1. Flame Rating: Class 1.
 - 2. Self-extinguishing according to ASTM D635.

2.2 SUPPORT SYSTEMS

- A. Steel Slotted Support Systems:
 - 1. Standard Features: Preformed steel channels and angles with minimum 13/32 inch (10 mm) diameter holes at a maximum of 8 inch (200 mm) on center in at least one surface.
 - a. Referenced Standard: MFMA-4 factory-fabricated components for field assembly.
 - b. Material for Channel, Fittings, and Accessories: Galvanized steel.
 - c. Channel Width: Selected for applicable load criteria.
 - d. Metallic Coatings: Hot-dip galvanized after fabrication and applied according to MFMA-4.
 - e. Protect finishes on exposed surfaces from damage by applying a strippable, temporary protective covering before shipping.
- B. Conduit and Cable Support Devices:
 - 1. Standard Features: Steel and malleable-iron hangers, clamps, and associated fittings, designed for types and sizes of raceway or cable to be supported.
- C. Structural Steel for Fabricated Supports and Restraints:
 - 1. Standard Features: ASTM A36/A36M steel plates, shapes, and bars; black and galvanized.

2.3 MOUNTING, ANCHORING, AND ATTACHMENT COMPONENTS

- A. Powder-Actuated Fasteners:
 - 1. Standard Features: Threaded-steel stud, for use in hardened portland cement concrete, steel, or wood, with tension, shear, and pullout capacities appropriate for supported loads and building materials where used.
- B. Mechanical-Expansion Anchors:
 - 1. Standard Features: Insert-wedge-type, zinc-coated steel, for use in hardened portland cement concrete, with tension, shear, and pullout capacities appropriate for supported loads and building materials where used.

- C. Concrete Inserts:
 - 1. Standard Features: Steel or malleable-iron, slotted support system units are similar to MSS Type 18 units and comply with MFMA-4 or MSS SP-58.
- D. Clamps for Attachment to Steel Structural Elements:
 - 1. Standard Features: MSS SP-58 units are suitable for attached structural element.
- E. Through Bolts:
 - 1. Standard Features: Structural type, hex head, and high strength. Comply with ASTM F3125/F3125M, **Grade A325** (**Grade A325M**).
- F. Toggle Bolts:
 - 1. Standard Features: All steel springhead type.
- G. Hanger Rods:
 - 1. Standard Features: Threaded steel.

PART 3 - EXECUTION

3.1 SELECTION OF HANGERS AND SUPPORTS

- A. Maximum Support Spacing and Minimum Hanger Rod Size for Raceways: Space supports for EMT, IMC, and ERMC as required by NFPA 70. Minimum rod size must be **1/4 inch** (**6 mm**) in diameter.
- B. Multiple Raceways or Cables: Install trapeze-type supports fabricated with steel slotted support system, sized so capacity can be increased by at least 25 percent in future without exceeding specified design load limits.
 - 1. Secure raceways and cables to these supports with two-bolt conduit clamps or single-bolt conduit clamps using spring friction action for retention in support channel.
- C. Spring-steel clamps designed for supporting single conduits without bolts may be used for **1-1/2 inch** (**38 mm**) and smaller raceways serving branch circuits and communication systems above suspended ceilings, and for fastening raceways to trapeze supports.

3.2 INSTALLATION OF HANGERS AND SUPPORTS

- A. Comply with manufacturer's published instructions.
- B. Reference Standards for Installation: Unless more stringent installation requirements are specified in the Contract Documents or manufacturer's published instructions, comply with the following:

1. Electrical Construction: ICC IBC, ICC IFC, NFPA 1, NFPA 70, and NECA NEIS 1.
2. Hot Work: NFPA 51B.
3. Work in Confined Spaces: NFPA 350.
4. Installation of Steel Conduit: NECA NEIS 101.

C. Special Installation Techniques:

1. Raceway Support Methods: In addition to methods described in NECA NEIS 1, EMT and ERMC may be supported by openings through structure members, in accordance with NFPA 70.
2. Strength of Support Assemblies: Where not indicated, select sizes of components so strength will be adequate to carry present and future static loads within specified loading limits. Minimum static design load used for strength determination must be weight of supported components plus 200 lb (90 kg).
3. Mounting and Anchorage of Surface-Mounted Equipment and Components: Anchor and fasten electrical items and their supports to building structural elements by the following methods unless otherwise indicated by code:
 - a. To Wood: Fasten with lag screws or through bolts.
 - b. To New Concrete: Bolt to concrete inserts.
 - c. To Masonry: Approved toggle-type bolts on hollow masonry units and expansion anchor fasteners on solid masonry units.
 - d. To Existing Concrete: Expansion anchor fasteners.
 - e. Instead of expansion anchors, powder-actuated driven threaded studs provided with lock washers and nuts may be used in existing standard-weight concrete 4 inch (100 mm) thick or greater. Do not use for anchorage to lightweight-aggregate concrete or for slabs less than 4 inch (100 mm) thick.
 - f. To Steel: Welded threaded studs complying with AWS D1.1/D1.1M, with lock washers and nuts.
 - g. To Light Steel: Sheet metal screws.
 - h. Items Mounted on Hollow Walls and Nonstructural Building Surfaces: Mount cabinets, panelboards, disconnect switches, control enclosures, pull and junction boxes, transformers, and other devices on slotted-channel racks attached to substrate by means that comply with seismic-restraint strength and anchorage requirements.
4. Drill holes for expansion anchors in concrete at locations and to depths that avoid the need for reinforcing bars.

D. Interfaces with Other Work:

1. Provide vibration and seismic controls with hangers and supports.
2. Touchup Finishes:
 - a. Clean field welds and abraded areas of shop paint. Paint exposed areas immediately after erecting hangers and supports. Use same materials as used for shop painting. Comply with SSPC-PA 1 requirements for touching up field-painted surfaces.
 - 1) Apply paint by brush or spray to provide minimum dry film thickness of 2.0 mils (0.05 mm).

- b. Cleaning and touchup painting of field welds, bolted connections, and abraded areas of shop paint on miscellaneous metal.
 - c. Galvanized Surfaces: Clean welds, bolted connections, and abraded areas and apply galvanizing-repair paint to comply with ASTM A780.
- 3. Installation of Fabricated Metal Supports:
 - a. Provide site-fabricated metal supports.
 - b. Cut, fit, and place miscellaneous metal supports accurately in location, alignment, and elevation to support and anchor electrical materials and equipment.
 - c. Field Welding: Comply with AWS D1.1/D1.1M. Submit welding certificates.

END OF SECTION 260529

SECTION 260533.13 - CONDUITS FOR ELECTRICAL SYSTEMS

PART 1 - GENERAL

1.1 SUMMARY

A. Section Includes:

1. Type EMT duct raceways and elbows.
2. Type ERMC duct raceways, elbows, couplings, and nipples.
3. Type FMC duct raceways.
4. Type LFMC duct raceways.
5. Fittings for conduit, tubing, and cable.
6. Joint compounds.
7. Solvent cements.

B. Related Requirements:

1. Section 018116 "Facility Environmental Requirements" specifies temperature, humidity, acoustical, and other field conditions applicable to the Work specified in this Section.
2. Section 018123 "Facility Seismic and Wind Criteria" specifies seismic hazard, wind hazard, other structural load design conditions applicable to the Work specified in this Section.
3. Section 078413 "Penetration Firestopping" specifies firestopping referenced by this Section.
4. Section 260529 "Hangers and Supports for Electrical Systems" specifies conduit hangers and supports referenced by this Section.
5. Section 260553 "Identification for Electrical Systems" specifies electrical equipment labels.

1.2 REFERENCES

A. Abbreviations and Acronyms for Electrical Raceway Types:

1. EMT: Electrical metallic tubing.
2. EMT-S: Steel electrical metallic tubing.
3. ERMC: Electrical rigid metal conduit.
4. ERMC-S: Steel electrical rigid metal conduit.
5. FMC: Flexible metal conduit.
6. FMC-S: Steel flexible metal conduit.
7. LFMC: Liquidtight flexible metal conduit.
8. LFMC-S: Steel liquidtight flexible metal conduit.
9. RGS: See ERMC-S-G.
10. RMC: See ERMC.

B. Definitions:

1. Conduit: A structure containing one or more duct raceways.
2. Duct Raceway: A single enclosed raceway for conductors or cable.

1.3 ACTION SUBMITTALS

- A. Product Data: For each type of product.
- B. Field quality-control reports.

1.4 INFORMATIONAL SUBMITTALS

- A. Manufacturer's published instructions.

PART 2 - PRODUCTS

2.1 PERFORMANCE REQUIREMENTS

- A. Regulatory Requirements: Products or components listed and labeled in accordance with NFPA 70, by qualified electrical testing laboratory recognized by authorities having jurisdiction, and marked for intended location and application.

2.2 TYPE EMT DUCT RACEWAYS AND ELBOWS

- A. UL FJMX - Steel Electrical Metal Tubing (EMT-S) and Elbows:
 1. Listing Criteria: Investigated, labeled, and marked by qualified electrical testing laboratory in accordance with guide information and standards specified for the following UL product categories:
 - a. UL CCN FJMX; including UL 797.
 2. Standard Features:
 - a. Material: Steel.
 - b. Exterior Coating: [**Zinc**] [**Alternate corrosion-resistant coating**].
 - c. Interior Coating: [**Zinc with organic top coating**] [**Zinc**] [**Organic coating**].
 - d. Minimum Trade Size: [**Metric designator 16 (trade size 1/2)**] [**Metric designator 21 (trade size 3/4)**].
 3. Other Available Features Required by the Project:
 - a. Colors: As indicated on the Drawings.

2.3 TYPE ERM C DUCT RACEWAYS, ELBOWS, COUPLINGS, AND NIPPLES

A. UL DYIX - Galvanized-Steel Electrical Rigid Metal Conduit (ERM C-S-G), Elbows, Couplings, and Nipples:

1. Listing Criteria: Investigated, labeled, and marked by qualified electrical testing laboratory in accordance with guide information and standards specified for the following UL product categories:
 - a. UL CCN DYIX; including UL 6.
2. Standard Features:
 - a. Exterior Coating: Zinc.
 - b. Interior Coating: Zinc.
 - c. Minimum Trade Size: Metric designator 16 (trade size 1/2)
3. Other Available Features Required by the Project:
 - a. Colors: As indicated on the Drawings.

2.4 TYPE FMC DUCT RACEWAYS

A. UL DXUZ - Steel Flexible Metal Conduit (FMC-S):

1. Listing Criteria: Investigated, labeled, and marked by qualified electrical testing laboratory in accordance with guide information and standards specified for the following UL product categories:
 - a. UL CCN DXUZ; including UL 1.
2. Standard Features:
 - a. Material: Steel.
 - b. Minimum Trade Size: [**Metric designator 16 (trade size 1/2)**] [**Metric designator 21 (trade size 3/4)**].
3. Other Available Features Required by the Project:
 - a. Colors: As indicated on the Drawings.

2.5 TYPE LFMC DUCT RACEWAYS

A. UL DXHR - Steel Liquidtight Flexible Metal Conduit (LFMC-S):

1. Listing Criteria: Investigated, labeled, and marked by qualified electrical testing laboratory in accordance with guide information and standards specified for the following UL product categories:
 - a. UL CCN DXHR; including UL 360.

2. Standard Features:
 - a. Material: Steel.
 - b. Minimum Trade Size: [**Metric designator 16 (trade size 1/2)**] [**Metric designator 21 (trade size 3/4)**].
3. Other Available Features Required by the Project:
 - a. Colors: As indicated on the Drawings.

2.6 FITTINGS FOR CONDUIT, TUBING, AND CABLE

A. UL DWTT - Fittings for Type ERM C Duct Raceways:

1. Listing Criteria: Investigated, labeled, and marked by qualified electrical testing laboratory in accordance with guide information and standards specified for the following UL product categories:
 - a. UL CCN DWTT; including UL 514B.
2. Standard Features:
 - a. Material: Steel.
 - b. Coupling Method: Compression coupling or Setscrew coupling. Setscrew couplings with only single screw per conduit are unacceptable.
 - c. Expansion and Deflection Fittings: UL 651 with flexible bonding jumper.

B. UL FKAV - Fittings for Type EMT Duct Raceways:

1. Listing Criteria: Investigated, labeled, and marked by qualified electrical testing laboratory in accordance with guide information and standards specified for the following UL product categories:
 - a. UL CCN FKAV; including UL 514B.
2. Standard Features:
 - a. Material: Steel.
 - b. Coupling Method: Compression coupling or Setscrew coupling. Setscrew couplings with only single screw per conduit are unacceptable.
 - c. Expansion and Deflection Fittings: UL 651 with flexible bonding jumper.

C. UL ILNR - Fittings for Type FMC Duct Raceways:

1. Listing Criteria: Investigated, labeled, and marked by qualified electrical testing laboratory in accordance with guide information and standards specified for the following UL product categories:
 - a. UL CCN ILNR; including UL 514B.

2.7 JOINT COMPOUNDS

A. UL FOIZ - Electrically Conductive Corrosion-Resistant Compound for Threaded Conduit:

1. Listing Criteria: Investigated, labeled, and marked by qualified electrical testing laboratory in accordance with guide information and standards specified for the following UL product categories:
 - a. UL CCN FOIZ; including UL Subject 2419.

PART 3 - EXECUTION

3.1 SELECTION OF CONDUITS FOR ELECTRICAL SYSTEMS

A. Unless more stringent requirements are specified in the Contract Documents or manufacturer's published instructions, comply with NFPA 70 for selection of duct raceways. Consult Architect for resolution of conflicting requirements.

B. Outdoors:

1. Exposed and Subject to Severe Physical Damage: ERMCM.
2. Exposed and Subject to Physical Damage: ERMCM or Corrosion-resistant EMT.
3. Exposed and Not Subject to Physical Damage: ERMCM or Corrosion-resistant EMT.
4. Concealed Aboveground: EMT.
5. Connection to Vibrating Equipment (Including Transformers and Hydraulic, Pneumatic, Electric Solenoid, or Motor-Driven Equipment): LFMC.

C. Indoors:

1. Exposed and Subject to Severe Physical Damage: ERMCM.
2. Exposed and Subject to Physical Damage: ERMCM or EMT.
3. Exposed and Not Subject to Physical Damage: EMT.
4. Concealed in Ceilings and Interior Walls and Partitions: EMT.
5. Damp or Wet Locations: Corrosion-resistant EMT.
6. Connection to Vibrating Equipment (Including Transformers and Hydraulic, Pneumatic, Electric Solenoid, or Motor-Driven Equipment): LFMC.

D. Duct Fittings: Select fittings in accordance with NEMA FB 2.10 guidelines.

1. ERMCM: Provide threaded-type fittings unless otherwise indicated.

3.2 INSTALLATION OF CONDUITS FOR ELECTRICAL SYSTEMS

A. Comply with manufacturer's published instructions.

B. Reference Standards for Installation: Unless more stringent installation requirements are specified in the Contract Documents or manufacturer's published instructions, comply with the following:

1. Electrical Construction: ICC IBC, ICC IFC, NFPA 1, NFPA 70, and NECA NEIS 1.
2. Electrical Safety: NFPA 70E.
3. Grounding and Bonding: NECA NEIS 331 and Article 250 of NFPA 70.
4. Work in Confined Spaces: NFPA 350.
5. Type EMT-S: Article 358 of NFPA 70 and NECA NEIS 101.
6. Type ERMC-S: Article 344 of NFPA 70 and NECA NEIS 101.
7. Type FMC-S: Article 348 of NFPA 70 and NECA NEIS 101.
8. Expansion Fittings: NEMA FB 2.40.
9. Consult Architect for resolution of conflicting requirements.

C. Special Installation Techniques:

1. General Requirements for Installation of Duct Raceways:
 - a. Complete duct raceway installation before starting conductor installation.
 - b. Install no more than equivalent of three 90-degree bends in conduit run. Support within **12 inch (300 mm)** of changes in direction.
 - c. Make bends in duct raceway using large-radius preformed ells except for parallel bends. Field bending must be in accordance with NFPA 70 minimum radii requirements. Provide only equipment specifically designed for material and size involved.
 - d. Conceal conduit within finished walls, ceilings, and floors unless otherwise indicated. Install conduits parallel or perpendicular to building lines.
 - e. Support conduit within **12 inch (300 mm)** of enclosures to which attached.
 - f. Install duct sealing fittings at accessible locations in accordance with NFPA 70 and fill them with listed sealing compound. For concealed duct raceways, install fitting in flush steel box with blank cover plate having finish similar to that of adjacent plates or surfaces. Install duct sealing fittings in accordance with NFPA 70.
 - g. Install devices to seal duct raceway interiors at accessible locations. Locate seals so no fittings or boxes are between the seal and the following changes of environments. Seal interior of duct raceways at the following points:
 - 1) Where conduits pass from warm to cold locations, such as boundaries of refrigerated spaces.
 - 2) Where an underground service duct raceway enters a building or structure.
 - 3) Conduit extending from interior to exterior of building.
 - 4) Conduit extending into pressurized duct raceway and equipment.
 - 5) Conduit extending into pressurized zones that are automatically controlled to maintain different pressure set points.
 - 6) Where otherwise required by NFPA 70.
 - h. Do not install duct raceways or electrical items on "explosion-relief" walls or rotating equipment.
 - i. Do not install conduits within **2 inch (50 mm)** of the bottom side of a metal deck roof.
 - j. Keep duct raceways at least **6 inch (150 mm)** away from parallel runs of flues and steam or hot-water pipes. Install horizontal duct raceway runs above water and steam piping.
 - k. Cut conduit perpendicular to the length. For conduits metric designator 53 (trade size 2) and larger, use roll cutter or a guide to make cut straight and perpendicular to the length. Ream inside of conduit to remove burrs.

- l. Install pull wires in empty duct raceways. Provide polypropylene or monofilament plastic line with not less than **200 lb (90 kg)** tensile strength. Leave at least **12 inch (300 mm)** of slack at both ends of pull wire. Cap underground duct raceways designated as spare above grade alongside duct raceways in use.
 - m. Install duct raceways square to the enclosure and terminate at enclosures without hubs with locknuts on both sides of enclosure wall. Install locknuts hand tight, plus one-quarter turn more.
 - 1) Termination fittings with shoulders do not require two locknuts.
 - n. Terminate threaded conduits into threaded hubs or with locknuts on inside and outside of boxes or cabinets. Install bushings on conduits up to metric designator 35 (trade size 1-1/4) and insulated throat metal bushings on metric designator 41 (trade size 1-1/2) and larger conduits terminated with locknuts..
2. Type ERM C:
 - a. Threaded Conduit Joints, Exposed to Wet, Damp, Corrosive, or Outdoor Conditions: Apply listed compound that maintains electrical conductivity to threads of duct raceway and fittings before making up joints. Follow compound manufacturer's published instructions.
 3. Type F M C:
 - a. Provide a maximum of **36 inch (915 mm)** of flexible conduit for equipment subject to vibration, noise transmission, or movement; and for transformers and motors.
 4. Duct Raceway Terminations at Locations Subject to Moisture or Vibration:
 - a. Provide insulating bushings to protect conductors, including conductors smaller than 4 AWG..
 5. Duct Fittings: Install fittings in accordance with NEMA FB 2.10 guidelines.
 - a. EMT: Provide setscrew or compression, steel fittings. Comply with NEMA FB 2.10.
 - b. Flexible Conduit: Provide only fittings listed for use with flexible conduit type. Comply with NEMA FB 2.20.
 6. Expansion-Joint Fittings:
 - a. Install in runs of aboveground PVC that are located where environmental temperature change may exceed **30 deg F (17 deg C)** and that have straight-run length that exceeds **25 ft (7.6 m)**. Install in runs of aboveground ERM C and EMT conduit that are located where environmental temperature change may exceed **100 deg F (55 deg C)** and that have straight-run length that exceeds **100 ft (30 m)**.
 - b. Install type and quantity of fittings that accommodate temperature change listed for the following locations:
 - 1) Outdoor Locations Not Exposed to Direct Sunlight: **125 deg F (70 deg C)** temperature change.

- 2) Outdoor Locations Exposed to Direct Sunlight: 155 deg F (86 deg C) temperature change.
 - 3) Indoor Spaces Connected with Outdoors without Physical Separation: 125 deg F (70 deg C) temperature change.
 - 4) Attics: 135 deg F (75 deg C) temperature change.
- c. Install fitting(s) that provide expansion and contraction for at least 0.000078 inch per foot of length of straight run per deg F (0.0115 mm per meter of length of straight run per deg C) of temperature change for metal conduits.
 - d. Install expansion fittings at locations where conduits cross building or structure expansion joints.
 - e. Install expansion-joint fitting with position, mounting, and piston setting selected in accordance with manufacturer's published instructions for conditions at specific location at time of installation. Install conduit supports to allow for expansion movement.
7. Duct Raceways Penetrating Rooms or Walls with Acoustical Requirements: Seal duct raceway openings on both sides of rooms or walls with acoustically rated putty or firestopping.
 8. Identification: Provide labels for conduit assemblies, duct raceways, and associated electrical equipment.
 - a. Provide warning signs.

D. Interfaces with Other Work:

1. Firestop penetrations of fire-rated floor and wall assemblies.
2. Provide conduit hangers and supports.

3.3 FIELD QUALITY CONTROL OF CONDUITS FOR ELECTRICAL SYSTEMS

A. Tests and Inspections:

1. Perform manufacturer's recommended tests and inspections.
2. Pull solid aluminum or wood test mandrel through duct to prove joint integrity and adequate bend radii, and test for out-of-round duct. Provide minimum 12 inch (300 mm) long mandrel equal to duct size minus 1/4 inch (6 mm). If obstructions are indicated, remove obstructions and retest.
3. Conduit Placement:
 - a. Verify that center-line location and offsets are in accordance with the Drawings.
 - b. Verify that hangers and supports for conduits are attached to structure.
 - c. Verify that nuts on bolts or hanger rods are secure.
 - d. Verify that space between raceways and cored holes are filled with non-shrinking grout or other approved material indicated on the Drawings and the Specifications.
 - e. Verify that expansion devices are installed at locations indicated on the Drawings and the Specifications.
 - f. Verify that ends are cut square to provide flush-butting surfaces when spliced and inside edges are free of burrs that could impede installation of cables.

- g. Verify minimum separation of utilities, or that approved mechanical protection has been provided to surrounding conduit(s) where minimum separation cannot be achieved.

4. Document all changes on Record Drawings.

B. Nonconforming Work:

- 1. Conduit will be considered defective if it does not pass tests and inspections.
- 2. Remove and replace defective units and retest.

C. Field Quality-Control Reports: Collect, assemble, and submit test and inspection reports.

3.4 CLEANING

- A. Verify that bentonite or other drilling fluids are contained and removed, and site is restored to its original or improved condition.

3.5 PROTECTION

- A. Protect coatings, finishes, and cabinets from damage and deterioration.

- 1. Repair damage to galvanized finishes with zinc-rich paint recommended by manufacturer.
- 2. Repair damage to PVC coatings or paint finishes with matching touchup coating recommended by manufacturer.

END OF SECTION 260533.13

SECTION 260533.16 - BOXES AND COVERS FOR ELECTRICAL SYSTEMS

PART 1 - GENERAL

1.1 SUMMARY

A. Section Includes:

1. Junction boxes and pull boxes.

B. Related Requirements:

1. Section 018116 "Facility Environmental Requirements" specifies temperature, humidity, acoustical, and other field conditions applicable to the Work specified in this Section.
2. Section 018123 "Facility Seismic and Wind Criteria" specifies seismic hazard, wind hazard, other structural load design conditions applicable to the Work specified in this Section.
3. Section 078413 "Penetration Firestopping" specifies materials and methods for sealing penetrations of rated walls and partitions referenced by this Section.
4. Section 260526 "Grounding and Bonding for Electrical Systems" specifies grounding and bonding referenced by this Section.
5. Section 260553 "Identification for Electrical Systems" specifies electrical equipment labels and warning signs installed by this Section.

1.2 DEFINITIONS

A. Communications Outlet: One or more communications jacks, or cables and plugs, mounted in a box or ring, with a suitable protective cover.

B. Enclosure: The case or housing of an apparatus, or the fence or wall(s) surrounding an installation, to prevent personnel from accidentally contacting energized parts or to protect the equipment from physical damage. Types of enclosures and enclosure covers include the following:

1. Cabinet: An enclosure that is designed for either surface mounting or flush mounting and is provided with a frame, mat, or trim in which a swinging door or doors are or can be hung.
2. Concrete Box: A box intended for use in poured concrete.
3. Conduit Body: A means for providing access to the interior of a conduit or tubing system through one or more removable covers at a junction or terminal point. In the United States, conduit bodies are listed in accordance with outlet box requirements.
4. Conduit Box: A box having threaded openings or knockouts for conduit, EMT, or fittings.
5. Cover Plate: A cover designed for protecting wiring devices installed in flush-mounted device boxes while permitting their safe operation; also called a faceplate or wallplate.
6. Cutout Box: An enclosure designed for surface mounting that has swinging doors or covers secured directly to and telescoping with the walls of the enclosure.

7. Device Box: A box with provisions for mounting a wiring device directly to the box.
 8. Extension Ring: A ring intended to extend the sides of an outlet box or device box to increase the box depth, volume, or both.
 9. Floor Box: A box mounted in the floor intended for use with a floor box cover and other components to complete the floor box enclosure.
 10. Floor-Mounted Enclosure: A floor box and floor box cover assembly with means to mount in the floor that is sealed against the entrance of scrub water at the floor level.
 11. Floor Nozzle: An enclosure used on a wiring system, intended primarily as a housing for a receptacle, provided with a means, such as a collar, for surface-mounting on a floor, which may or may not include a stem to support it above the floor level, and is sealed against the entrance of scrub water at the floor level.
 12. Junction Box: A box with a blank cover that joins different runs of raceway or cable and provides space for connection and branching of the enclosed conductors.
 13. Outlet Box: A box that provides access to a wiring system having pryout openings, knockouts, threaded entries, or hubs in either the sides or the back, or both, for the entrance of conduit, conduit or cable fittings, or cables, with provisions for mounting an outlet box cover, but without provisions for mounting a wiring device directly to the box.
 14. Pedestal Floor Box Cover: A floor box cover that, when installed as intended, provides a means for typically vertical or near-vertical mounting of receptacle outlets above the floor's finished surface.
 15. Pull Box: A box with a blank cover that joins different runs of raceway and provides access for pulling or replacing the enclosed cables or conductors.
 16. Raised-Floor Box: A floor box intended for use in raised floors.
 17. Recessed Access Floor Box: A floor box with provisions for mounting wiring devices below the floor surface.
 18. Recessed Access Floor Box Cover: A floor box cover with provisions for passage of cords to recessed wiring devices mounted within a recessed floor box.
 19. Ring: A sleeve, which is not necessarily round, used for positioning a recessed wiring device flush with the plaster, concrete, drywall, or other wall surface.
 20. Ring Cover: A box cover, with raised center portion to accommodate a specific wall or ceiling thickness, for mounting wiring devices or luminaires flush with the surface.
 21. Termination Box: An enclosure designed for installation of termination base assemblies consisting of bus bars, terminal strips, or terminal blocks with provision for wire connectors to accommodate incoming or outgoing conductors, or both.
- C. Receptacle: A fixed connecting device arranged for insertion of a power cord plug. Also called a power jack.
- D. Receptacle Outlet: One or more receptacles mounted in a box with a suitable protective cover.

1.3 ACTION SUBMITTALS

- A. Product Data: For each type of product.

1.4 INFORMATIONAL SUBMITTALS

- A. Manufacturer's published instructions.

PART 2 - PRODUCTS

2.1 PERFORMANCE REQUIREMENTS

- A. Regulatory Requirements: Products or components listed and labeled in accordance with NFPA 70, by qualified electrical testing laboratory recognized by authorities having jurisdiction, and marked for intended location and application.

2.2 JUNCTION BOXES AND PULL BOXES

A. UL BGUZ - Indoor Sheet Metal Junction and Pull Boxes:

- 1. Listing Criteria: Investigated, labeled, and marked by qualified electrical testing laboratory in accordance with guide information and standards specified for the following UL product categories:
 - a. UL CCN BGUZ; including UL 50 and UL 50E.
- 2. Standard Features:
 - a. Box with a blank cover that serves the purpose of joining different runs of raceway or cable.
 - b. Degree of Protection: Type 1.

B. UL BGUZ - Indoor Cast-Metal Junction and Pull Boxes:

- 1. Listing Criteria: Investigated, labeled, and marked by qualified electrical testing laboratory in accordance with guide information and standards specified for the following UL product categories:
 - a. UL CCN BGUZ; including UL 50 and UL 50E.
- 2. Standard Features:
 - a. Box with a blank cover that serves the purpose of joining different runs of raceway or cable.
 - b. Degree of Protection: Type 1.

C. UL BGUZ - Outdoor Sheet Metal Junction and Pull Boxes:

- 1. Listing Criteria: Investigated, labeled, and marked by qualified electrical testing laboratory in accordance with guide information and standards specified for the following UL product categories:
 - a. UL CCN BGUZ; including UL 50 and UL 50E.
- 2. Standard Features:
 - a. Box with a blank cover that serves the purpose of joining different runs of raceway or cable.

- b. Degree of Protection: Type 3R.

D. UL BGUZ - Outdoor Cast-Metal Junction and Pull Boxes:

1. Listing Criteria: Investigated, labeled, and marked by qualified electrical testing laboratory in accordance with guide information and standards specified for the following UL product categories:
 - a. UL CCN BGUZ; including UL 50 and UL 50E.
2. Standard Features:
 - a. Box with a blank cover that serves the purpose of joining different runs of raceway or cable.
 - b. Degree of Protection: Type 3R.

PART 3 - EXECUTION

3.1 SELECTION OF BOXES AND COVERS FOR ELECTRICAL SYSTEMS

- A. Unless more stringent requirements are specified in Contract Documents or manufacturers' published instructions, comply with NFPA 70 for selection of boxes and enclosures. Consult Architect for resolution of conflicting requirements.
- B. Degree of Protection:
 1. Outdoors:
 - a. Type 3R unless otherwise indicated.
 2. Indoors:
 - a. Type 1 unless otherwise indicated.
- C. Exposed Boxes Installed Less Than 2.5 m (8 ft) Above Floor:
 1. Provide cast-metal boxes Boxes with knockouts or unprotected openings are prohibited.

3.2 INSTALLATION OF BOXES AND COVERS FOR ELECTRICAL SYSTEMS

- A. Comply with manufacturer's published instructions.
- B. Reference Standards for Installation: Unless more stringent installation requirements are specified in Contract Documents or manufacturers' published instructions, comply with the following:
 1. Electrical Construction: ICC IBC, ICC IFB, NFPA 1, NFPA 70, and NECA NEIS 1.
 2. Electrical Safety: NFPA 70E.
 3. Grounding and Bonding: NECA NEIS 331 and Article 250 of NFPA 70.

4. Work in Confined Spaces: NFPA 350.
5. Outlet, Device, Pull, and Junction Boxes: Article 314 of NFPA 70.

C. Special Installation Techniques:

1. Provide boxes in wiring and raceway systems wherever required for pulling of wires, making connections, and mounting of devices or fixtures.
2. Fasten junction and pull boxes to, or support from, building structure. Do not support boxes by conduits.
3. Do not rely on locknuts to penetrate nonconductive coatings on enclosures. Remove coatings in the locknut area prior to assembling conduit to enclosure to ensure a continuous ground path.

D. Interfaces with Other Work:

1. Identification: Provide labels for boxes and associated electrical equipment.
 - a. Identify field-installed conductors, interconnecting wiring, and components.
 - b. Label each enclosure with engraved metal or laminated-plastic nameplate.
 - c. Provide warning signs and arc-flash hazard warning labels for electrical equipment.

3.3 FIELD QUALITY CONTROL OF BOXES AND COVERS

A. Tests and Inspections:

1. Perform manufacturer's recommended tests and inspections.
2. Perform tests and inspections recommended by standards listed in "Reference Standards for Installation" Paragraph.

B. Nonconforming Work:

1. Boxes and covers will be considered defective if they do not pass tests and inspections.
2. Remove and replace defective units and retest.

C. Field Quality-Control Reports: Collect, assemble, and submit test and inspection reports.

3.4 CLEANING

- A. Remove construction dust and debris from boxes before installing cover plates, covers, and hoods.

3.5 PROTECTION

- A. After installation, protect boxes from construction activities. Remove and replace items that are contaminated, defaced, damaged, or otherwise caused to be unfit for use prior to acceptance by Owner.

END OF SECTION 260533.16

SECTION 260553 - IDENTIFICATION FOR ELECTRICAL SYSTEMS

PART 1 - GENERAL

1.1 SUMMARY

A. Section Includes:

1. Labels.
2. Bands.
3. Signs.

1.2 ACTION SUBMITTALS

A. Product Data: For each type of product.

B. Identification Schedule: For each piece of electrical equipment and electrical system components to be index of nomenclature for electrical equipment and system components used in identification signs and labels. Use same designations indicated on Drawings.

PART 2 - PRODUCTS

2.1 LABELS

A. Performance Criteria:

1. Regulatory Requirements: Listed and labeled in accordance with NFPA 70, by qualified electrical testing laboratory recognized by authorities having jurisdiction, and marked for intended location and application.
2. Listing Criteria: UL CCN PGDQ2 for components; including UL 969.

B. UL PGDQ2 - Vinyl Wraparound Labels: Preprinted, flexible labels laminated with clear, weather- and chemical-resistant coating and matching wraparound clear adhesive tape for securing label ends.

C. UL PGDQ2 - Self-Adhesive Wraparound Labels: Preprinted or Write-on, 3 mil (0.08 mm) thick, polyester or vinyl flexible label with acrylic pressure-sensitive adhesive.

1. Self-Lamination: Clear; UV-, weather-, and chemical-resistant; self-laminating, with protective shield over legend. Size labels such that clear shield overlaps entire printed legend.
2. Marker for Labels:
 - a. Permanent, waterproof, black ink marker recommended by tag manufacturer.
 - b. Machine-printed, permanent, waterproof, black ink recommended by printer manufacturer.

- D. UL PGDQ2 - Self-Adhesive Labels: Polyester or Vinyl, thermal, transfer-printed, **3 mil (0.08 mm)** thick, multicolor, weather- and UV-resistant, pressure-sensitive adhesive labels, configured for intended use and location.

1. Minimum Nominal Size:

- a. **1-1/2 by 6 inch (37 by 150 mm)** for raceway and conductors.
- b. **3-1/2 by 5 inch (76 by 127 mm)** for equipment.
- c. As required by authorities having jurisdiction.

2.2 BANDS

- A. Snap-Around Labels: Slit, pretensioned, flexible, preprinted, color-coded acrylic sleeves, with diameters sized to suit diameters and that stay in place by gripping action.
- B. Snap-Around, Color-Coding Bands: Slit, pretensioned, flexible, solid-colored acrylic sleeves, **2 inch (50 mm)** long, with diameters sized to suit diameters and that stay in place by gripping action.

2.3 SIGNS

- A. Laminated Acrylic or Melamine Plastic Signs:

1. Engraved legend.
2. Thickness:
 - a. For signs up to **20 sq. inch (129 sq. cm)**, minimum **1/16 inch (1.6 mm)** thick.
 - b. For signs larger than **20 sq. inch (129 sq. cm)**, **1/8 inch (3.2 mm)** thick.
 - c. Engraved legend with black letters on white face.
 - d. Self-adhesive.
 - e. Framed with mitered acrylic molding and arranged for attachment at applicable equipment.

PART 3 - EXECUTION

3.1 PREPARATION

- A. Self-Adhesive Identification Products: Before applying electrical identification products, clean substrates of substances that could impair bond, using materials and methods recommended by manufacturer of identification product.

3.2 SELECTION OF COLORS AND IDENTIFICATION MARKINGS

- A. Comply with 29 CFR 1910.144 for color identification of hazards, and the following:
1. Ceiling-mounted hangers, supports, cable trays, and raceways must be finished, painted, or suitably marked safety yellow where less than **7.7 ft (2.3 m)** above finished floor.

- B. Pipe and Conduit Labeling: Comply with ASME A13.1.
- C. Color-Coding for Phase- and Voltage-Level Identification, 1000 V or Less: Use colors listed below for ungrounded feeder and branch-circuit conductors.
 - 1. Color must be factory applied or field applied for sizes larger than 6 AWG when permitted by authorities having jurisdiction.
 - 2. Colors for 208Y/120 V Circuits:
 - a. Phase A: Black.
 - b. Phase B: Red.
 - c. Phase C: Blue.
 - 3. Colors for 240 V Circuits:
 - a. Phase A: Black.
 - b. Phase B: Red.
 - 4. Colors for 480Y/277 V Circuits:
 - a. Phase A: Brown.
 - b. Phase B: Orange.
 - c. Phase C: Yellow.
 - 5. Color for Neutral (Grounded Conductor): White.
 - 6. Color for Equipment Ground: **[Bare copper] [Green] [Green with yellow stripe]**.
- D. Color-Coding Instructional Signs: Self-adhesive labels, including color code for grounded and ungrounded conductors.
- E. Identify conductors, cables, and terminals in enclosures and at junctions, terminals, pull points, and locations of high visibility. Identify by system and circuit designation.
- F. Pull and Junction Boxes, 1000 V or Less: For conductors in pull and junction boxes, use snap-around color-coding bands to identify phase.
 - 1. Locate identification at changes in direction, at penetrations of walls and floors, at **50 ft (15 m)** maximum intervals in straight runs, and at **25 ft (7.6 m)** maximum intervals in congested areas.
 - 2. Identify system voltage with black letters on orange field.
- G. Accessible Raceways, 1000 V or Less, for Feeder, and Branch Circuits, More Than 30 A and 120 V to Ground: Identify with self-adhesive raceway labels.
 - 1. Locate identification at changes in direction, at penetrations of walls and floors, at **50 ft (15 m)** maximum intervals in straight runs, and at **25 ft (7.6 m)** maximum intervals in congested areas.
 - 2. Identify system voltage with black letters on orange field.

H. Equipment Identification Labels:

1. Black letters on white field.
2. Indoor Equipment: Self-adhesive label.
3. Outdoor Equipment: Laminated acrylic or melamine sign.
4. Equipment to Be Labeled:
 - a. Enclosures and electrical cabinets.
 - b. Access doors and panels for concealed electrical items.
 - c. Enclosed switches.
 - d. Uninterruptible power supplies.
 - e. Battery-inverter units.
 - f. Battery racks.
 - g. Power-generating units.
 - h. Monitoring and control equipment.

3.3 SELECTION OF SIGNS AND HAZARD MARKINGS

A. Comply with 29 CFR 1910.145 for danger, caution, warning, and safety instruction signs.

B. Signs, labels, and tags required for personnel safety must comply with the following standards:

1. Safety Colors: NEMA Z535.1.
2. Facility Safety Signs: NEMA Z535.2.
3. Safety Symbols: NEMA Z535.3.
4. Product Safety Signs and Labels: NEMA Z535.4.
5. Safety Tags and Barricade Tapes for Temporary Hazards: NEMA Z535.5.

C. Electrical Hazard Warnings:

1. Arc-Flash Hazard Warning: Self-adhesive labels. Comply with NFPA 70E requirements for arc-flash hazard warning labels.
2. Raceways and Cables Carrying Circuits at More Than 1000 V:
 - a. Black letters on orange field.
 - b. Legend: "DANGER - CONCEALED HIGH VOLTAGE WIRING."
3. Multiple Power Sources Warning Legend: "DANGER - ELECTRICAL SHOCK HAZARD - EQUIPMENT HAS MULTIPLE POWER SOURCES."
4. OSHA Workspace Clearance Warning Legend: "WARNING - OSHA REGULATION - AREA IN FRONT OF ELECTRICAL EQUIPMENT MUST BE KEPT CLEAR FOR 3 FEET MINIMUM."

D. Warning Labels for Indoor Cabinets, Boxes, and Enclosures for Power and Lighting: Self-adhesive labels.

1. Apply to exterior of door, cover, or other access.
2. For equipment with multiple power or control sources, apply to door or cover of equipment, including, but not limited to, the following:

- a. Power-transfer switches.
 - b. Controls with external control power connections.
- E. Operating Instruction Signs: Self-adhesive labels.

3.4 INSTALLATION

- A. Install identification materials and devices at locations for most convenient viewing without interference with operation and maintenance of equipment. Install access doors or panels to provide view of identifying devices.
- B. Thermal Movements: Allow for thermal movements from ambient and surface temperature changes typical for electrical equipment environments specified in Section 260011 "Facility Performance Requirements for Electrical."
- C. Paint: Comply with requirements in painting Sections for paint materials and application requirements. Retain paint system applicable for surface material and location (exterior or interior).
- D. Fasteners for Labels and Signs: Self-tapping, stainless steel screws or stainless steel machine screws with nuts and flat and lock washers.
- E. Verify and coordinate identification names, abbreviations, colors, and other features with requirements in other Sections requiring identification applications, Drawings, Shop Drawings, manufacturer's wiring diagrams, and operation and maintenance manual. Use consistent designations throughout Project.
- F. Install identifying devices before installing acoustical ceilings and similar concealment.
- G. Verify identity of item before installing identification products.
- H. Coordinate identification with Project Drawings, manufacturer's wiring diagrams, and operation and maintenance manual.
- I. Apply identification devices to surfaces that require finish after completing finish work.
- J. Install signs with approved legend to facilitate proper identification, operation, and maintenance of electrical systems and connected items.
- K. Elevated Components: Increase sizes of labels, signs, and letters to those appropriate for viewing from floor.
- L. Vinyl Wraparound Labels:
 - 1. Secure tight to surface of raceway or cable at location with high visibility and accessibility.
 - 2. Attach labels that are not self-adhesive type with clear vinyl tape, with adhesive appropriate to location and substrate.
- M. Snap-Around Labels: Secure tight to surface at location with high visibility and accessibility.

- N. Self-Adhesive Wraparound Labels: Secure tight to surface at location with high visibility and accessibility.
- O. Snap-Around Color-Coding Bands: Secure tight to surface at location with high visibility and accessibility.
- P. Heat-Shrink, Preprinted Tubes: Secure tight to surface at location with high visibility and accessibility.
- Q. Marker Tapes: Secure tight to surface at location with high visibility and accessibility.
- R. Self-Adhesive Vinyl Tape: Secure tight to surface at location with high visibility and accessibility.
 - 1. Field-Applied, Color-Coding Conductor Tape: Apply in half-lapped turns for minimum distance of 6 inch (150 mm) where splices or taps are made. Apply last two turns of tape with no tension to prevent possible unwinding.
- S. Tape and Stencil: Comply with requirements in painting Sections for surface preparation and paint application.
- T. Baked-Enamel Signs: Attach signs that are not self-adhesive type with mechanical fasteners appropriate to location and substrate.
- U. Metal-Backed Butyrate Signs: Attach signs that are not self-adhesive type with mechanical fasteners appropriate to location and substrate.
- V. Laminated Acrylic or Melamine Plastic Signs: Attach signs that are not self-adhesive type with mechanical fasteners appropriate to location and substrate.

END OF SECTION 260553

SECTION 26 05 75 – ELECTRICAL CONNECTIONS TO EQUIPMENT

PART 1 - GENERAL

1.1 SUMMARY

- A. Section Includes: Electrical connections to Owner-furnished equipment, and equipment specified under other Divisions and Sections.

1.2 REFERENCES

- A. NFPA 70 - National Electrical Code
- B. NEMA WD 1 - General Purpose Wiring Devices
- C. NEMA WD 6 - Wiring Device Configurations

1.3 SUBMITTALS

- A. Conform to Division 1 requirements.
- B. Product Data: Provide wiring device manufacturer's catalog information showing dimensions, configurations, and construction.
- C. Manufacturer's Instructions: Indicate application conditions and limitations of use stipulated by Product testing agency specified under Regulatory Requirements. Include instructions for storage, handling, protection, examination, preparation, installation, and starting of Product.

1.4 REGULATORY REQUIREMENTS

- A. Electrical: Conform to NFPA 70.
- B. Furnish products listed and classified by Underwriters Laboratories, Inc. as suitable for purpose specified and shown.

PART 2 - PRODUCTS

2.1 CORDS AND CAPS

- A. Attachment Plug Construction: Conform to NEMA WD 1.
- B. Configuration: NEMA WD 6; match receptacle configuration at outlet provided for equipment.

- C. Cord Construction: NFPA 70; Type SJO multiconductor flexible cord with identified equipment grounding conductor, suitable for use in damp locations.
- D. Size: Suitable for connected load of equipment, length of cord, and rating of branch circuit overcurrent protection.

PART 3 - EXECUTION

3.1 COORDINATION

- A. Coordinate work with systems and other trades under provisions of Division 1.
- B. Obtain and review shop drawings, product data, and manufacturer's instructions for Owner-Furnished equipment, and equipment furnished under other Divisions and Sections.
- C. Determine connection locations and requirements prior to roughing in.
- D. Sequence rough-in of electrical connections to coordinate with installation schedule for equipment.
- E. Sequence electrical connections to coordinate with start-up schedule for equipment.

3.2 EXAMINATION

- A. Verify that equipment is ready for electrical connection, wiring, and energization.
- B. Verify that equipment supplied matches submittal data.

3.3 CONNECTIONS

- A. Make electrical connections in accordance with equipment manufacturer's instructions.
- B. Make conduit connections to equipment using flexible conduit. Use liquidtight flexible conduit with watertight connectors in damp or wet locations.
- C. Make wiring connections using wire and cable with insulation suitable for temperatures encountered in heat producing equipment.
- D. Make wiring connections using wire and cable with insulation suitable for temperatures encountered in heat producing equipment.
- E. Provide suitable strain-relief clamps and fittings for cord connections at outlet boxes and equipment connection boxes.
- F. Install disconnect switches, controllers, control stations, and control devices as indicated.
- G. Modify equipment control wiring with terminal block jumpers as indicated.

- H. Provide interconnecting conduit and wiring between devices and equipment where indicated or required.

END OF SECTION 26 05 75

SECTION 262816 - ENCLOSED SWITCHES AND CIRCUIT BREAKERS

PART 1 - GENERAL

1.1 SUMMARY

A. Section Includes:

1. Nonfusible switches.
2. Enclosures.

B. Related Requirements:

1. Section 018116 "Facility Environmental Requirements" specifies basis-of-design environmental conditions and performance criteria that are applicable to product selection and installation of the Work on the Project.
2. Section 018123 "Facility Seismic and Wind Criteria" specifies basis-of-design seismic and wind criteria for nonstructural components on the Project.

1.2 DEFINITIONS

- A. GFEP: Ground-fault circuit-interrupter for equipment protection.
- B. GFLS: Ground-fault circuit-interrupter for life safety.
- C. SPDT: Single pole, double throw.

1.3 ACTION SUBMITTALS

A. Product Data:

1. For each type of enclosed switch, circuit breaker, accessory, and component indicated. Include nameplate ratings, dimensioned elevations, sections, weights, and manufacturers' technical data on features, performance, electrical characteristics, ratings, accessories, and finishes.
2. Enclosure types and details for types other than UL 50E, Type 1.
3. Current and voltage ratings.

B. Shop Drawings: For enclosed switches and circuit breakers.

1. Include plans, elevations, sections, details, and attachments to other work.

C. Field Quality-Control Submittals:

1. Field quality-control reports.

1.4 INFORMATIONAL SUBMITTALS

- A. Sample warranties.

1.5 CLOSEOUT SUBMITTALS

- A. Operation and Maintenance Data: For enclosed switches to include in emergency, operation, and maintenance manuals.
 - 1. In addition to items specified in Section 017823 "Operation and Maintenance Data," include the following:
 - a. Manufacturer's written instructions for testing and adjusting enclosed switches.
- B. Warranty documentation.

1.6 WARRANTY

- A. Special Installer Extended Warranty: Installer warrants that fabricated and installed enclosed switches perform in accordance with specified requirements and agrees to repair or replace components or products that fail to perform as specified within extended-warranty period.
 - 1. Extended-Warranty Period: Two years from date of Substantial Completion; full coverage for labor, materials, and equipment.
- B. Special Manufacturer Extended Warranty: Manufacturer warrants that enclosed switches perform in accordance with specified requirements and agrees to provide repair or replacement of components or products that fail to perform as specified within extended-warranty period.
 - 1. Extended-Warranty Period: Three years from date of Substantial Completion; full coverage for labor, materials, and equipment.

PART 2 - PRODUCTS

2.1 GENERAL REQUIREMENTS

- A. Source Limitations: Obtain products from single manufacturer.
- B. Product Selection for Restricted Space: Drawings indicate maximum dimensions for enclosed switches and circuit breakers, including clearances between enclosures, and adjacent surfaces and other items. Comply with indicated maximum dimensions.
- C. Electrical Components, Devices, and Accessories: Listed and labeled in accordance with NFPA 70, by qualified electrical testing laboratory recognized by authorities having jurisdiction, and marked for intended location and application.

2.2 NONFUSIBLE SWITCHES

- A. Subject to conformance with requirements, provide products by one of the following:
 - 1. Siemens ITE.
 - 2. Eaton Cutler hammer.
 - 3. General Electric.
 - 4. Square D.
- B. Type GD, General Duty, Three Pole, Single Throw, 240 V(ac), 600 A and Smaller: UL 98 and NEMA KS 1, horsepower rated, lockable handle with capability to accept two padlocks, and interlocked with cover in closed position.
- C. Accessories:
 - 1. Equipment Ground Kit: Internally mounted and labeled for copper and aluminum ground conductors.
 - 2. Neutral Kit: Internally mounted; insulated, capable of being grounded and bonded; labeled for copper and aluminum neutral conductors.
 - 3. Lugs: Mechanical type, suitable for number, size, and conductor material.

2.3 ENCLOSURES

- A. Enclosed Switches: UL 489, NEMA KS 1, UL 50E, and UL 50, to comply with environmental conditions at installed location.
- B. Enclosure Finish: Enclosure must be finished with gray baked enamel paint, electrodeposited on cleaned, phosphatized steel (UL 50E Type 1) or a brush finish on Type 304 stainless steel (UL 50E Type 4-4X stainless steel).

PART 3 - EXECUTION

3.1 EXAMINATION

- A. Examine elements and surfaces to receive enclosed switches for compliance with installation tolerances and other conditions affecting performance of the Work.
- B. Proceed with installation only after unsatisfactory conditions have been corrected.
 - 1. Commencement of work will indicate Installer's acceptance of areas and conditions as satisfactory.

3.2 SELECTION OF ENCLOSURES

- A. Indoor, Dry and Clean Locations: UL 50E, Type 1.
- B. Outdoor Locations: UL 50E, Type 4X.

3.3 INSTALLATION

- A. Comply with manufacturer's published instructions.
- B. Special Techniques:
 - 1. Coordinate layout and installation of switches, circuit breakers, and components with equipment served and adjacent surfaces. Maintain required workspace clearances and required clearances for equipment access doors and panels.
 - 2. Install individual wall-mounted switches and circuit breakers with tops at uniform height unless otherwise indicated.
 - 3. Comply with mounting and anchoring requirements specified in Section 260548.16 "Seismic Controls for Electrical Systems."
 - 4. Temporary Lifting Provisions: Remove temporary lifting of eyes, channels, and brackets and temporary blocking of moving parts from enclosures and components.
 - 5. Install fuses in fusible devices.

3.4 IDENTIFICATION

- A. Comply with requirements in Section 260553 "Identification for Electrical Systems."
 - 1. Identify field-installed conductors, interconnecting wiring, and components; provide warning signs.
 - 2. Label each enclosure with engraved metal or laminated-plastic nameplate.

3.5 FIELD QUALITY CONTROL

- A. Tests and Inspections for Switches:
 - 1. Visual and Mechanical Inspection:
 - a. Inspect physical and mechanical condition.
 - b. Inspect anchorage, alignment, grounding, and clearances.
 - c. Verify that unit is clean.
 - d. Verify blade alignment, blade penetration, travel stops, and mechanical operation.
 - e. Inspect bolted electrical connections for high resistance using one of the following methods:
 - 1) Use low-resistance ohmmeter.
 - a) Compare bolted connection resistance values to values of similar connections. Investigate values that deviate from those of similar bolted connections by more than 50 percent of lowest value.
 - 2) Verify tightness of accessible bolted electrical connections by calibrated torque-wrench method in accordance with manufacturer's published data or NETA ATS Table 100.12.

- a) Bolt-torque levels must be in accordance with manufacturer's published data. In absence of manufacturer's published data, use NETA ATS Table 100.12.
 - f. Verify correct phase barrier installation.
 - g. Verify lubrication of moving current-carrying parts and moving and sliding surfaces.
- B. Nonconforming Work:
 - 1. Enclosed switches will be considered defective if they do not pass tests and inspections.
 - 2. Remove and replace defective units and retest.
- C. Collect, assemble, and submit test and inspection reports.
 - 1. Test procedures used.
 - 2. Include identification of each enclosed switch tested and describe test results.
 - 3. List deficiencies detected, remedial action taken, and observations after remedial action.

3.6 ADJUSTING

- A. Adjust moving parts and operable components to function smoothly, and lubricate as recommended by manufacturer.

3.7 PROTECTION

- A. After installation, protect enclosed switches and circuit breakers from construction activities. Remove and replace items that are contaminated, defaced, damaged, or otherwise caused to be unfit for use prior to acceptance by Owner.

END OF SECTION 262816